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**Modern statistical inference in non-identifiable models: the
case of probabilistic PCA**

par

Arghya Datta

Department of Mathematics and Statistics
Faculté des arts et des sciences

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Modern statistical inference in non-identifiable models: the case of probabilistic PCA

présentée par

Arghya Datta

a été évaluée par un jury composé des personnes suivantes :

Guy Wolf

(président-rapporteur)

Philippe Gagnon

(directeur de recherche)

Florian Maire

(codirecteur)

Janie Coulombe

(membre du jury)

David A. Stephens

(examineur externe)

(représentant du doyen de la FESP)

Résumé

L'analyse en composantes principales (ACP) est une technique statistique fondamentale ayant de nombreuses applications en apprentissage automatique, en biologie, en finance quantitative et en science actuarielle. En science actuarielle, par exemple, l'ACP peut être utilisée pour réduire la dimension de données de courbes de taux, ce qui permet de résumer l'information essentielle contenue dans des ensembles de données de grande dimension tout en facilitant leur exploration et leur interprétation en vue de la prévision des prix futurs des actifs et des mouvements économiques. D'un point de vue statistique, l'ACP apparaît comme la solution du maximum de vraisemblance du modèle d'analyse en composantes principales probabiliste (ACPP) dans la limite où le bruit gaussien tend vers zéro. Malgré son usage répandu, une compréhension théorique approfondie de la solution du maximum de vraisemblance du modèle ACPP faisait défaut, principalement en raison de la non-identifiabilité du modèle. Dans cette thèse, nous proposons un cadre général, inspiré d'idées issues de la topologie quotient, afin de traiter de tels problèmes de non-identifiabilité.

Cependant, le modèle ACPP avec bruit gaussien est bien connu pour sa sensibilité aux observations extrêmes, ou valeurs aberrantes, pouvant conduire à des conclusions trompeuses. Nous proposons ensuite une alternative bayésienne robuste à l'ACPP, qui en préserve les principaux avantages tout en offrant une meilleure fiabilité en présence de valeurs aberrantes. Notre modèle d'analyse en composantes principales bayésienne (ACPB) remplace le bruit gaussien par des distributions à queues lourdes, telles que la loi de Student, afin de limiter l'influence des valeurs aberrantes en attribuant des probabilités non négligeables aux événements de queue. En nous appuyant sur des travaux antérieurs en régression linéaire robuste, nous montrons théoriquement que le modèle robuste d'ACPB proposé atteint la robustesse, en estimant adéquatement les paramètres du modèle sous des hypothèses raisonnables sur le nombre de valeurs aberrantes. Ici, la robustesse renvoie à un impact borné ou asymptotiquement nul des valeurs aberrantes sur les estimateurs lorsque celles-ci deviennent de plus en plus extrêmes. Nous étayons nos résultats théoriques par des expériences numériques et utilisons l'inférence variationnelle par ascension coordonnée (IVAC) pour calculer des estimateurs robustes, fournissant ainsi une

solution complète. Par souci d'exhaustivité, nous proposons également, dans ce contexte, une étude comparative de l'inférence variationnelle et de l'échantillonnage de Gibbs.

Enfin, à l'aide d'outils issus des systèmes dynamiques et en tirant parti de développements récents dans la littérature sur l'inférence variationnelle, nous établissons des garanties de convergence pour l'IVAC appliquée à l'ACPB avec erreurs gaussiennes.

Mots-clés : consistance, inférence variationnelle par ascension coordonnée, réduction de dimension, échantillonneur de Gibbs, distributions à queues lourdes, valeurs aberrantes, itération de puissance, espace quotient, robustesse, résolution de conflit

Abstract

Principal Component Analysis (PCA) is a fundamental statistical technique with broad applications in machine learning, biology, quantitative finance, and actuarial science. For example, in actuarial science, PCA can be used to reduce the dimensionality of yield curve data, thereby summarizing key information in high-dimensional datasets while facilitating their exploration and interpretation for predicting future asset prices and economic movements. Statistically, PCA arises as the Maximum Likelihood (ML) solution to the Probabilistic PCA (PPCA) model in the limit of vanishing Gaussian noise. Despite its widespread use, a strong theoretical understanding of the ML solution to the PPCA model was missing, primarily because the model is non-identifiable. In this thesis, we propose a general framework, inspired by ideas from quotient topology, to address such non-identifiability issues arising in statistics.

However, PPCA with Gaussian noise is well known to be sensitive to extreme observations, or outliers, which can lead to misleading conclusions. We subsequently propose a robust Bayesian alternative to PPCA that preserves its main advantages while offering improved reliability in the presence of outliers. Our robust Bayesian PCA (BPCA) model replaces Gaussian noise with heavy-tailed distributions, such as the Student's t distribution, in order to limit the influence of outliers by assigning non-negligible probabilities to tail events. Building on prior work in robust linear regression, we theoretically show that proposed robust BPCA achieves robustness, accurately estimating model parameters under reasonable assumptions on the number of outliers. Here, robustness refers to a bounded or vanishing impact on parameter estimates as outliers become more extreme. We support our theoretical findings with numerical experiments and use the Coordinate Ascent Variational Inference (CAVI) algorithm to compute robust estimators, thus providing an end-to-end solution. For completeness, we also provide a comparative study of variational inference and Gibbs sampling in this context.

Finally, using tools from dynamical systems and leveraging recent advances in the variational inference literature, we establish convergence guarantees for CAVI applied to BPCA with Gaussian errors.

Keywords: Consistency, coordinate ascent variational inference, dimension reduction, Gibbs sampler, heavy-tailed distributions, outliers, power iteration, quotient space, robustness, resolution of conflict

To Baba, Didi, and Ma

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List of Acronyms and Abbreviations

ACP	Analyse en composantes principales
ACPB / BPCA	Analyse en composantes principales bayésienne, de l’anglais <i>Bayesian Principal Component Analysis</i>
IVAC / CAVI	Inférence variationnelle par ascension coordonnée, de l’anglais <i>Coordinate Ascent Variational Inference</i>
MCMC	Monte Carlo par chaînes de Markov, de l’anglais <i>Markov Chain Monte Carlo</i>
MSE	Erreur quadratique moyenne, de l’anglais <i>Mean Square Error</i>
PCA	Analyse en composantes principales, de l’anglais <i>Principal Component Analysis</i>
ACPP / PPCA	Analyse en composantes principales probabiliste, de l’anglais <i>Probabilistic Principal Component Analysis</i>

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... To be written in the final version...

Introduction

We begin this thesis with an informal discussion of non-identifiable statistical models since they are the central objects of study in this work. In simple words, a statistical model is identifiable if two different model parameter values always produce two different probability distributions for the observed data. When identifiability fails, different choices of parametrization may give rise to exactly the same statistical model, and thus the data cannot distinguish between them.

To make this precise, we now give the formal definition. Let Θ be a parameter space and let

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

be a parametric statistical model, where P_θ denotes the distribution of the observed data under the parameter value θ . $\mathcal{P}$ is said to be identifiable if the map

$$\theta \longmapsto P_\theta$$

from Θ into $\mathcal{P}$ is injective; equivalently, whenever $P_{\theta_1} = P_{\theta_2}$, one must have $\theta_1 = \theta_2$ [10]. When this map is not injective, the model is non-identifiable. A classical example is a finite mixture model, where permuting the labels of the mixture components leaves the distribution unchanged. Similar ambiguities also arise in factor analysis and in principal component analysis (PCA) and its probabilistic variants. Statistical inference is therefore more challenging in non-identifiable models: estimators, posterior distributions, and optimization algorithms may only recover an entire equivalence class rather than a unique parameter value. This phenomenon will be a recurring theme in the upcoming chapters, where we will repeatedly see how non-identifiability affects both theory and computation.

Although Chapter 1 provides a fairly broad treatment of non-identifiability and of the mathematical issues it creates, the main focus of the thesis is centered on different probabilistic variants of PCA. In particular, we focus on how non-identifiability appears in probabilistic and Bayesian formulations of PCA, how it affects consistency, robustness, and variational inference, and how one can still carry out meaningful statistical inference despite these ambiguities.

0.1. Principal component analysis

PCA is one of the most fundamental tools in statistics and data analysis. Given a data matrix

$$\mathbf{X} \in \mathbb{R}^{n \times d},$$

whose rows represent n observations in $\mathbb{R}^d$, PCA aims to construct a low-dimensional representation of the data that preserves as much variability as possible. Since its origin, PCA has become a standard dimension reduction technique in statistics [4], machine learning [2], genomics [7], finance [6], and many other areas where high-dimensional data must be summarized.

Throughout this thesis, PCA plays *the* role of the central statistical method. At its core, PCA provides a deterministic low-rank approximation to a data matrix $\mathbf{X}$ (see Theorem 1). PCA leads naturally to probabilistic principal component analysis (PPCA) [12], a probabilistic model where the observations are generated from latent variables defined in chapter 1. This probabilistic viewpoint is especially useful when one wishes to study uncertainty quantification and robustness when the data is contaminated with outliers as we shall explore in the later chapters of the thesis.

0.1.1. Singular value decomposition and best rank k approximation

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$, and let

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

be a singular value decomposition (SVD) of $\mathbf{X}$ (see [1]), where

$$\mathbf{U} \in \mathbb{R}^{n \times r}, \quad \mathbf{V} \in \mathbb{R}^{d \times r}, \quad \mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_r),$$

with

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0,$$

and $r = \text{rank}(\mathbf{X})$. Writing $\mathbf{U} = (u_1, \dots, u_r)$ and $\mathbf{V} = (v_1, \dots, v_r)$, the vectors u_j and v_j denote the j th columns of orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$, respectively. The columns of $\mathbf{V}$ are called the right singular vectors of $\mathbf{X}$, and the columns of $\mathbf{U}$ are the left singular vectors.

For a fixed integer $k \leq r$, define the truncated SVD

$$\mathbf{X}_{(k)} := \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top = \sum_{j=1}^k \sigma_j u_j v_j^\top,$$

where $\mathbf{U}_k$, $\mathbf{\Sigma}_k$, and $\mathbf{V}_k$ are obtained by retaining only the first k columns, respectively. $\mathbf{X}_{(k)}$ is the classical PCA approximation of rank k . The reason PCA is important is that $\mathbf{X}_{(k)}$ is not merely a convenient rank- k approximation; it is in fact the *best* rank- k approximation to $\mathbf{X}$ as made precise in the following theorem.

Theorem 1 (Eckart–Young). *Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ have singular value decomposition $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. Then for every $1 \leq k < \text{rank}(\mathbf{X})$,*

$$\mathbf{X}_{(k)} = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$$

solves

$$\arg \min_{\text{rank}(\mathbf{M})=k} \|\mathbf{X} - \mathbf{M}\|_F.$$

Moreover,

$$\|\mathbf{X} - \mathbf{X}_{(k)}\|_F^2 = \sum_{j=k+1}^r \sigma_j^2.$$

Proof. The singular value decomposition allows us to write:

$$\mathbf{X} = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^\top,$$

where each $\mathbf{u}_j \mathbf{v}_j^\top$ is a rank one matrix. Moreover, these rank one matrices are orthonormal since, for all i, j ,

$$\text{tr}((\mathbf{u}_i \mathbf{v}_i^\top)^\top (\mathbf{u}_j \mathbf{v}_j^\top)) = \text{tr}(\mathbf{v}_i \mathbf{u}_i^\top \mathbf{u}_j \mathbf{v}_j^\top) = (\mathbf{u}_i^\top \mathbf{u}_j)(\mathbf{v}_i^\top \mathbf{v}_j) = \delta_{ij},$$

where

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

Therefore, if we keep only the first k terms in the expansion of $\mathbf{X}$, then the squared Frobenius error becomes:

$$\|\mathbf{X} - \mathbf{X}_{(k)}\|_F^2 = \left\| \sum_{j=k+1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^\top \right\|_F^2 = \sum_{j=k+1}^r \sigma_j^2.$$

It remains to show that no other matrix of rank k can do strictly better. Let $\mathbf{M}$ be an arbitrary matrix with $\text{rank}(\mathbf{M}) = k$, and let $\tau_1 \geq \dots \geq \tau_k > 0$ denote its singular values. Using

$$\|\mathbf{X} - \mathbf{M}\|_F^2 = \text{tr}((\mathbf{X} - \mathbf{M})^\top (\mathbf{X} - \mathbf{M})) = \|\mathbf{X}\|_F^2 + \|\mathbf{M}\|_F^2 - 2\text{tr}(\mathbf{X}^\top \mathbf{M}),$$

standard trace inequalities (see [3]), and $2ab \leq a^2 + b^2$, we get

$$\begin{aligned} 2\text{tr}(\mathbf{X}^\top \mathbf{M}) &\leq 2 \sum_{j=1}^k \sigma_j \tau_j \\ &\leq \sum_{j=1}^k \sigma_j^2 + \sum_{j=1}^k \tau_j^2 \\ &= \sum_{j=1}^k \sigma_j^2 + \|\mathbf{M}\|_F^2. \end{aligned}$$

Substituting this bound into the previous identity yields

$$\|\mathbf{X} - \mathbf{M}\|_F^2 \geq \|\mathbf{X}\|_F^2 - \sum_{j=1}^k \sigma_j^2 = \sum_{j=k+1}^r \sigma_j^2.$$

Thus, no rank- k matrix can have smaller Frobenius error than $\mathbf{X}_{(k)}$. $\square$

Theorem 1 shows that PCA is the canonical solution to the problem of approximating a high-dimensional dataset by a lower-dimensional object. In particular, the first k right singular vectors define the k -dimensional directions that capture the highest variation in the data.

0.1.2. Geometric interpretation of dimension reduction

PCA may be viewed geometrically as the orthogonal projection of the data onto a k -dimensional subspace that minimizes the total squared reconstruction error. If the data is centered, the principal directions are the eigenvectors associated with the largest eigenvalues of the empirical covariance matrix

$$\frac{1}{n} \mathbf{X}^\top \mathbf{X}.$$

Thus PCA can be interpreted either through low-rank matrix approximation or through a variance-maximizing orthogonal basis.

In the literature, PCA is often introduced as a dimension-reduction method, but its importance goes well beyond that role in practice. It serves as a basic tool for visualization, feature extraction, compression, and exploratory data analysis. In many modern applications, the dimension of dataset is large and strong dependencies exist among features; in such settings, a low-rank representation is advantageous for further computation.

At the same time, the classical PCA framework has limitations. It is linear, and it is well known to be sensitive to atypical or extreme observations. Moreover, classical PCA is deterministic and does not directly provide a framework for uncertainty quantification.

0.2. Beyond linearity

Although the thesis focuses on different variants of PPCA in Euclidean space, there exist variants of PCA belonging to a much broader family of dimension reduction methods [4]. In particular, for nonlinear extensions, one may consider methods such as kernel PCA, manifold learning, and nonlinear latent variable models [9, 11, 5]. For infinite-dimensional data, one can consider variants such as functional PCA [8]. In high-dimensional and modern statistical settings, sparse PCA has also found applications [13].

0.3. Overview of the thesis

This thesis is organized as a thesis by articles as follows:

Chapter 1 studies the consistency of maximum likelihood of PPCA model. The main objective is to provide a rigorous theoretical understanding of PPCA from a topological point of view. Although PCA and PPCA are classical tools, subtle issues arising from the model's non-identifiability, such as the metric space structure of the quotient topological space, require careful treatment to prove consistency results.

Chapter 2 proposes a robust Bayesian version of PPCA which is meant to be more reliable when the dataset contains extreme observations (outliers). The key idea is to replace the Gaussian noise assumption by a heavier-tailed alternative, leading to a model that preserves the usefulness of PCA while improving robustness. This chapter investigates both the modeling aspects and the practical statistical behavior of the robust estimators.

Chapter 3 studies the convergence of coordinate ascent variational inference (CAVI) for Bayesian PCA (BPCA). In Chapter 2, CAVI is used to compute robust estimators efficiently, which is a widely used algorithm in variational inference (VI). Although CAVI often converges quickly in practice, characterizing its convergence properties remains challenging. This chapter studies convergence speed and other related properties of CAVI for BPCA using dynamical system and some recent modern development in VI literature.

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Chapter 1

On the Consistency of Maximum Likelihood Estimation of Probabilistic Principal Component Analysis

This chapter is based on joint work with my coauthor, Sayak Chakrabarty, currently a final year Ph.D. student in computer science at Northwestern University. This work is already published in Advances in Neural Information Processing Systems, with me as first author [8].

Résumé

L'analyse en composantes principales probabiliste (ACPP) constitue aujourd'hui l'un des outils statistiques les plus utilisés pour réduire la dimension ambiante des données. De la mise à l'échelle multidimensionnelle à l'imputation de données manquantes, ce modèle couvre un large spectre d'applications, allant des sciences et de l'ingénierie à la finance quantitative.

Malgré cette large applicabilité, il existe encore peu de garanties théoriques permettant de justifier la validité de la solution du maximum de vraisemblance (MV) pour ce modèle. En effet, il est bien connu que l'estimateur du maximum de vraisemblance (EMV) ne permet de retrouver les vrais paramètres qu'à une rotation près. L'obstacle principal provient du défaut d'identifiabilité intrinsèque du modèle ACPP, induit par la symétrie rotationnelle de la paramétrisation. Pour lever cette ambiguïté, nous proposons une approche nouvelle fondée sur les espaces topologiques quotients et, plus précisément, nous montrons que la solution du maximum de vraisemblance est consistante dans un espace euclidien quotient approprié. En outre, nos résultats de consistance couvrent une classe plus générale d'estimateurs contenant l'EMV. Enfin, sous une hypothèse de compacité, nous établissons également la consistance forte de l'EMV et, par conséquent, la consistance forte de l'estimation de covariance dans le modèle ACPP.

Abstract

Probabilistic principal component analysis (PPCA) is currently one of the most used statistical tools to reduce the ambient dimension of the data. From multidimensional scaling to the imputation of missing data, PPCA has a broad spectrum of applications ranging from science and engineering to quantitative finance.

Despite this wide applicability in various fields, hardly any theoretical guarantees exist to justify the soundness of the maximum likelihood (ML) solution for this model. In fact, it is well known that the maximum likelihood estimation (MLE) can only recover the true model parameters up to a rotation. The main obstruction is posed by the inherent identifiability nature of the PPCA model resulting from the rotational symmetry of the parameterization. To resolve this ambiguity, we propose a novel approach using quotient topological spaces and in particular, we show that the maximum likelihood solution is consistent in an appropriate quotient Euclidean space. Furthermore, our consistency results encompass a more general class of estimators beyond the MLE. Strong consistency of the ML estimate and consequently strong covariance estimation of the PPCA model have also been established under a compactness assumption.

Keywords: Covariance estimation, identifiability, latent variable models, quotient space, rotational invariance.

1.1. Introduction

In the era of big data, principal component analysis (PCA) is a standard dimension reduction tool frequently used in exploratory data analysis. PCA is however not a statistical model, implying that uncertainty quantification is not available. Uncertainty quantification is desirable in some situations, like when principal components have an interpretation. Probabilistic principal component analysis (PPCA) is a model introduced by [25] to remedy the situation. It consists of an additive model with a normal noise term. Let $p > q$ be two given positive integers. The PPCA model can be written as $x = \mathbf{W}z + \varepsilon$, where $x \in \mathbb{R}^p$ denotes a random vector that is observed after a linear transformation $\mathbf{W}$ is applied to a latent (hidden) variable $z \in \mathbb{R}^q$ up to a gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_p)$. We assume that our data is centered, meaning x has zero mean. In practice, the latent variable z is not observed and is conventionally assumed to be normally distributed. Given a set of n data points or observations $\{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^p$, the two unknown parameters $\mathbf{W}$ and σ^2 are estimated by maximum likelihood estimation (MLE).

PPCA is a very well-known and pervasive technique used in dimension reduction of high-dimensional data and used in figuring out the relevant dimensions in large-scale data mining. It must be said that the primary goal of PPCA is not always to perform better than traditional PCA but open up a series of possibilities for future extensions for further analysis. PPCA also enables

comparison with other probabilistic techniques. Namely, it is possible to quantify the uncertainty in parameters $\mathbf{W}$ and σ^2 through a Bayesian reformulation [25]. One can also consider modelling the error terms using heavy-tailed distributions (such as Student's t) to achieve robustness [10]. These models have appealing properties; in particular, the quality of the information retained from the original data is similar to that of the original PCA with added flexibility.

Despite the wide applicability, nothing is known about the consistency of the maximum likelihood (ML) estimation of the PPCA. The main obstruction is posed by the identifiability of the PPCA model over the parameter space $(\mathbf{W}, \sigma^2) \in \mathbb{M}(p, q) \times \mathbb{R}_+$, where $\mathbb{M}(p, q)$ and $\mathbb{R}_+$ denote the set of all $p \times q$ matrices and the set of positive real numbers respectively. One can restrict the attention only to rank q matrices, but as we shall see, doing so will not significantly make our goal any harder (or easier) to achieve as the ML estimation of $\mathbf{W}$ has rank q . It can be readily seen that our model induces a Gaussian distribution on the data points $x \sim \mathcal{N}(0, \mathbf{W}\mathbf{W}^T + \sigma^2 I_p)$, which is to be used for the maximum likelihood estimation. From a frequentist point of view, we now assume that the unknown parameters $(\mathbf{W}, \sigma^2)$ has a true value which is equal to $(\mathbf{W}_0, \sigma_0^2)$. Additionally, we observe that the marginal distribution of the data points $\mathcal{N}(0, \mathbf{W}\mathbf{W}^T + \sigma^2 I_p)$ used for the ML estimation remains invariant under any translation of $\mathbf{W}$ by an orthogonal matrix $\mathbf{R}$ since $(\mathbf{W}\mathbf{R})(\mathbf{W}\mathbf{R})^T = \mathbf{W}\mathbf{W}^T$. As can be seen in the literature it is only possible to recover the maximum likelihood estimate of $\mathbf{W}$ up to rotations [25]. This lack of identifiability puts an immediate roadblock to directly applying Wald's consistency theorems [27], a well-known method to guarantee the consistency of maximum likelihood estimate.

It turns out that the quotients of the Euclidean spaces and in particular, the quotient of the space $\mathbb{M}(p, q) \times \mathbb{R}_+ \subset \mathbb{R}^{p \times q} \times \mathbb{R}$ is the natural topological object to consider to talk about the consistency results of the PPCA model. In a simplified situation when σ^2 is known, heuristically, the quotient space treats all the points $\mathbf{W}_0$ and its rotational translates $\mathbf{W}_0\mathbf{R}$ as a single entity by labelling them equivalent and therefore throwing all of them into a single equivalence class. Therefore, the parameter space for $\mathbf{W}$, i.e. $\Sigma = \mathbb{M}(p, q)$ becomes a set of equivalence classes $\tilde{\Sigma} = \{[\mathbf{W}] : \mathbf{W} \in \mathbb{M}(p, q)\}$ through an equivalence relation $\mathbf{W}' \in [\mathbf{W}]$ iff $\mathbf{W}'\mathbf{W}'^T = \mathbf{W}\mathbf{W}^T$ or equivalently $\mathbf{W}' = \mathbf{W}\mathbf{R}$, for some orthogonal matrix $\mathbf{R}$. In this setting, we thus get rid of the identifiability issue by passing through the quotient of $\mathbb{R}^{p \times q} \times \mathbb{R}$ via the proposed equivalence relation. We then study the consistency of the maximum likelihood in this space. The formal details about the definition of quotient space by a given equivalence relation can be found in [section 1.4](#).

The above approach is inspired by [21] where the author outlines a way to extend Wald's consistency theorems [27] under the non-identifiability assumption. We would like to mention that, even though passing through the quotient topological space eliminates the rotational

ambiguity of the ML estimate by definition, it introduces many further technical issues. To the best of our knowledge, all the subtleties have not been properly addressed in Redner’s work even though the author applies the proposed methodology to analyze the consistency of ML estimates of mixture models. One of the key things that is not explicitly mentioned in their work is that the underlying metric space structure of the quotient topological space can be substantially different from the natural topological structure of the quotient space itself and the topology generated by the open balls with the induced metric in quotient space is in general, not same as that of the usual quotient topology. Thus related geometric properties, for instance, the notion of convergence and other regularity properties of the parameter space become more delicate to analyze. We address all these underlying issues carefully later in [section 1.4](#).

To the best of our knowledge, no previous research work has attempted to justify the wide usage of the ML estimate of the PPCA model because of the inherent identifiability nature of the model parameters. To resolve this challenge we propose a novel framework to see the PPCA model through the lens of an appropriate quotient of an Euclidean space. This perspective not only allows us to formalize the problem in precise mathematical terms but also helps us resolve the dispute caused by the non-identifiability.

Considering from the point of novelty, the contributions of this paper are two-fold: we prove that the maximum likelihood estimates $(\hat{\mathbf{W}}, \hat{\sigma}^2)$ are consistent, i.e., $(\hat{\mathbf{W}}, \hat{\sigma}^2) \rightarrow (\mathbf{W}_0, \sigma_0^2)$ in probability in the appropriate quotient Euclidean space. In fact, our consistency results are valid for a general family of estimators that in particular, contains the maximum likelihood estimator. Furthermore, strong consistency, i.e., an almost sure convergence of the ML estimates to the true parameter is also deduced when the parameter space is compact.

The rest of the paper is organized as follows: [section 1.2](#) outlines the necessary background and a comprehensive discussion of related works. In [section 1.3](#), we formally introduce the PPCA model, the ML estimates of the parameters of interest, and the main question addressed in this paper. In [section 1.4](#), we provide key definitions and results from the theory of quotient topological spaces in a self-contained way. [section 1.5](#) contains our main contributions, followed by detailed proofs in [section 1.6](#). In [section 1.7](#), we state Wald’s conditions and interpret them in the quotient parameter space. Finally, [section 1.8](#) discusses impact and open questions before we conclude in [section 1.9](#).

1.2. Background and Previous Work

The abstract formalism concerning quotient topological spaces is not foreign to the theory of applied statistics. For an in-depth discussion about the application of quotient space into various statistical models, see [\[18\]](#). Even though, the necessary tools and definition of topological

quotients are introduced in [section 1.4](#), an inquisitive reader is certainly encouraged to consult the classic [\[19\]](#). The main line of attack used in this paper has been previously introduced in [\[21\]](#), where the author proves the consistency of mixture models under the standard regularity conditions of Wald’s consistency theorems [\[27\]](#). It is worthwhile to mention that the theoretical justification of using the maximum likelihood estimates in practice has gained a lot of recent interest. For instance, [\[6\]](#) has a rigorous treatment of several consistency results with an application to non-parametric mixture models.

In [section 1.1](#), we have already mentioned that PPCA offers many advantages over traditional PCA, particularly in terms of model flexibility and the potential for generalization to accommodate various statistical reformulations. Building upon the recognition of its strength by [\[1\]](#), [\[11\]](#), and [\[22\]](#) as one of the most notable advancements in probabilistic models, subsequent research endeavours have leveraged PPCA to achieve remarkable results in various applications. Within the realm of cryo-electron microscopy (cryo-EM) in the field of biology, extensive discussions in the literature have explored its efficacy, as evidenced by the works of [\[13, 20, 24\]](#). Similarly, in the domain of computer vision, the probabilistic model has proven its superiority, as demonstrated by [\[23\]](#) and others. While the precise reasons underlying its success in certain situations remain elusive, PPCA has found widespread applicability in diverse domains. It has found great applications in incremental learning for visual tracking [\[17\]](#), in the study of Gaussian processes [\[3, 15\]](#), orthogonal signal correction (OSC) [\[16\]](#), weighted nuclear norm minimization [\[12\]](#), and in outlier detection algorithms like [\[9\]](#). Recent advancements have showcased its potential in areas such as few-shot learning [\[28\]](#), data mining techniques [\[30\]](#), and most notably, in finance for exploiting market integration for pure alpha investments [\[26\]](#). The broad spectrum of applications where PPCA has exhibited remarkable performance underscores its versatility and its value as a powerful tool in numerous domains.

As we have already noted, the theoretical justification behind the ML estimates used for the PPCA model in practice is not always straightforward and has recently attracted the attention of several authors. In a fairly recent work [\[7\]](#), the authors provide a consistent estimator of the quantity $\mathbf{W}\mathbf{W}^T$ assuming σ^2 is known. Their work is primarily concerned with the model selection questions in statistics using a variational approach. Even though in our work we assume the rank of the matrix $\text{rank}(\mathbf{W}) = q$ to be known and is fixed by the user (this assumption can be removed as long as q does not grow with the sample size n), it is of independent interest to want to estimate q . In their work [\[7\]](#), the authors constructed an estimator of the quantity q to demonstrate how closely their estimated model recovers the true model corresponding to the true value of $\mathbf{W}$, they were also able to estimate the marginal covariance of the data points and thereby came up with a consistent estimator of $\mathbf{W}\mathbf{W}^T$. One may notice that, in this case, estimating

$\mathbf{W}\mathbf{W}^T$ instead of $\mathbf{W}$ eliminates the non-identifiability issue. It is important to mention that their work does not use the maximum likelihood approach and instead uses variational techniques to construct an estimator of $\mathbf{W}\mathbf{W}^T$. In this regard, it is relevant to mention that another work [4] was able to provide a consistent estimator of the parameter q using maximum likelihood estimation. Finally, we restate that no related previous work attempts to justify the commonly used ML estimates $(\hat{\mathbf{W}}, \hat{\sigma}^2)$ of the PPCA model, perhaps due to ambiguity caused by the inherent non-identifiable parameter space. In the following section, we begin by presenting the PPCA model and concerned ML estimates originally developed in [25]

1.3. Problem Statement

Notation In what follows, we shall always let capital letters denote random variables and their deterministic realizations will be denoted by lowercase letters. Let $X_1, X_2, \dots$ be a sequence of independent identically distributed random variables where the p.d.f. of the distribution of X_1 is known except a parameter $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$ in some parameter space $\Theta \subset \mathbb{R}^{p \times q} \times [\epsilon_*, \infty)$ for some matrix $\mathbf{W}_\theta$ and positive integers p, q and some $\epsilon_* > 0$. We assume the distribution of X_1 is generated from a true parameter θ_0 . For a given parameter $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$, we let $f(x; \theta)$ denote the normal p.d.f. $\mathcal{N}(x; 0, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I_p)$. Henceforth, the notation $\|\mathbf{W}\|$ will mainly refer to the spectral norm, which is defined as the largest eigenvalue of the matrix $\mathbf{W}\mathbf{W}^T$. It is worth noting that in some rare instances, we have employed the Frobenius norm, but such usage has been explicitly mentioned.

Unless otherwise stated, all the probabilities and expectations are taken with respect to the true parameter θ_0 . Finally, for a $m \times m$ matrix $\mathbf{A}$, we denote its characteristic polynomial $\det(\lambda I_m - \mathbf{A})$ by $\mathcal{P}_A(\lambda)$.

The PPCA model: Let us consider that we have access to a structured dataset taking the form of a matrix $\mathbf{X}$ whose lines are given by $x_1, \dots, x_n \in \mathbb{R}^p$ with p a large positive integer. We assume that the columns of $\mathbf{X}$ are centered (meaning a mean of 0). We assume

$$x_i = \mathbf{W}z_i + \varepsilon_i, \quad (1)$$

where $z_1, \dots, z_n, \varepsilon_1, \dots, \varepsilon_n$ are i.i.d. the realizations of independent random variables with $z \sim \mathcal{N}(0, I_q)$ and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_p)$, and $\mathbf{W}$ is an unknown fixed $p \times q$ matrix, $\sigma > 0$ being an unknown fixed scale parameter and I_q and I_p being identity matrices of size q and p , respectively. The random variable z plays the role of a hidden (latent) variable and ε that of an error term. We recall that $\text{rank}(\mathbf{W}) = q$. To estimate the parameters $\mathbf{W}$ (often called the loading matrix) and the unknown variance σ^2 of the homoscedastic error term ε , [25] employs the maximum likelihood

(ML) procedure and solutions are given as follows

$$\widehat{\mathbf{W}} = \mathbf{U} (\Delta_q - \sigma^2 I_q)^{1/2} \mathbf{R} \text{ and } \widehat{\sigma}^2 = \frac{1}{p-q} \sum_{j=q+1}^p \delta_j,$$

where $\mathbf{R}$ is any $q \times q$ rotation matrix, the columns of $\mathbf{U}$ are given by the first q eigen vectors of the sample covariance matrix $S_x = 1/n \sum_{i=1}^n x_i x_i^T$ and $\Delta_q = \text{diag}(\delta_1, \dots, \delta_q)$ are the q eigenvalues of S_x sorted in the descending order. We note that the solution $\widehat{\mathbf{W}}$, in this case, is not unique since $\mathbf{R}$ could be any orthogonal matrix which in practice, is often set to be I_q . For now, we restrict our discussion assuming the rank q is fixed. For this model, given the parameters $(\mathbf{W}, \sigma^2)$, we have the following marginal density for the generated data $\{x_i\}$

$$p(x|\mathbf{W}, \sigma) = \int p(x|z, \mathbf{W}, \sigma) p(z) dz = \mathcal{N}(x; 0, \mathbf{W}\mathbf{W}^T + \sigma^2 I_p),$$

where we used the fact that the latent variable z is distributed as $\sim \mathcal{N}(0, I_q)$. Thus for our problem, the parameter space is

$$\Theta := \{\theta : \theta = (\mathbf{W}_\theta, \sigma_\theta^2) \in \mathbb{R}^{p \times q} \times [\epsilon_*, \infty)\}.$$

In the above definition there is a slight abuse of notation as the parameter θ is self-referencing to itself. Nonetheless, we retain the notation for the sake of clarity and the ease of exposition. We assume the data points $\{x_i\}$ are generated from a true parameter $\theta_0 = (\mathbf{W}_0, \sigma_0^2)$, i.e., they are i.i.d. samples from normal distribution whose p.d.f. is given by $f(x; \theta_0)$. Finally, we note that assuming the existence of such ϵ^* in the definition of Θ is not prohibitive since $\liminf \widehat{\sigma}^2 > 0$ if the population covariance is positive symmetric definite. In this work, we aim to prove the consistency of maximum likelihood estimates, i.e. we want to discuss whether $(\widehat{\mathbf{W}}, \widehat{\sigma}^2) \rightarrow (\mathbf{W}_0, \sigma_0^2)$ holds P_{θ_0} almost surely or in probability.

Nearly all consistency proofs of well-known statistical models take identifiability for granted. Unfortunately, the PPCA model is not identifiable, since there may be matrices $\mathbf{W}_\theta$ and $\mathbf{W}_\phi$ such that $\mathbf{W}_\theta \mathbf{W}_\theta^T = \mathbf{W}_\phi \mathbf{W}_\phi^T$, in which case $f(x; \theta) = f(x; \phi)$ if $\sigma_\theta = \sigma_\phi$. It is not difficult to see in such cases, $\mathbf{W}_\phi$ will be a rotational translate of the matrix $\mathbf{W}_\theta$, i.e., $\mathbf{W}_\phi = \mathbf{W}_\theta \mathbf{R}'$ for some orthogonal matrix $\mathbf{R}'$. This rotational symmetry in parameterization results in a lack of identifiability. In the current work we adopt a similar approach as outlined in [21] and subsequently work in a quotient topological space of Θ . Next, we rigorously introduce this quotient space framework and develop analytical tools to prove our consistency results.

1.4. A primer on quotient topological spaces

In this section, we give the definition of a quotient of a topological space by a given equivalence relation and discuss several important results pertinent to our work.

Definition 1 (Equivalence relation). *Let S be a set. A binary relation $\sim$ on S is said to be an equivalence iff it is (i) reflexive: $x \sim x, \forall x \in S$, (ii) symmetric: $x \sim y \iff y \sim x, \forall x, y \in S$ and (iii) transitive: $x \sim y, y \sim z \implies x \sim z, \forall x, y, z \in S$.*

*It is evident that an equivalence relation on a set induces a partition of the set and vice-versa. The disjoint sets in this partition are called the equivalence classes. If $s \in S$ is given, then we write $[s] := \{y \in S \mid y \sim s\}$ for the equivalence class that contains s and finally the set of equivalent classes $(S/\sim) := \{[s] \mid s \in S\}$ are called the **quotient** of the set S by the equivalence relation $\sim$.*

1.4.1. Quotient topological spaces and its metrizability

Given a space X and an equivalence relation $\sim$ on X , the set-theoretic quotient $X/\sim$ (the set of equivalence classes) inherits a topology from X , called the quotient topology [19]. It is well known that the surjective map $\pi : X \rightarrow X/\sim$ defined as $x \rightarrow [x]$ is continuous. We now record a useful result from [19], which will be important in proving our main results in the forthcoming sections.

Theorem 2. *If $X, X/\sim$ are topological spaces stated as above, then for another topological space Y and for a continuous map $f : X \rightarrow Y$ with the additional property that $x \sim x'$ implies $f(x) = f(x')$, there exists unique continuous map $\tilde{f} : (X/\sim) \rightarrow Y$ such that $f = \tilde{f} \circ \pi$*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow \pi & \nearrow \tilde{f} & \\ X/\sim & & \end{array}$$

Figure 1. Illustration of Theorem 2 using a commutative diagram.

Throughout this paper, we have $Y = \mathbb{R}$ and X is a subset of $\mathbb{R}^p$ for some positive integer p , which is a complete metric space and Euclidean distance serves as a natural metric on $\mathbb{R}^p$. The key idea is to extend this metric from X to $X/\sim$ and this job is accomplished through an intermediate state, by introducing a suitable pseudometric on X . A pseudometric δ is a function on $X \times X$ to non-negative real numbers and the only difference between a pseudometric and a

metric is that $\delta(x,y) = 0$ need not imply $x = y$ for $x,y \in X$. To construct the most commonly used metric on the quotient space $X/\sim$ we first introduce:

let $X \subset \mathbb{R}^p$ be a complete subspace of the Euclidean space and d be the standard Euclidean distance on $\mathbb{R}^p$. For $x,y \in X$, we let

$$d_1(x,y) := \inf\{d(x_1,y_1) + \dots + d(x_n,y_n) \mid x \sim x_1, y_i \sim x_{i+1} (1 \leq i \leq n-1), y_n \sim y\}.$$

Then it can be checked that d_1 defines a pseudometric on X . We now declare $x \approx y$ whenever $d_1(x,y) = 0$, and it is not difficult to verify the relation $\approx$ is an equivalence. Then we can endow the set-theoretic quotient $X/\approx$ with the metric $\bar{d}$, which we define as $\bar{d}([x],[y]) = d_1(x,y)$ where $[x],[y]$ are the classes of x,y in $X/\approx$. Furthermore, we let $X_\approx$ to be the completion of the above space under the metric $\bar{d}$, meaning all the Cauchy sequences converge.

As the reader may have noticed already, there are two equivalence relations at play on X , namely $\sim$ and $\approx$. Although the construction of the space $X/\approx$ is dependent on the equivalence $\sim$ through the definition of d_1 , it is not straightforward to identify the relationship between the two abstract quotient spaces $X/\sim$ and $X/\approx$. Now is a good time to pause and unveil the primary reason to justify the development of these technical tools. *Our initial goal is to metrize the topological quotient $X/\sim$ for a given equivalence relation $\sim$ so that we could extend Wald's conditions [27] in $X/\sim$ to prove consistency results in this space.* The procedure outlined in the previous paragraph provides a recipe for constructing a metric $\bar{d}$ on the space $X/\approx$. The next crucial result, [Lemma 1](#) shows that for certain nice equivalence relations $\sim$, the two spaces $X/\approx$ and $X/\sim$ are identical. This result is a key fact that allows us to maneuver the metric construction into $X/\sim$. We now state the result from [\[29\]](#). The proof of this result is skipped as it is deemed to be technical and it serves a little purpose to our main goal. For a thorough treatment of metrizability of quotient topological spaces, we refer the inquisitive reader to the comprehensive book on Lipschitz algebras [\[29\]](#).

Lemma 1. *Let C be a closed subset of a complete metric space X . Let X/C denote the quotient space of X by the equivalence defined by $x \sim y$ if either $x = y$ or $x,y \in C$ then the underlying set $X_\approx$ and the quotient X/C are the same. The metric on X/C in this case, will take the form:*

$$\bar{d}([x],[y]) = \min(d(x,y), d(x,C) + d(y,C)) \quad \text{for any } [x],[y] \in X/C.$$

In the above statement, $d(x,C)$ denotes the distance between the point x and the closed subset C defined by $d(x,C) := \inf_{c \in C} d(x,c)$ and likewise for $d(y,C)$. An immediate follow-up question of the above result would be: what is the space X and the closed subset C we use in our context? We now recall two things from [section 1.4](#): first, the parameter space for the PPCA model, namely $\Theta \subset \mathbb{R}^{pq+1}$; and second, the issue of identifiability, which arose because there could exist two parameters $\theta, \phi \in \Theta$ such that their corresponding densities are equal, which is

equivalent to $\mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I_p = \mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I_p$. In what follows, using Redner's idea [21] we consider the closed subset defined by

$$C := \{\theta \in \Theta : \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I_p = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p\} \subset \Theta,$$

which allows us to get rid of identifiability since in the quotient topological space Θ/C , $f(x; [\theta]) = f(x; [\theta_0])$ if and only if $[\theta] = [\theta_0]$ or equivalently $\theta \sim \theta_0$ by construction. We can now hope for checking Wald's consistency conditions in Θ/C to prove $[\hat{\theta}]$ converges to $[\theta_0]$ in probability, where $\hat{\theta} = (\hat{W}, \hat{\sigma}^2)$ denotes the MLE.

In the introduction, we pointed out that despite the extension of Wald's work [27] to quotient topological spaces by [21], an inadequate amount of attention was paid to the comprehensive understanding of the interplay between the quotient topology and the topology generated by the metric $\bar{d}$ in that treatment. For instance, a priori it is, by no means obvious how one would interpret their assumption 3 in the quotient space Θ/C without invoking Lemma 1. Also, their assumption 5 does not clearly mention what mode of convergence has been used since there are two available notions of convergence, one coming from the quotient topology and another coming from the metric topology generated by $\bar{d}$, and those two topologies are not same in general. It is a nontrivial fact that the latter is contained in the former (for a discussion on this topic see [14]). However, in most statistical applications, for example in the applications to mixture distributions and clustering [21], we tend to quotient the parameter space by a suitable closed subspace depending on the context, and subsequently the quotient metric takes an easier form as given in Lemma 1. Therefore, several technical constructions get greatly simplified in these applications.

1.5. Theoretical Results

In this section, we state our main results. Let $\{x_i\}_{i=1}^n \subset \mathbb{R}^p$ be a sequence of data points that are generated from latent observations $\{z_i\}_{i=1}^n \in \mathbb{R}^q$ via the PPCA model defined in (1) and recall from section 1.3 that the data points are assumed to be generated from the true parameter $\theta_0 := (\mathbf{W}_0, \sigma_0^2) \in \Theta$ of the underlying model. Our data points x_i can therefore be viewed as a sequence of realizations of i.i.d. random variables whose true marginal density is given by $f(x; \theta_0) = \mathcal{N}(x; 0, \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p)$. We note a crucial point that for two parameters $\theta, \phi \in \Theta$, $f(x; \theta) = f(x; \phi)$ holds if and only if $[\theta] = [\phi]$ in Θ/C . Therefore, we write $f(x; \theta)$ for simplicity, instead of cumbersome $f(x; [\theta])$ when stating results in the quotient space Θ/C . In the spirit of [27] and [21], two of our main results are the following:

Theorem 3. *Let S is any closed subset of Θ not intersecting C then*

$$\mathbb{P} \left(\limsup_n \sup_{\theta \in S} \prod_{i=1}^n \frac{f(x_i; \theta)}{f(x_i; \theta_0)} = 0 \right) = 1.$$

Theorem 4. *Let $\theta_n(x_1, \dots, x_n)$ be any measurable function of the observations $x_1, \dots, x_n$ such that*

$$\prod_{i=1}^n \frac{f(x_i; \theta_n)}{f(x_i; \theta_0)} \geq c > 0 \text{ for all } n,$$

then $\mathbb{P} \left(\lim_n [\theta_n] = [\theta_0] \right) = 1$, that is θ_n converges to θ_0 in Θ/C almost surely, and therefore in probability.

The above results have elegant geometric interpretations. In particular, [Theorem 3](#) implies that if U_C is any open neighbourhood containing C then almost surely all but finitely many terms of the sequence $\{\theta_n\}$ land inside U_C . As an immediate consequence of [Theorem 4](#) when $c = 1$, we obtain:

Corollary 1 (Strong consistency of the MLE in the quotient space). *Let θ_n denote the sequence of maximum likelihood estimates of the PPCA model. We then have that*

$$\mathbb{P} \left(\lim_n [\theta_n] = [\theta_0] \right) = 1$$

that is maximum likelihood estimates converge to the true parameter θ_0 almost surely, and therefore in probability in the quotient space Θ/C .

Additionally, using the above corollary we obtain that the covariance for the PPCA process can be consistently (strong) estimated under a compactness assumption.

Theorem 5. *Let Θ_0 be a compact subset of Θ containing the point $\theta_0 = (\mathbf{W}_0, \sigma_0^2)$. Let $\theta_n = (\hat{\mathbf{W}}, \hat{\sigma}^2)$ denote the sequence of maximum likelihood estimates of the PPCA model. Then we have*

$$\mathbb{P} \left(\hat{\mathbf{W}}\hat{\mathbf{W}}^T + \hat{\sigma}^2 I_p \rightarrow \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p \right) = 1.$$

Now we talk about possible generalizations above results in light of [\[31\]](#), where the observations $\{X_i\}_{i=1}^n$ are assumed to be identically distributed but *need not be independent*. As noted in Wolfowitz's work that [\[27\]](#) proves strong consistency (i.e. almost sure convergence) and his proof techniques can be extended for dependent random variables $\{X_i\}_{i=1}^n$ as long as the sequence $\{X_i\}$ satisfies the strong law of large numbers. In the spirit of Wald's work, [\[31\]](#) proves the consistency of ML estimates (i.e. convergence in probability) only under the assumption that the sequence $\{X_i\}$ satisfies weak law of large numbers. The assumption that the sequence $\{X_i\}$ satisfies the weak law of large numbers may seem technical but it offers a great deal of flexibility in real life applications due to the following result of Bernstein [\[5\]](#).

Theorem 6 (Bernstein). *Let $\{X_i\}$ be a sequence of centered random variables. If there exists a constant $\kappa > 0$ such that for every $i \in \mathbb{N}$ we have $\text{Var}(X_i) \leq \kappa$ and if the following condition is true:*

$$\lim_{|i-j| \rightarrow \infty} \text{Cov}(X_i, X_j) = 0,$$

then the weak law of large numbers holds.

We may note that the above result could be extremely relevant in real life applications as intuitively it accommodates all statistical models which allows local dependence among observations, in the sense that X_i can significantly correlate with X_j as long as j remains sufficiently close to i (i.e., $|i - j|$ is small), but the correlation gradually vanishes as $|i - j|$ becomes large. Temporal stochastic processes or time series data with appropriate assumptions could be tailored into the category of those statistical models where Bernstein's result applies. We therefore state our next set of results on the consistency of maximum likelihood estimates of the PPCA model but only under a weak law assumption. The following result is an immediate consequence of [31].

Theorem 7. *Let the data points $\{x_i\}_{i=1}^n$ be realizations of identically distributed random variables $\{X_i\}_{i=1}^n$ whose density is given by $f(x; \theta_0)$. Let the sequence of random variables $\{X_i\}$ satisfy the weak law of large numbers. Given $\eta > 0$ and a closed subset S of Θ not intersecting C , there exists a quantity $h(S) \in (0, 1)$ which depends only on the closed set S and an integer $N(\eta, S)$ such that*

$$\mathbb{P} \left(\sup_{\theta \in S} \prod_{i=1}^n \frac{f(x_i; \theta)}{f(x_i; \theta_0)} > h^n(S) \right) < \eta \text{ for every } n \geq N(\eta, S).$$

Theorem 8. *Let the data points $\{x_i\}_{i=1}^n$ be realizations of identically distributed random variables $\{X_i\}_{i=1}^n$ whose density is given by $f(x; \theta_0)$. Let the sequence of random variables $\{X_i\}$ satisfy the weak law of large numbers. Let $\theta_n(x_1, \dots, x_n)$ be any measurable function of the observations $x_1, \dots, x_n$ such that*

$$\prod_{i=1}^n \frac{f(x_i; \theta_n)}{f(x_i; \theta_0)} \geq c > 0 \text{ for all } n,$$

then $[\theta_n] \xrightarrow{\mathbb{P}} [\theta_0]$, that is θ_n converges to θ_0 in θ/C in probability.

Furthermore we also have the two following results in spirit of [Corollary 1](#) and [Theorem 5](#).

Corollary 2 (Consistency of the MLE in the quotient space). *Let the data points $\{x_i\}_{i=1}^n$ be realizations of identically distributed random variables $\{X_i\}_{i=1}^n$ whose density is given by $f(x; \theta_0)$. Let the sequence of random variables $\{X_i\}$ satisfy the weak law of large numbers. Let*

θ_n denote the sequence of maximum likelihood estimates of the PPCA model. We then have that

$$[\theta_n] \xrightarrow{\mathbb{P}} [\theta_0],$$

that is maximum likelihood estimates converge to the true parameter θ_0 in probability in the quotient space Θ/C .

Additionally, using the above corollary we can obtain prove consistent covariance estimation is possible for the PPCA model under a compactness assumption.

Theorem 9 (Consistent estimation of covariance). *Let the data points $\{x_i\}_{i=1}^n$ be realizations of identically distributed random variables $\{X_i\}_{i=1}^n$ whose density is given by $f(x; \theta_0)$. Let the sequence of random variables $\{X_i\}$ satisfy the weak law of large numbers. Let Θ_0 be a compact subset of Θ containing the point $\theta_0 = (\mathbf{W}_0, \sigma_0^2)$. Let $\theta_n = (\widehat{\mathbf{W}}, \widehat{\sigma}^2)$ denote the sequence of maximum likelihood estimates of the PPCA model. Then we have*

$$\widehat{\mathbf{W}}\widehat{\mathbf{W}}^T + \widehat{\sigma}^2 I_p \xrightarrow{\mathbb{P}} \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p.$$

1.6. Proof of theoretical results

We only present the proofs of [Theorem 8](#) and [Theorem 9](#). [Corollary 2](#) follows from [Theorem 7](#) [\[31\]](#). The proofs of [Theorem 4](#), [Corollary 1](#) and [Theorem 5](#) will be analogous.

Proof of [Theorem 8](#). We fix two quantities $\eta_1, \eta_2 > 0$. We need to show that there exists $N_0(\eta_1, \eta_2)$ such that

$$\mathbb{P}(\bar{d}([\theta_n], [\theta_0]) > \eta_1) \leq \eta_2$$

holds for all $n \geq N_0(\eta_1, \eta_2)$. We consider the following closed subset of the parameter space Θ which does not intersect C

$$S_0 := \{\theta \in \Theta : \bar{d}([\theta], [\theta_0]) \geq \eta_1\}.$$

Using [Theorem 7](#) we select $0 < h(\eta_2) < 1$ such that

$$\mathbb{P}\left(\sup_{\theta \in S_0} \prod_{i=1}^n \frac{f(x_i; \theta)}{f(x_i; \theta_0)} > h^n(\eta_2)\right) < \eta_2 \text{ for every } n \geq N(\eta_1, \eta_2),$$

for some $N(\eta_1, \eta_2)$. The dependence on η_1 comes from the choice of the set S_0 . Next, we choose N_0 such that $h^n(\eta_2) < c$ holds for every $n \geq N_0$. Let $N_0(\eta_1, \eta_2) = \max(N_0, N(\eta_1, \eta_2))$. Therefore for all $n \geq N_0(\eta_1, \eta_2)$, we have that

$$\mathbb{P}(\bar{d}([\theta_n], [\theta_0]) > \eta_1) \leq \mathbb{P}\left(\sup_{\theta \in S_0} \prod_{i=1}^n \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \geq c\right)$$

$$\begin{aligned} &\leq \mathbb{P} \left(\sup_{\theta \in S_0} \prod_{i=1}^n \frac{f(x_i; \theta)}{f(x_i; \theta_0)} > h^n(\eta_2) \right) \\ &\leq \eta_2. \end{aligned}$$

□

To prove [Theorem 9](#) we need the following lemma.

Lemma 2. *Let Y be a complete metric space, and let $\psi : \Theta \rightarrow Y$ be a Lipschitz map such that $\psi(\theta) = \psi(\theta')$ whenever $\theta \sim \theta'$. Then ψ lifts to a unique Lipschitz map from Θ/C to Y .*

Proof. The proof of the statement follows from proposition 1.4.4 and proposition 1.4.3 in [\[29\]](#). □

Proof of Theorem 9. We consider the function $\psi : \Theta \rightarrow \mathbb{R}^{p \times p}$ defined by $\psi(\mathbf{W}, \sigma^2) = \mathbf{W}\mathbf{W}^T + \sigma^2 I_p$. Since $\theta_0 \in \text{int}(\Theta_0)$, we can therefore consider the restriction $\psi|_{\Theta_0} : \Theta_0 \rightarrow \mathbb{R}^{p \times p}$, which is a Lipschitz function as ψ is C^1 and Θ_0 is compact. Therefore the lift $\tilde{\psi}|_{\Theta_0} : \Theta_0/C \rightarrow \mathbb{R}^{p \times p}$ is Lipschitz and hence continuous with respect to the topology generated by the metric $\bar{d}$.

Invoking [Corollary 2](#), we have $[\widehat{\mathbf{W}}, \widehat{\sigma}^2] \xrightarrow{\mathbb{P}} [\mathbf{W}_0, \sigma_0^2]$. Since $\tilde{\psi}|_{\Theta_0}$ is continuous with respect to the topology generated by the quotient metric $\bar{d}$ from the previous lemma, using standard results from probability theory we infer that $\tilde{\psi}|_{\Theta_0}([\widehat{\mathbf{W}}, \widehat{\sigma}^2]) \xrightarrow{\mathbb{P}} \tilde{\psi}|_{\Theta_0}([\mathbf{W}_0, \sigma_0^2])$. □

As we saw in the demonstration above, [Lemma 2](#) plays a crucial role. We would like to *emphasize* that [Theorem 2](#) would not be sufficient to ensure the continuity of the lift of ψ with respect to the metric $\bar{d}$ as the quotient topology needs to be same that of the topology generated by the metric $\bar{d}$ on the quotient space Θ/C (see the counterexamples subsection in the appendices).

1.7. Auxiliary Lemmas

Our goal for this section will be to state Wald's conditions [\[27\]](#) to establish the consistency of MLE of the PPCA model and interpret those in the quotient parameter space Θ/C as stated in [\[21\]](#). Unless otherwise stated, throughout this section we will be working in the quotient parameter space Θ/C introduced in [section 1.4](#). We recall that $\theta_0 = (\mathbf{W}_0, \sigma_0^2)$ denotes the unknown parameter such that the true marginal distribution of the data points $\{x_i\}_{i=1}^n$ is given by the density $f(x; \theta_0) = \mathcal{N}(0, \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p)$. For a given $r > 0$, we let $N_r([\theta])$ denote the closed ball of radius r around $[\theta]$ in Θ/C . Following [\[21\]](#) we begin by introducing the quantities:

$$f(x, \theta, r) = \sup_{[\phi] \in N_r([\theta])} f(x; \phi) \text{ and } f^*(x, \theta, r) = \max\{1, f(x, \theta, r)\},$$

$$h(x,s) = \sup_{[\phi] \notin N_s([\theta_0])} f(x;\phi) \text{ and } h^*(x,s) = \max\{1, h(x,s)\}.$$

We are now in a position to state Wald's conditions as a series of following lemmas whose proofs will imply [Theorem 3](#) and [Theorem 7](#) as a consequence of the results stated in [\[27\]](#) and [\[31\]](#), respectively.

Lemma 3. *The parameter space $(\Theta/C, \bar{d})$ is a metric space with the property that every closed and bounded subset of Θ/C is compact.*

Lemma 4. *For each parameter $[\theta] \in \Theta/C$ and for sufficiently small r and sufficiently large s , $f(\cdot, \theta, r)$ is measurable and the following expectations are bounded*

$$\mathbb{E}_{\theta_0}[\log f^*(x, \theta, r)] < \infty \text{ and } \mathbb{E}_{\theta_0}[\log h^*(x, s)] < \infty.$$

Lemma 5. *Let $\{[\theta_i]\} \subset \Theta/C$ be a sequence. If $\bar{d}([\theta_i], [\theta_0]) \rightarrow \infty$ then $f(x; \theta_i) \rightarrow 0$ except on a P_{θ_0} -null set which does not depend on the sequence $\{[\theta_i]\}$.*

Lemma 6. *For each $([\theta], [\phi]) \in \Theta/C \times \Theta/C$ We have*

$$\mathbb{E}_{\phi}[\log f(x; \theta)] < \infty.$$

Lemma 7. *If $[\theta_i] \rightarrow [\theta]$ in Θ/C then $f(x; \theta_i) \rightarrow f(x; \theta)$ except on a P_{θ_0} -null set which does not depend on the sequence $\{[\theta_i]\}$.*

The above conditions are fairly technical and primarily deal with the geometry of the quotient parameter space Θ/C . One also needs to be careful as there are two different topologies available in this quotient space which need not be the same. For instance, in [Lemma 7](#), the convergence under the if condition $[\theta_i] \rightarrow [\theta]$ is required to hold in the topology generated by the metric $\bar{d}$ which is a stronger requirement than merely asking for convergence in the quotient topology. The proofs of the above results are deferred to the relevant subsection in the appendices.

1.8. Limitations and Broader Impact

In this work, we were primarily focused to establish the consistency of maximum likelihood estimates of the PPCA model in a quotient Euclidean space. However, we were neither able to provide a rate of convergence nor we could say something definitive about the asymptotic distribution of the MLE, and this opens up two avenues for future research in [chapter 4](#). Nonetheless, it is worth noting that our research does not present any discernible negative societal implications.

1.9. Conclusion

In this paper, we proposed a novel topological framework to provide rigorous justification for the theoretical validity of the maximum likelihood estimation of the PPCA model. Although our

consistency results hold within a quotient space, implying that the true parameter is recoverable up to a closed subset of the parameter space, our results establish that this represents the optimal achievable outcome due to the challenges caused by identifiability inherent in the underlying model. Our result is stated in terms of the abstract framework of quotient topological spaces, but an immediate (and concrete) consequence of our work is (strong) consistent covariance estimation of the PPCA process through maximum likelihood estimates, which wasn't known before. Our rigorous framework opens doors for applications in many other statistical models where rotational ambiguity (or more generally ambiguity due to a closed space or symmetric parametrization) is present. One such example is the matrix factorization problem where one is interested in the estimation of two matrices $\mathbf{A}$, $\mathbf{B}$ such that the data matrix $\mathbf{X} = \mathbf{AB} + \text{noise}$ is one such problem. Our methodology is highly flexible in the sense that it does not depend on the statistical model much as long as there are some regularities in the geometry of the parameter space Θ , which makes it even more usable, even for non-linear models such as nonlinear independent component analysis where the observed data is generated as $x = \xi(z|\theta) + \text{noise}$, where z is a latent random vector and θ is a parameter and the function ξ is non linear. The methodology we developed in our work for the PPCA model could be readily applied in this case as our quotient space construction barely depends on the fact that ξ is linear for PPCA. The primary focus for us was to build a strong theoretical foundation which was missing in the relevant literature. Furthermore, our study offers a concise and comprehensive treatment of the theory of quotient topological spaces, with an eye on statistical applications.

1.10. Appendix of Chapter 1

1.10.1. Counterexample: Quotient and metric topologies on the quotient parameter space are different.

It was pointed out in the introduction that special care is needed when talking about continuity and convergence in probability within the quotient parameter space Θ/C as there are two different topologies at play. In fact, a substantial challenge of our work was to meticulously address the technical issues that arise from the interplay between these two topologies. In [21], it was claimed that the MLE converges to the true parameter in the quotient topological space and this follows from the theory of quotient spaces which is not true in general, as it is untrue that the quotient metric behaves in a similar fashion to that of the original Euclidean metric and therefore it would take a great deal of effort to unambiguously define the right notion of convergence to be used for assumption 5 (and 3) in [21]. The interaction between the metric topology and the quotient topology depends highly on the equivalence relation $\sim$ on the parameter space Θ . Here is a counter-example highlighting a situation where the quotient topology is nontrivial but the topology generated by the quotient metric $\bar{d}$ is degenerate:

Consider the space $X = \{(x,y) : x,y \geq 0\} - \{(0,0)\}$ with the usual Euclidean norm, i.e., the first quadrant with axes except the origin and define the equivalence relation $\sim$ on X given by: $p_1 \sim p_2 \iff p_1 = \lambda p_2$ for some $\lambda > 0$, i.e., the points p_1, p_2 are equivalent if they are on the same ray. The projection map here is $\pi : X \rightarrow X/\sim$ is given by $\pi(p) = [p]$. Geometrically, it is helpful to visualize the space $X/\sim$ as a north-east part of the unit circle S^1 , which is a subspace of a full circle. Under this equivalence relation it is true that $d_1([x],[y]) = 0$ (from the definition of pseudo metric d_1 in section 1.4) for any two distinct $[x],[y] \in X/\sim$, which can be easily argued by taking $x_1 = x/n, y_1 = y/n$ with $n \in \mathbb{N}$, so that, $x_1 \sim x, y_1 \sim y$ and letting $n \rightarrow \infty$. Therefore the usual metric structure totally breaks down in the quotient space as any two different equivalence class has a distance zero between them while in this case the quotient topology is non-trivial and we can still talk about convergence with regards to the quotient topology. But to extend Wald's condition in the quotient, the space $X/\sim$ must be endowed with a non-trivial metric space structure, which is impossible in a situation like this. We also note that the quotient space *had to* pass through a pseudometric to achieve the desired metric space structure in order to set up the ground for Wald's results to extend. In general, it is very hard to interpret something meaningful out of this intangible metric from a statistical point of view. However, when the quotient is done with respect to a closed set, one can provide a concrete tangible geometrical description of the quotient metric space through Lemma 1, which in our work has been done in the context of PPCA. The model PPCA is *not* important here, rather

the *key* part is that: the quotient has to be with respect to a *nice enough equivalence* (for instance with respect to a closed set, which is the case in Redner's work, our work and numerous other statistical models where identifiability arises). This *key technical but unavoidable* construction seems to have gone unnoticed in earlier works. Note that, it was imperative to come up with an explicit and tangible description of the metric in the quotient space as that plays a crucial role towards the proof of assumption 5 ([Lemma 7](#) in our work) in [\[21\]](#), whose proof was assumed straightforward.

1.10.2. Counterexample: [Theorem 2](#) does not ensure continuity of the lift under $\bar{d}$.

We pointed out in [section 1.6](#) that [Lemma 2](#) is not sufficient to guarantee the continuity of the lift of the map ψ used in the deduction of [Theorem 9](#). Indeed it is possible that a continuous (and *not* Lipschitz) function $g : \Theta \rightarrow Y$ lifts to a function $\tilde{g} : \Theta/C \rightarrow Y$ which is no longer continuous with respect to the metric topology generated by $\bar{d}$ on the space Θ/C . Here is a counterexample which highlights the situation:

Let X be a subset of the unit square $[0,1]^2$ consisting of the origin and the points $a_n = (\frac{1}{n}, 0)$, $b_n = (\frac{1}{n}, 1 - \frac{1}{n})$, and $c_n = (\frac{1}{n}, 1)$ for $n \geq 2$. We endow X with the Euclidean metric inherited from $[0,1]^2$. We declare the following equivalence relation $\sim$ on X : we let $a_n \sim b_n$ for all n , and let $g : X \rightarrow \mathbb{R}$ be the function such that $g(0) = g(a_n) = g(b_n) = 0$, and $g(c_n) = 1$. In the quotient space $X/\sim$, the sequence $\{[c_n]\}$ converges to the origin, so the lift $\tilde{g}$ isn't continuous, as $\tilde{g}([c_n]) = 1$ for every n and $g(0) = 0$. One may note that the function g is not Lipschitz in this example since the points $\{b_n\}$ and $\{c_n\}$ will become arbitrarily close as $n \rightarrow \infty$ while $|g(b_n) - g(c_n)| = 1$. Even though the domain of the function g , in this case, is discrete, one can extend g continuously to $[0,1]^2$, using the Tietz extension theorem (see [\[19\]](#)) and provide a counterexample where the domain is an Euclidean space.

We note that ψ (used in the deduction of [Theorem 9](#)) is not Lipschitz in the parameter space Θ due to the presence of a quadratic term in $\mathbf{W}$. One way to ensure that ψ is Lipschitz is to assume the existence of a compact subset Θ_0 which contains the point θ_0 in its interior.

1.10.3. Proofs of auxiliary lemmas

Proof of [Lemma 3](#). This is a direct consequence of the properties discussed in [section 1.4](#) and in particular, the fact that the topology generated from the metric $\bar{d}$ is weaker than the standard quotient topology on Θ/C . Let K be a closed and bounded set in $(\Theta/C, \bar{d})$ and $\{U_\alpha\}_{\alpha \in \Lambda}$ be

an open cover for K . Since the set $\pi^{-1}(K)$ is closed (by the continuity of quotient map) and bounded in Θ , it is compact. Therefore the open cover $\{\pi^{-1}(U_\alpha)\}_{\alpha \in \Lambda}$ can be reduced to a finite sub-cover of $\pi^{-1}(K)$ in Θ . Using the surjectivity of the quotient map, we then see that there is a finite subset $F \subset \Lambda$ such that

$$K = \pi(\pi^{-1}(K)) \subseteq \pi\left(\bigcup_{\alpha \in F} \pi^{-1}(U_\alpha)\right) = \bigcup_{\alpha \in F} U_\alpha.$$

□

Proof of Lemma 4. To ensure the measurability conditions so that the above integrals are well defined we include all the P_{θ_0} null sets to the associated Borel sigma algebra as discussed in [2] (p.167, Corollary 7.42.1).

Given a parameter $\phi = (\mathbf{W}_\phi, \sigma_\phi^2) \in \Theta$, and $r > 0$, we note the function $\phi \mapsto -p \log(2\pi)/2 - \log \det(\mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I)/2$ is continuous functions on Θ . If $\phi \sim \phi'$, then the function evaluates to the same value for ϕ and ϕ' . Therefore the function factors through the quotient topological space Θ/C by Theorem 2. In this case, the map $\phi \mapsto -p \log(2\pi)/2 - \log \det(\mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I)/2$ can be realized as a continuous function on the quotient space Θ/C . For $[\theta] \in \Theta/C$ fixed, we consider the ball $N_r([\theta])$, which is compact. Thus by the virtue of continuous functions, the map will have finite maximum in $N_r([\theta])$. We call these maximum $M_1(r)$. The log-likelihood $\log f(x; \phi)$ is given by

$$\log f(x; \phi) = -\frac{p \log(2\pi)}{2} - \frac{1}{2} \log \det(\mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I_p) - \frac{1}{2} x^T (\mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I_p)^{-1} x,$$

and thus we have that

$$\begin{aligned} \mathbb{E}_{\theta_0}[\log f^*(x, \theta, r)] &= \int_{\mathbb{R}^p} \log \max \left\{ 1, \sup_{[\phi] \in N_r([\theta])} f(x; \phi) \right\} dP_{\theta_0}(x) \\ &\leq \int_{\mathbb{R}^p} \sup_{[\phi] \in N_r([\theta])} \log f(x; \phi) dP_{\theta_0}(x) \\ &= \int_{\mathbb{R}^p} M_1(r) dP_{\theta_0}(x) - \frac{1}{2} \mathbb{E}_\theta \left[\inf_{[\phi] \in N_r([\theta])} x^T (\mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I)^{-1} x \right] \\ &\leq M_1(r) \quad (\text{since } (\mathbf{W}_\phi \mathbf{W}_\phi^T + \sigma_\phi^2 I)^{-1} \text{ is positive definite}). \end{aligned}$$

Bounding $h^*(x, s)$ is a similar task. First we note there exists $M^* > 0$ such that $f(x; \phi) < 1$ (using Wald's assumption 3 [27] or the next lemma) whenever $\bar{d}([\phi], 0) = \|\phi\| > M^*$. Pick s sufficiently large such that $[\phi] \notin N_s([\theta_0])$ implies $\|\phi\| \geq M^*$ (can be ensured using triangle inequality $\|\phi\| \geq \|\theta_0\| - \|\phi\| - \|\theta_0\|$) and thus in this case $h^*(x, s) = 1$ and $\mathbb{E}[\log h^*(x, s)]$ evaluates to 0. □

Proof of Lemma 5. Since $\bar{d}([\theta_i], [\theta_0]) \rightarrow \infty$ in Θ/C implies $d(\theta_i, \theta_0) \rightarrow \infty$ in Θ (by definition of $\bar{d}$ in section 1.4), we have $\|\theta_i\| \rightarrow \infty$ by triangle inequality. We aim to show $\log f(x; \theta_i) \rightarrow -\infty$. The log-likelihood, therefore, is bounded by

$$\begin{aligned}
\log f(x; \theta_i) &= -\frac{p \log(2\pi)}{2} - \frac{1}{2} \log \det(\mathbf{W}_{\theta_i} \mathbf{W}_{\theta_i}^T + \sigma_{\theta_i}^2 I_p) - \frac{1}{2} x^T (\mathbf{W}_{\theta_i} \mathbf{W}_{\theta_i}^T + \sigma_{\theta_i}^2 I_p)^{-1} x \\
&\leq -\frac{1}{2} \log \det(\mathbf{W}_{\theta_i} \mathbf{W}_{\theta_i}^T + \sigma_{\theta_i}^2 I_p) \\
&= -\frac{1}{2} \log \mathcal{P}_{-\mathbf{W}_{\theta_i} \mathbf{W}_{\theta_i}^T}(\sigma_{\theta_i}^2) \quad \left(\mathcal{P}_{-\mathbf{W}_{\theta_i} \mathbf{W}_{\theta_i}^T} \text{ is the characteristic polynomial} \right) \\
&= -\frac{1}{2} \sum_{j=1}^p \log(\sigma_{\theta_i}^2 + \lambda_{ij}) \quad \left(\{\lambda_{ij}\}_{j=1}^p \text{ eigenvalues of } \mathbf{W}_{\theta_i} \mathbf{W}_{\theta_i}^T \right) \\
&\leq \frac{(p-1)}{2} \log \left(\frac{1}{\epsilon_*} \right) - \frac{1}{2} \log(\sigma_{\theta_i}^2 + \|\mathbf{W}_{\theta_i}\|).
\end{aligned}$$

We observe that if $\|\theta_i\| \rightarrow \infty$, either $\|\mathbf{W}_{\theta_i}\|$ or $\|\sigma_{\theta_i}^2\|$ (or both) must become arbitrarily large, which in turn implies the required result. $\square$

Proof of Lemma 6.

$$\begin{aligned}
\mathbb{E}_\phi[|\log f(x; \theta)|] &= \int_{\mathbb{R}^p} |\log f(x; \theta)| dP_\phi(x) \\
&= \int_{\mathbb{R}^p} \left(\frac{p \log(2\pi)}{2} + \frac{1}{2} \log \det(\mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I) \right) dP_\phi(x) \\
&\quad + \frac{1}{2} \mathbb{E}_\phi[x^T (\mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I)^{-1} x] \\
&\leq c_1(\theta) + \frac{1}{2} \mathbb{E}_\phi[\|x\|_2 \|(\mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I)^{-1} x\|_2] \\
&\leq c_1(\theta) + \|(\mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 I)^{-1}\| \mathbb{E}_\phi[\|x\|_2^2].
\end{aligned}$$

$\square$

Proof of Lemma 7. Suppose, $\theta \in C$. We observe $\bar{d}([\theta_i], [\theta]) \rightarrow 0$ in Θ/C implies $d(\theta_i, C) \rightarrow 0$ in Θ . Since C is a closed set and Θ is complete, therefore the sequence $\{\theta_i\}$ converges to some $\theta_0 \in C$. As $\theta \mapsto f(x; \theta)$ is continuous, we have $f(x; \theta_i) \rightarrow f(x; \theta_0) = f(x; \theta)$, in this case. If $\theta \notin C$, then there is a neighbourhood $U_\theta \subset \Theta$ of θ such that $U_\theta \cap C = \emptyset$ and consequently all but finitely many terms in the sequence $\{\theta_i\}$ belong to U_θ . This implies, in this case, $\bar{d}([\theta_i], [\theta]) = d(\theta_i, \theta)$ for all but finitely many i and the conclusion readily follows. $\square$

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Chapter 2

A Bayesian robust approach to probabilistic PCA

This chapter is based on joint work with my supervisors Philippe Gagnon and Florian Maire. This work will be submitted soon.

Résumé

L'analyse en composantes principales bayésienne (ACPB) constitue l'une des classes de modèles statistiques les plus populaires pour la réduction de dimension. Elle est utilisée, par exemple, pour extraire une structure de faible dimension à partir d'ensembles de données de grande dimension en génomique, en imagerie et en finance. Un avantage de cette version probabiliste de l'analyse en composantes principales est qu'elle permet la quantification de l'incertitude. Le modèle ACPB repose toutefois typiquement sur une hypothèse de normalité. Un défaut de ce modèle est donc son manque de robustesse aux valeurs aberrantes (c'est-à-dire aux observations erronées ou extrêmes). Une divergence entre les tendances présentes dans la masse des données et celles associées aux valeurs aberrantes entraîne ainsi une inférence biaisée et une réduction de dimension déformée. Pour remédier à ce problème, nous étendons de manière substantielle les modèles existants à queues lourdes en considérant des distributions à variation régulière ou log-régulière. Nous établissons un résultat théorique caractérisant la robustesse. Nous montrons que l'implantation est simple à l'aide d'une augmentation par variables latentes et d'un échantillonneur de Gibbs.

Abstract

Bayesian principal component analysis (BPCA) forms one of the most popular classes of statistical models for dimensionality reduction. It is used, for instance, to extract low-dimensional structure from high-dimensional data sets in genomics, imaging, and finance. An advantage of this probabilistic version of PCA is that it allows for uncertainty quantification. BPCA typically

relies on a normality assumption. A flaw of this model is thus that it is not robust against outliers (i.e., against erroneous or extreme data points). A difference in trends in the bulk of the data and the outliers thus yields skewed inference and dimension reduction. To address this problem, we significantly extend the existing heavy-tailed models by considering regularly or log-regularly varying distributions. We provide a theoretical result characterizing the robustness. We show that the implementation is straightforward using latent variable augmentation through variational inference and a Gibbs sampler.

Keywords: dimension reduction, Gibbs sampler, heavy-tailed distributions, outliers, regularly varying functions, resolution of conflict.

2.1. Introduction

In the era of big data, we face a number of challenges when analyzing data sets, as they are often comprised of observations involving a massive number of variables. Principal component analysis (PCA) is a standard dimension reduction tool frequently used in exploratory data analysis. It has a broad spectrum of applications ranging from neuroscience [21] to quantitative finance [22]. It is for instance used in [22] to reduce the dimensionality of yield curve data, which are of central importance in assessing economic growth and in pricing theory. In this application, three principal components (PCs) are retained for further analysis, and they roughly correspond to the level, slope and curvature of the yield curves. PCA is however not a statistical model, implying that uncertainty quantification is not available. Uncertainty quantification is desirable in some situations, for example when PCs admit an interpretation as in the previous example. Probabilistic principal component analysis (PPCA) is a model introduced by [25] to remedy this issue. It consists of a low rank matrix additive model with a normal noise term. An analogous Bayesian reformulation has also been discussed in [2]. These models have appealing properties; in particular, the information retained from the original data is of comparable quality to that obtained by classical PCA. There is however a drawback to using models with normal errors: they are not adapted to the presence of outliers (erroneous or extreme data points). This is a consequence of the light tails of the normal distribution. Models based on heavy-tailed distributions have proved to be successful at achieving robustness (see [6] for a location–scale model and [10] for the linear regression model). The rationale is that such distributions are better adapted to the eventual presence of outliers by assigning non-negligible probabilities to extreme values. To retain as much as possible of the appealing properties of the normal models, the strategy is to use a distribution that remains close to the normal distribution while having heavier tails, making Student’s t distribution a natural candidate. This approach was adopted in [4], and an analogous Bayesian approach was considered in [11]. In this paper, we significantly extend the model of [11] by considering a broader family of Bayesian PCA (BPCA) models with

error distributions that are either regularly varying or log-regularly varying, in the spirit of [6]. Even though such approaches have been applied in several contexts [12, 16, 30], not much is known about their theoretical properties. One of the main contributions of this paper is therefore to provide, in section 2.3, a theoretical characterization of the robustness of the proposed class of models.

In section 2.4, we evaluate both the computational efficiency of exact inference via Gibbs sampling and approximate inference via coordinate ascent variational inference (CAVI), as well as the estimation performance of the proposed class of models in the commonly used Student's t setting. Finally, since BPCA based models are inherently non-identifiable, we identify a low dimensional quantity for which valid statistical inference is possible.

2.2. Robust Bayesian PCA

2.2.1. Model and assumptions

In this section, we introduce a class of robust Bayesian PCA, significantly expanding the work of [11]. In the proposed class of models, the low rank matrix has sub-Gaussian entries and the noise has either a regularly varying distribution or a log-regularly varying distribution [6]. Recall that a random variable is sub-Gaussian if its tails decay at least as fast as Gaussian tails [26, Section 2.5].

We consider a data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with standardized columns (meaning each column has a sample mean of 0 and standard deviation of 1). Let us denote the rows of the data matrix by $X_1, \dots, X_n \in \mathbb{R}^d$. In this section, we consider a heavy-tailed formulation of the BPCA model as:

$$X_i = \mathbf{W}Z_i + \tau^{-1/2}\varepsilon_i, \quad i = 1, \dots, n, \quad (2)$$

where $\mathbf{W} = (w_{\ell j}) \in \mathbb{R}^{d \times k}$ is a loading matrix, $Z_i = (z_{ij}) \in \mathbb{R}^k$ is a standardized latent variable with $\ell = 1, \dots, d$ and $j = 1, \dots, k$, and $\varepsilon_i = (\varepsilon_{i1}, \dots, \varepsilon_{id}) \in \mathbb{R}^d$ is a standardized error vector. The latent variables Z_i play an analogous role to the PCs in PCA. It is assumed that $\varepsilon_1, \dots, \varepsilon_n$ are independent and identically distributed random variables, each having a probability density function (PDF) denoted by f_ε . In a Bayesian setting, we assume that the groups of random variables $(w_{\ell j})$, (z_{ij}) , τ , and ε_i are all independent. To continue the model definition, we specify the class of prior distributions we allow. This class of prior distributions reflects the fact that, in PCA, the principal direction vectors are orthonormal and the PCs are orthogonal, with norms proportional to their explained variances. We also assume the priors on these random variables: $w_{\ell j} \sim f_w$, $z_{ij} \sim \frac{1}{\sigma_j} f_z(\cdot/\sigma_j)$, and $\tau \sim f_\tau$ where $\sigma_1, \dots, \sigma_k > 0$ are hyperparameters specified by the user. We now present sufficient conditions on f_ε , f_w , f_z and f_τ essential for robustness of the proposed Bayesian model.

Assumption 1. *The prior distributions of $w_{\ell j}$ and z_{ij} are proper and sub-Gaussian. The distribution of τ is proper and has finite non-negative moments, that is, $\int \tau^\kappa f_\tau(\tau) d\tau < \infty$ for all $\kappa \in \mathbb{R}_{\geq 0}$, where $\mathbb{R}_{\geq 0}$ denotes the set of all non-negative real numbers.*

A natural choice for f_τ is the gamma distribution, with shape parameter $c_\tau > 0$ and scale parameter $d_\tau > 0$, where c_τ and d_τ are hyperparameters specified by the user.

To complete the definition of the model, we make assumptions on the PDF f_ε . We also let $\mathbb{R}_{>0}$ denote the set of all strictly positive real numbers.

Assumption 2. *The PDF f_ε is strictly positive, and there exist a function $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{>0}$ and a constant C_ε such that, for all $y \in \mathbb{R}^d$, we have $f_\varepsilon(y) = C_\varepsilon f(\|y\|)$ and, for any $x_2 \geq x_1$, $f(x_2) \leq f(x_1)$. Also, f is bounded, that is, there exists a constant $C > 0$ such that $f \leq C$. Finally, either of the following holds:*

(i) *regularly varying function: there exists a constant $\alpha > 0$ such that*

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x^{-(\alpha+d)}} = 1;$$

(ii) *log-regularly varying function: there exists a constant $\alpha > 0$ such that*

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x^{-d}(\log x)^{-(\alpha+1)}} = 1.$$

We will call f_ε regularly varying or log-regularly varying depending on the corresponding f .

In [Assumption 2](#), we thus make a contribution by extending the notion of log-regularly varying distributions proposed in [\[6\]](#) and used in robustness contexts in, for example, [\[10, 15\]](#) to multivariate settings.

In this work, we consider Student's t distributions as a primary example, since they are a commonly used model for robustness in linear regression [\[10\]](#) and in machine-learning settings such as robust Bayesian PCA [\[11\]](#), robust probabilistic PCA [\[4\]](#), mixtures of robust probabilistic principal component analyzers [\[1\]](#), robust supervised probabilistic principal component analysis [\[29\]](#), and robust bilinear probabilistic principal component analysis [\[20\]](#), so they will be the focus of the numerical experiments in [section 2.4](#), while we leave for future work the development of an implementation strategy that makes this multivariate extension of log-regularly varying distributions useful in practice.

We also note that the Student's t distribution satisfies [Assumption 2](#) as a regularly varying distribution and that one of the reasons for its popularity is that it can be represented as a scale mixture of normal distributions, which significantly facilitates the implementation of numerical methods for statistical inference, as explained in [subsection 2.2.2](#), and these relatively well-known characteristics of the Student's t distribution are formally stated in the following proposition, whose proof is provided in [subsection 2.6.2](#) for completeness.

Proposition 1. Let f_ε be a d -dimensional Student's t PDF with $\nu > 0$ degrees of freedom, that is

$$f_\varepsilon(y) = \frac{\Gamma((\nu + d)/2)}{\Gamma(\nu/2) (\nu\pi)^{d/2}} \left(1 + \frac{\|y\|^2}{\nu}\right)^{-(\nu+d)/2}, \quad y \in \mathbb{R}^d,$$

where Γ is the gamma function. Then $f_\varepsilon(y) = C_\varepsilon f(\|y\|)$, where

$$C_\varepsilon = \frac{\Gamma((\nu + d)/2) \nu^{\nu/2}}{\Gamma(\nu/2) \pi^{d/2}} \quad \text{and} \quad f(x) = \nu^{-(\nu+d)/2} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+d)/2}, \quad x \in \mathbb{R}_{\geq 0}.$$

Moreover, f satisfies [Assumption 2](#) with $\alpha = \nu$. Finally, $\varepsilon \sim f_\varepsilon$ admits the scale mixture of normals representation

$$\varepsilon \mid U = u \sim \mathcal{N}(0, u^{-1} I_d), \quad U \sim \text{Gamma}(\nu/2, 2/\nu),$$

where the gamma distribution is parameterized by shape and scale.

2.2.2. Posterior inference

For estimation using the Student's t model in the numerical experiments in [section 2.4](#), we exploit the latent variable augmentation of [Proposition 1](#). More precisely, we have that,

$$\varepsilon_i \mid u_i \sim \mathcal{N}(0, u_i^{-1} I_d) \quad \text{and} \quad u_i \sim \text{Gamma}(\nu/2, 2/\nu), \quad i = 1, \dots, n.$$

Additionally, to the random variables forming the model in [subsection 2.2.1](#), the random variable $U = (u_1, \dots, u_n)$ is now included and u_i 's are assumed to be independent among themselves and of all other random variables (except that u_i is not independent of ε_i).

For the numerical experiments, we specialize the prior densities introduced in [subsection 2.2.1](#) by choosing

$$f_w(w) = \mathcal{N}(w; 0, 1), \quad f_z(z) = \mathcal{N}(z; 0, 1), \quad f_\tau(\tau) = \text{Gamma}(\tau; c_\tau, d_\tau).$$

Under the notation of [subsection 2.2.1](#), this means that

$$w_{\ell j} \sim \mathcal{N}(0, 1), \quad z_{ij} \sim \mathcal{N}(0, \sigma_j^2), \quad \tau \sim \text{Gamma}(c_\tau, d_\tau).$$

Let $\mathbf{Z}$ denote the matrix whose rows are $z_1, \dots, z_n$, and define $\Lambda := \text{diag}(\sigma_1^{-2}, \dots, \sigma_k^{-2})$. Rewriting the corresponding prior densities in matrix form (see [section 3.4.3.2](#)) gives:

$$\begin{aligned} \pi(\mathbf{W}) &= (2\pi)^{-dk/2} \exp\left(-\frac{1}{2} \text{tr}(\mathbf{W}^\top \mathbf{W})\right), \\ \pi(\mathbf{Z}) &= (2\pi)^{-nk/2} |\Lambda|^{n/2} \exp\left(-\frac{1}{2} \text{tr}(\mathbf{Z} \Lambda \mathbf{Z}^\top)\right), \\ \pi(U) &= \prod_{i=1}^n \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} u_i^{\nu/2-1} \exp\left(-\frac{\nu}{2} u_i\right), \end{aligned}$$

$$\pi(\tau) := f_\tau(\tau) = \frac{\tau^{c_\tau-1} \exp(-\tau/d_\tau)}{d_\tau^{c_\tau} \Gamma(c_\tau)}.$$

From now on, for consistency, we denote the prior density of τ by $\pi(\tau)$ rather than $f_\tau(\tau)$; more generally, $\pi(\cdot)$ denotes a generic prior distribution and $\pi(\cdot|\mathbf{X})$ a generic posterior distribution. Let x_i denote the observed i th row of $\mathbf{X}$. With the Student's t model and the latent augmentation, the posterior is given by

$$\begin{aligned} \pi(\mathbf{W}, \mathbf{Z}, \tau, U | \mathbf{X}) &\propto p(\mathbf{X} | \mathbf{W}, \mathbf{Z}, \tau, U) \pi(\mathbf{W}) \pi(\mathbf{Z}) \pi(\tau) \pi(U) \\ &= \left[\prod_{i=1}^n p(x_i | \mathbf{W}, z_i, \tau, u_i) \right] \pi(\mathbf{W}) \pi(\mathbf{Z}) \pi(\tau) \pi(U). \end{aligned} \quad (3)$$

where $p(\cdot | \mathbf{W}, \mathbf{Z}, \tau, U)$ denotes the conditional distribution. Conditionally on $(\mathbf{W}, z_i, \tau, u_i)$, the random vector X_i is Gaussian with mean $\mathbf{W}z_i$ and covariance matrix $(\tau u_i)^{-1} I_d$. We also define $\Xi := \text{diag}(\tau u_1, \dots, \tau u_n)$, a $n \times n$ diagonal matrix. By [Proposition 1](#), integrating out U recovers the posterior under the original Student's t error model:

$$\pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}) = \int_{(0, \infty)^n} \pi(\mathbf{W}, \mathbf{Z}, \tau, U | \mathbf{X}) dU.$$

Thus, estimation based on the augmented model is equivalent to estimation based on the original model as long as the statistical quantity of interest does not depend on U . With the latent augmentation, we can implement a simple Gibbs sampler [[13](#)] to draw from the posterior distribution; the algorithm is presented in [subsection 2.6.1](#). It is important to note that latent augmentation is a computational trick that resembles data augmentation [[19](#)]. Because the number of unknowns $(n(k+1) + dk + 1)$ is very high, Gibbs sampling may converge slowly, and Monte Carlo estimates may be inaccurate unless the sampler is run for a very long time. We perform a numerical experiment in [section 2.4](#) to illustrate this.

We thus also consider variational inference (VI) as an alternative to solve this problem approximately. In VI, we want to find an approximate solution to the following optimization problem:

$$q^*(\mathbf{W}, \mathbf{Z}, \tau, U) = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\mathbf{W}, \mathbf{Z}, \tau, U) || \pi(\mathbf{W}, \mathbf{Z}, \tau, U | \mathbf{X})), \quad (4)$$

where we consider a mean field variational family defined as:

$$\mathcal{Q} = \left\{ q(\mathbf{W}, \mathbf{Z}, \tau, U) \left| \begin{array}{l} q(\mathbf{W}, \mathbf{Z}, \tau, U) = q_{\mathbf{W}} q_{\tau} \prod_{i=1}^n q_{z_i} q_{u_i} \quad \text{with} \\ q_{\mathbf{W}} = \mathcal{N}(\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}} \otimes I_d), \\ q_{z_i} = \mathcal{N}(\mu_{z_i}, \Sigma_{z_i}), \quad q_{u_i} = \text{Gamma}(a_i, b_i) \text{ for } 1 \leq i \leq n, \\ \text{and } q_{\tau} = \text{Gamma}(c_{\tau}, d_{\tau}), \end{array} \right. \right\}.$$

We solve (4) using CAVI, a standard optimization scheme in VI used for Bayesian PCA [[3, 5](#)].

To implement the CAVI scheme, we successively update $q_{\mathbf{W}}$, then $q_{z_1}, \dots, q_{z_n}$, then $q_{u_1}, \dots, q_{u_n}$, and finally q_{τ} .

The resulting algorithm is given in [Algorithm 1](#), and its derivation is provided in [subsubsection 2.6.1.2](#). In [section 2.4](#), we use it in particular to estimate $\mathbf{Z}\mathbf{W}'$ which represents the low rank reconstruction of $\mathbf{X}$. Since VI may underestimate posterior uncertainty, Gibbs sampling may still be preferable when uncertainty quantification is the main objective [\[18\]](#).

Algorithm 1 An explicit CAVI algorithm for tBPCA

- (1) Initialization: for $1 \leq i \leq n$, set the parameters $\mu_{z_i}^{(0)}, \Sigma_{z_i}^{(0)}, a_i^{(0)}, b_i^{(0)}$ and $c_{\tau}^{(0)}, d_{\tau}^{(0)}$, and let $q^{(0)}(z_i) = \mathcal{N}(z_i; \mu_{z_i}^{(0)}, \Sigma_{z_i}^{(0)})$, $q^{(0)}(u_i) = \text{Gamma}(u_i; a_i^{(0)}, b_i^{(0)})$, and $q^{(0)}(\tau) = \text{Gamma}(\tau; c_{\tau}^{(0)}, d_{\tau}^{(0)})$. Fix $\varepsilon > 0$.
- (2) Update the parameters of the distribution of $\mathbf{W}$:

$$\Sigma_{\mathbf{W}}^{(t+1)} = ((\mu_{\mathbf{Z}}^{(t)})' \widehat{\Xi}^{(t)} \mu_{\mathbf{Z}}^{(t)} + \sum_{i=1}^n c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} \Sigma_{z_i}^{(t)} + I_k)^{-1},$$

$$\mu_{\mathbf{W}}^{(t+1)} = \mathbf{X}' \widehat{\Xi}^{(t)} \mu_{\mathbf{Z}}^{(t)} \Sigma_{\mathbf{W}}^{(t+1)},$$

where $\mu_{\mathbf{Z}}^{(t)} := (\mu_{z_1}^{(t)}, \dots, \mu_{z_n}^{(t)})'$ and $\widehat{\Xi}^{(t)} := \text{diag} \left(c_{\tau}^{(t)} d_{\tau}^{(t)} a_1^{(t)} b_1^{(t)}, \dots, c_{\tau}^{(t)} d_{\tau}^{(t)} a_n^{(t)} b_n^{(t)} \right)$.

- (3) Update the parameters of the distribution of the latent variables $\mathbf{Z}$:

$$\Sigma_{z_i}^{(t+1)} = (c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} (d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)}) + \Lambda)^{-1},$$

$$\mu_{z_i}^{(t+1)} = c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} \Sigma_{z_i}^{(t+1)} (\mu_{\mathbf{W}}^{(t+1)})' x_i,$$

for $1 \leq i \leq n$.

- (4) Update the parameters of the distribution of the latent variables $u_1, \dots, u_n$:

$$a_i^{(t+1)} = \left(\frac{\nu}{2} + \frac{d}{2} \right),$$

$$[b_i^{(t+1)}]^{-1} = \frac{c_{\tau}^{(t)} d_{\tau}^{(t)}}{2} \left[x_i' x_i - x_i' \mu_{\mathbf{W}}^{(t+1)} \mu_{z_i}^{(t+1)} - (\mu_{z_i}^{(t+1)})' (\mu_{\mathbf{W}}^{(t+1)})' x_i \right. \\ \left. + \text{tr}[(\mu_{z_i}^{(t+1)} (\mu_{z_i}^{(t+1)})' + \Sigma_{z_i}^{(t+1)}) (d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)})] \right] + \frac{\nu}{2},$$

for $1 \leq i \leq n$.

- (5) Update the parameters of the distribution of τ :

$$c_{\tau}^{(t+1)} = \frac{dn}{2} + c_{\tau},$$

$$[d_{\tau}^{(t+1)}]^{-1} = \sum_{i=1}^n \left[\frac{a_i^{(t+1)} b_i^{(t+1)}}{2} \left(x_i' x_i - x_i' \mu_{\mathbf{W}}^{(t+1)} \mu_{z_i}^{(t+1)} - (\mu_{z_i}^{(t+1)})' (\mu_{\mathbf{W}}^{(t+1)})' x_i \right. \right. \\ \left. \left. + \text{tr}[(\mu_{z_i}^{(t+1)} (\mu_{z_i}^{(t+1)})' + \Sigma_{z_i}^{(t+1)}) (d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)})] \right) \right] + \frac{1}{d_{\tau}}.$$

- (6) Calculate the relative increase in ELBO:

$$\Delta^{(t)} \text{ELBO} := \frac{\text{ELBO}(q^{(t+1)}) - \text{ELBO}(q^{(t)})}{\text{ELBO}(q^{(t)})}.$$

- (7) If $\Delta^{(t)} \text{ELBO} > \varepsilon$, return to Step 2.
-

2.2.3. Inference on principal components under non-identifiability

The model in (2) is non-identifiable, and this lack of identifiability is unrelated to the robust Bayesian PCA; it is already present in classical probabilistic and Bayesian formulations of PCA, where the matrices $\mathbf{Z}$ and $\mathbf{W}$ are only determined up to scale changes or orthogonal transformations. Indeed, for any observation $i \in \{1, \dots, n\}$, we have that $x_i = \mathbf{W}z_i = (2\mathbf{W})(z_i/2)$. Also, if $\mathbf{R} \in \mathbb{R}^{k \times k}$ is orthogonal, then $\mathbf{W}z_i = (\mathbf{WR})(\mathbf{R}^\top z_i)$. Therefore, without additional identifying constraints, separate statistical inference on $\mathbf{W}$ and $\mathbf{Z}$ is not possible.

In Bayesian PCA, one is nevertheless interested in the inference on principal components. Because separate inference on $\mathbf{W}$ and $\mathbf{Z}$ is not possible under the model's inherent non-identifiability, we instead focus on the low-rank reconstruction

$$\widehat{\mathbf{X}}_k := \mathbf{Z}\mathbf{W}^\top \in \mathbb{R}^{n \times d},$$

which is identifiable and has rank k . The goal is to perform PCA on $\widehat{\mathbf{X}}_k$ and recover the corresponding principal components.

A common way to bypass the identifiability issue in the PCA literature and in several robust extensions is to target the principal subspace spanned by the principal directions, or to impose orthogonality constraints directly, rather than interpret $\mathbf{W}$ and $\mathbf{Z}$ separately [25, 2, 31]. However, we are interested in the inference of the principal components and hence, we must consider a different approach.

We work directly with the identifiable reconstruction $\widehat{\mathbf{X}}_k$, rather than on $\mathbf{Z}$ and $\mathbf{W}$ separately. For any rank- k matrix $\mathbf{U} \in \mathbb{R}^{n \times d}$, let $\mathbf{R}_k(\mathbf{U}) \in \mathbb{R}^{d \times k}$ denote the matrix whose columns are the eigenvectors associated with the k non-zero eigenvalues of $\mathbf{U}^\top \mathbf{U}$, arranged in decreasing order of the eigenvalues. We then define the map

$$f_{\text{PC}}(\mathbf{U}) = \mathbf{U}\mathbf{R}_k(\mathbf{U}).$$

Thus, in our setting,

$$\mathbf{Z}_{\text{PC}} := f_{\text{PC}}(\widehat{\mathbf{X}}_k) = (\widehat{\mathbf{X}}_k)\mathbf{R}_k(\widehat{\mathbf{X}}_k).$$

In this work, the columns of the matrix $\mathbf{Z}_{\text{PC}}$ denote the principal components obtained from the low-rank reconstruction $\widehat{\mathbf{X}}_k$.

In section 2.4, we use the estimation techniques described in subsection 2.2.2 to sample from the posterior or variational posterior. For each draw, we evaluate f_{PC} , which provides uncertainty quantification for the principal components, as illustrated in the numerical experiments in subsection 2.4.3.

2.3. Robustness characterization result

To characterize the robustness of the model in (2), we study it under an asymptotic framework where the outliers move further and further away from the bulk of the data. In our setting, we consider that an outlying observation is produced by an extreme error term ε_i , which makes the corresponding observation x_i far away from the bulk of the data. This motivated the use of a heavy-tailed distribution f_ε , which is able to accommodate such extreme observations.

Under such an asymptotic framework, the observations $x_i \in \mathbb{R}^d$ are indexed by a variable $\omega \in \mathbb{N}$ and move away from the bulk of the data as $\omega \rightarrow \infty$ in a mathematically precise sense. Inspired from linear regression [10, 15], we consider that the observations x_i are such that

$$\|x_i\| = a_i + b_i\omega, \quad i = 1, \dots, n,$$

with $a_i > 0$ a constant, $b_i = 0$ if the observation is non-outlier, and $b_i \geq 1$ if it is outlier, and then we let $\omega \rightarrow \infty$. This mathematical representation allows for a general definition of outliers, that is, observations x_i whose magnitude is incompatible with the bulk of the data.

Under this asymptotic framework, we obtain a sequence of posterior distributions, indexed by ω , and we want to understand what a posterior distribution in this sequence looks like when ω is large. The asymptotic framework thus allows us to study how the posterior distribution of $(\mathbf{W}, \mathbf{Z}, \tau)$ behaves when some observations, namely the outlying observations, become more and more extreme.

The posterior density defined in (3) is thus now indexed such that

$$\pi_\omega(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}) \propto \pi(\mathbf{W}, \mathbf{Z}, \tau) \left[\prod_{i=1}^n \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right]. \quad (5)$$

The normalizing constant is therefore given by:

$$m_\omega(\mathbf{X}) = \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau) \left[\prod_{i=1}^n \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] d\tau d\mathbf{Z} d\mathbf{W}.$$

The density $\pi_\omega(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X})$ is the one used for computing probabilities and expectations for $(\mathbf{W}, \mathbf{Z}, \tau)$. Central to the characterization of the robustness of our model is the limiting behaviour of the PDF f_ε evaluated at an outlying point, the following proposition is about this.

Proposition 2. *Suppose Assumption 2 holds. For any outlier $\|x_i\| = a_i + b_i\omega$ with $b_i \geq 1$ and fixed $(\mathbf{W}, \mathbf{Z}, \tau) \in \mathbb{R}^{d \times k} \times \mathbb{R}^{n \times k} \times \mathbb{R}_{>0}$,*

$$\lim_{\omega \rightarrow \infty} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau) f_\varepsilon(x_i)} = 1, \quad (6)$$

where $g(\tau) = \tau^{-\alpha/2}$ if f_ε is a regularly varying function or $g(\tau) = 1$ if f_ε is a log-regularly varying function.

The proof of [Proposition 2](#) follows similarly from [\[8\]](#) and is given in [subsection 2.6.2](#). [Proposition 2](#) suggests that the PDF term of an outlier in the posterior density behaves in the limit like $g(\tau)f_\varepsilon(x_i)$. This is confirmed in [Theorem 10](#).

An outlier can be seen as a source of information that is in conflict with others. The sources with which it is in conflict represent, among others, are the non-outliers. Here, we consider that the prior distribution is not in conflict with the non-outliers. In the presence of conflicting sources of information and when using the proposed robust model, conflicting information is partially rejected as a trace remains, that is $g(\tau)$. Ideally, conflicting information is wholly rejected as its source becomes increasingly remote, which translates into a PDF which behaves asymptotically like $f_\varepsilon(x_i) \propto 1$ [\[27\]](#). The resulting model when f_ε is regularly varying is thus said to be *partially robust*. Recent research on resolution of conflict in linear-regression frameworks has shown that in order to attain whole robustness, it is required to assume that the error PDF f_ε has heavier tails than regularly varying density functions [\[7, 10, 9, 15\]](#); in [\[10\]](#), it is proved that it is sufficient to assume that the error PDF has tails which are log-regularly varying, a concept introduced in [\[6\]](#).

From [Proposition 2](#), when f_ε is log-regularly varying the robust approach thus acts automatically as a practitioner would, and excludes the outliers when they are far enough from the bulk of the data and there is no doubt that they truly are outliers. Note that, in both the regularly varying and log-regularly varying cases, the impact of outlying observations gradually diminishes as they are moved further away from the bulk of the data. Observations that are only moderately far from the bulk therefore still retain a certain influence.

We now introduce definitions that will be used in [Theorem 10](#). We define the set that records the indices of the outlying observations: $O := \{i : b_i \geq 1\}$. The index set of non-outlying observations is thus given by $O^c := \{1, 2, \dots, n\} \setminus O$. We also define the set of non-outlying observations: $\mathbf{X}_{O^c} := \{x_i : i \in O^c\}$. In our result, we demonstrate a convergence of the posterior distribution towards $\pi(\cdot, \cdot, \cdot | \mathbf{X}_{O^c})$, which has a density defined as follows

$$\pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}_{O^c}) := \pi(\mathbf{W}, \mathbf{Z}, \tau) g(\tau)^{|O|} \left[\prod_{i \in O^c} \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] / m(\mathbf{X}_{O^c}), \quad (7)$$

where $|O|$ denotes the cardinality of the set O and

$$m(\mathbf{X}_{O^c}) := \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau) g(\tau)^{|O|} \left[\prod_{i \in O^c} \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] d\tau d\mathbf{Z} d\mathbf{W}.$$

There is a slight abuse of notation because $\pi(\cdot, \cdot, \cdot | \mathbf{X}_{O^c})$ is not the posterior distribution given $\mathbf{X}_{O^c}$ only; there is an additional term in the density, $g(\tau)^{|O|}$, coming from densities evaluated at outlying observations (see [\(6\)](#)). To establish [Theorem 10](#), we need to guarantee that $\pi_\omega(\cdot, \cdot, \cdot | \mathbf{X})$ and $\pi(\cdot, \cdot, \cdot | \mathbf{X}_{O^c})$ are well defined, which is the case under a condition that we now present.

Assumption 3. *In the case where f_ε is regularly varying, the assumption is that $d|O^c| > \alpha|O| \iff |O|/n < d/(\alpha + d)$. In the case where f_ε is log-regularly varying, the assumption is that $d|O^c| \geq |O| \iff |O|/n \leq d/(d + 1)$.*

Proposition 3. *Suppose Assumptions 1, 2 and 3 hold. Then we have that $m(\mathbf{X}_{O^c}) < \infty$ and $m_\omega(\mathbf{X}) < \infty$ for all ω .*

Note also that the result of Proposition 3 can actually be obtained without Assumption 3 in the case where f_ε is log-regularly (follows similarly from the proof of Proposition 4 in [8]). Assumption 3 is, in this case, used for the robustness characterization result. We now present this result.

Theorem 10. *Suppose Assumptions 1, 2 and 3 hold. As $\omega \rightarrow \infty$,*

- (a) *the asymptotic behaviour of the marginal distribution is: $m_\omega(\mathbf{X}) / \prod_{i \in O} f_\varepsilon(x_i) \rightarrow m(\mathbf{X}_{O^c})$;*
- (b) *the posterior density converges pointwise: for all $\mathbf{W} \in \mathbb{R}^{d \times k}, \mathbf{Z} \in \mathbb{R}^{n \times k}, \tau > 0$, $\pi_\omega(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}) \rightarrow \pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}_{O^c})$;*
- (c) *the posterior distribution converges: $\pi_\omega(\cdot, \cdot, \cdot | \mathbf{X}) \rightarrow \pi(\cdot, \cdot, \cdot | \mathbf{X}_{O^c})$.*

An attractive feature of Theorem 10, which is typical of recent robustness characterization results, is the simplicity of its assumptions. These assumptions are easy to interpret. Assumptions 1 and 2 concern choices made by the user of the model, namely the specification of the prior distributions and of the error density f_ε , and can therefore be assessed directly. Assumption 3 concerns the proportion of outliers $|O|/n$ in the data set and is connected to the notion of *breakdown point*, usually defined as the largest proportion of outliers that an estimator can tolerate. See [8] for a detailed discussion of the optimality of a similar assumption in the context of robust Bayesian linear regression.

2.4. Numerical experiments

In this section, we present numerical experiments to illustrate several aspects of the proposed robust Bayesian PCA models. In subsection 2.4.1, we compare the computational cost of the CAVI algorithm and the Gibbs sampler in the robustness context. In subsection 2.4.2, we evaluate the performance of the proposed robust Bayesian PCA model with Student's t errors. Finally, in subsection 2.4.3, we perform statistical inference on the principal components obtained from the identifiable low-rank reconstruction.

Throughout all experiments, we fix the values $d = 10$ and $k = 3$. The sample size n is allowed to vary across experiments. Varying d is related to changing the number of unknown quantities in the model, and in this sense it plays a similar role to changing the value of n . We also performed additional experiments by varying d and k , and observed similar conclusions.

All experiments are based on synthetic datasets generated from a low-rank signal plus noise model defined in (2). We first generate two matrices

$$\mathbf{Z}_0 \in \mathbb{R}^{n \times k}, \quad \mathbf{W}_0 \in \mathbb{R}^{d \times k},$$

whose entries are independently sampled from a standard normal distribution. These matrices define the true low-rank signal

$$\mathbf{X}_k^0 = \mathbf{Z}_0 \mathbf{W}_0^\top.$$

The matrices $\mathbf{Z}_0$ and $\mathbf{W}_0$ are kept fixed for a given experiment, and we only change the noise distribution in order to create datasets that are clean or contaminated with outliers.

The observed data matrix is generated as $\mathbf{X} = \mathbf{X}_k^0 + \mathbf{E}$, where the rows of the error matrix $\mathbf{E}$ are generated independently. More precisely, if E_i denotes the i -th row then

$$E_i = \frac{1}{\sqrt{\tau_0}} \eta_i, \quad i = 1, \dots, n,$$

where $\tau_0 > 0$ is the true noise precision and η_i follows a standardized error distribution g . We consider the following two scenarios:

- **Clean case.** The standardized errors are generated from

$$g = \mathcal{N}_d(0, I_d).$$

- **Contaminated case.** A proportion p of the rows are contaminated. In this case, the standardized error distribution is

$$g = (1 - p) \mathcal{N}_d(0, I_d) + p \mathcal{N}_d(\omega v, I_d),$$

where $p \in (0, 1)$ controls the fraction of outlying rows, $\omega > 0$ controls the size of the contamination, and

$$v = \frac{1}{\sqrt{d}}(1, \dots, 1)^\top \in \mathbb{R}^d.$$

In the contaminated setting, non-outlying observations are generated from the first mixture component, whereas outlying observations are generated from the second component. Thus, the hyper-parameter p can be interpreted as the percentage of contaminated rows in the dataset which is fixed by the user, while ω controls how far these rows are shifted away from the clean data-generating mechanism. In the contaminated component, the standardized error has mean shift ωv . Since $\|v\|_2 = 1$, the contaminated rows are shifted by an amount ω in Euclidean norm relative to the clean component on average, and the shift is uniform across every direction.

In the following subsections, we compare Bayesian PCA models fitted to the same datasets, both in the clean and contaminated settings. The first one is the non-robust normal error model, and the second one is the robust Student's t error model proposed in this work. In subsections 2.4.1 and 2.4.3 the value of ν is set to 4 and in subsection 2.4.2 we justify why we make this choice among other things. For both models, we report results based on the same underlying

synthetic data-generating mechanism, so that the comparison focuses on the effect of the error distribution used for inference.

2.4.1. A comparison between CAVI and Gibbs

In this subsection, the main goal is to compare the computational cost of running a Gibbs sampler and CAVI in the context of Bayesian PCA. The comparison is important because the two methods have rather different notions of convergence, and they are competing methods that offer different advantages as mentioned in [subsection 2.2.2](#).

For CAVI, convergence means that the algorithm has reached a fixed point of the optimization problem in (4), up to a prescribed tolerance. Equivalently, the algorithm finds an approximate local minimizer of the KL divergence between the variational distribution and the posterior. In our experiments, we stop CAVI when the relative change in the ELBO becomes smaller than 10^{-4} . Decreasing this tolerance to 10^{-6} , or even 10^{-8} , did not improve the quality of the approximation in our experiments, but only increased the computational cost. In the literature a tolerance level of 10^{-4} , 10^{-5} or even 10^{-2} is considered standard (see [\[28\]](#)).

For the Gibbs sampler, the notion of convergence is different. The Gibbs sampler produces a Markov chain whose stationary distribution is the posterior. Thus, one has to run the chain long enough so that it reaches stationarity and mixes sufficiently well. In this work, we measure this using the time required to enter the central limit theorem (CLT) regime for ergodic averages. More precisely, we monitor a relevant scalar functional of the chain and identify when their empirical averages behave according to the Markov chain CLT, thus providing a practical way of measuring how many iterations are needed before the Gibbs sampler can be reliably used for sampling. The details of this diagnostic procedure are given in [section 2.7](#).

The solution obtained by the two methods should therefore be interpreted differently. The output of CAVI is a variational approximation at a fixed point. On the other hand, the Gibbs sampler gives samples from the posterior distribution after the chain has reached stationarity. This distinction is especially important when uncertainty quantification is the main goal as pointed out at the end of [subsection 2.2.2](#). However, when point estimation is the primary objective, for example estimating the noiseless low-rank matrix $\mathbf{X}_k^0$, CAVI can be a very competitive alternative. Indeed, in our experiments, the point estimates obtained from CAVI were very close to the ones obtained from long Gibbs runs at convergence.

A key computational difference is that sampling from the variational posterior is independent sampling once the variational approximation has been computed whereas Gibbs sampling produces dependent samples from a Markov chain. Therefore, in applications where the goal is mainly dimension reduction or reconstruction, a practitioner has two possible choices. If point estimation is needed, CAVI gives a very fast deterministic approximation, followed by cheap sampling from the variational posterior, if required. If accurate uncertainty quantification is of

the main interest, then Gibbs sampling is more appropriate, but it can be much more expensive as illustrated in the following tables. In that case, a practical strategy is to initialize the Gibbs sampler using the CAVI solution (see [17]).

We first report the number of iterations required for the Gibbs sampler to enter the CLT regime in the setting where a clean dataset is used in Table 1.

Method	$n = 50$	$n = 500$	$n = 5000$
Normal Gibbs	50	200	6400
Student's t Gibbs	800	1600	204800

Table 1. Convergence time for the Gibbs sampler as n varies in the clean setting.

The results show that the number of Gibbs iterations increases substantially with n . For the Normal model, the number of iterations increases from 50 when $n = 50$, to 200 when $n = 500$, and then to 6400 when $n = 5000$. For the Student's t model, the increase is much higher: the required number of iterations goes from 800, to 1600, and then jumps to 204800. This suggests that, in this experiment, the computational cost of reaching the CLT regime increases very rapidly (exponentially) with n .

We now present the number of iterations required by CAVI to converge in Table 2.

Method	$n = 50$	$n = 500$	$n = 5000$
Normal CAVI	14	17	21
Student's t CAVI	18	18	35

Table 2. Convergence time for CAVI as n varies in the clean setting.

In contrast with the Gibbs sampler, the number of CAVI iterations remains relatively stable as n increases. For the Normal model, the number of iterations only increases from 14 to 21 when n goes from 50 to 5000. For the Student's t model, the number of iterations increases from 18 to 35. Thus, while the Gibbs sampler requires a much larger number of iterations as n increases, CAVI remains *almost independent* of n in this experiment.

Similar behaviour was observed in the contaminated setting. For example, when the contamination proportion is $p = 0.05$ and the contamination size is $\omega = 20$, the same qualitative conclusion holds: the Gibbs sampler requires many more iterations to reach the CLT regime, whereas CAVI converges in a very small number of iterations. Moreover, as far as point estimation is concerned, the estimates of the noiseless low-rank reconstruction $\mathbf{Z}\mathbf{W}^\top$ obtained from CAVI and from long Gibbs runs were checked to be very close to each other at convergence.

These results suggest that CAVI decisively wins from the point of view of computational efficiency when the main objective is point estimation. This motivates the use of CAVI in the next subsections, where we evaluate the performance of the Gaussian and Student's t Bayesian PCA models.

2.4.2. Performance evaluation

In this section, we compare the estimation performance of the Normal model and the Student's t model under different contamination regimes. The goal is to understand how much efficiency is gained by using a robust model when outliers are present, and how much efficiency is lost when the data are clean.

We consider three scenarios. Scenario 0 corresponds to the clean case, where the contamination proportion is $p = 0$. Scenario 1 corresponds to a mildly contaminated setting, where $p = 0.05$. Scenario 2 corresponds to a more severely contaminated setting, where $p = 0.10$. In the contaminated scenarios, the outlying rows are generated according to the contamination mechanism described at the beginning of [section 2.4](#), with $\omega = 20$. Thus, Scenarios 1 and 2 allow us to study how the behavior of the two models changes as the proportion of contaminated observations increases.

The object of interest is the estimation of the underlying low-rank signal matrix $\mathbf{X}_k^0$. We conduct a Monte Carlo experiment. We simulate a total of $2N$ datasets of which N are clean and N are contaminated (either through Scenario 1 or 2) with $N = 100$. For each simulated dataset, we fit both the Normal model and the Student's t model, and we compare the recovered low-rank reconstruction with the true low-rank matrix $\mathbf{X}_k^0$. The mean estimation error is measured using the Frobenius norm between the estimated reconstruction and $\mathbf{X}_k^0$.

To summarize the results, we use the notions of *premium* and *protection*. The premium measures the price one pays for using the robust Student's t model when there are no outliers. In other words, it measures the relative increase in mean estimation error of the robust model compared with the Normal model in Scenario 0. A small premium means that the robust model loses little efficiency in the clean case. On the other hand, the protection measures the gain obtained by using the robust model when outliers are present. It measures the relative decrease in mean estimation error of the Student's t model compared with the Normal model in the contaminated scenarios. A large protection means that the robust model gives a substantial improvement over the Normal model in the presence of outliers.

Therefore, Scenario 0 is used to evaluate the premium, while Scenarios 1 and 2 are used to evaluate the protection. This gives a direct way to compare robustness and efficiency. Ideally, a robust model should have a small premium in the clean case and a large protection in the contaminated cases.

We consider several Student's t models by varying the degrees of freedom parameter ν . In particular, we take

$$\nu \in \{1, 2, 4, 6, 8, 10\}.$$

For each value of ν , and for each sample size n , we compute the corresponding premium and protection for the corresponding scenario. The results are presented using premium–protection

plots. In these plots, each robust alternative is represented by a point whose x-coordinate is the premium and whose y-coordinate is the protection. Thus, points corresponding to the upper-right corner of the plot (see Figure 2) points to robust alternatives that lose little efficiency in clean data while gaining substantial accuracy under contamination.

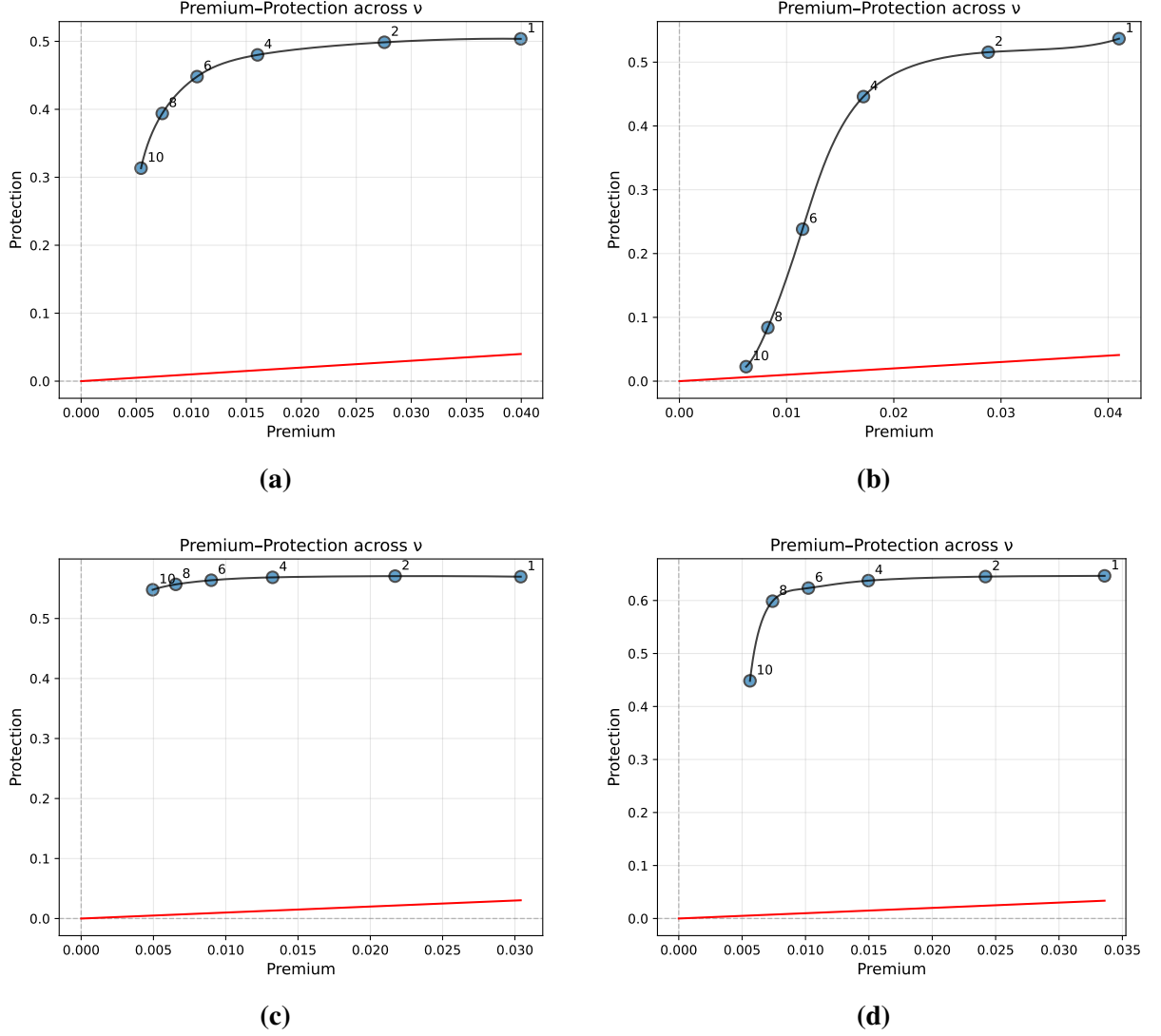


Figure 2. Premium versus protection for different Student's t models: (a) $p = 0.05$ for $n = 50$ (Scenario 1), (b) $p = 0.10$ for $n = 50$ (Scenario 2), (c) $p = 0.05$ for $n = 100$ (Scenario 1), and (d) $p = 0.10$ for $n = 100$ (Scenario 2).

In Figure 2, a particularly useful reference is the line $y = x$. Points above this line correspond to robust alternatives for which the protection is larger than the premium. In other words, the gain obtained under contamination is larger than the loss paid in the clean case. These are the most desirable models from the point of view of robustness. On the other hand, points below this line correspond to alternatives for which the efficiency loss in the clean case is larger than the protection obtained under contamination.

The results in [Figure 2](#) show a premium–protection trade-off. For very small values of ν , the Student’s t model is very robust, but it can pay a higher premium in the clean scenario. For larger values of ν , the Student’s t model becomes closer to the Normal model, and therefore the premium decreases, but the protection under contamination also becomes weaker. Among the values considered, $\nu = 4$ provides the best overall compromise. It gives strong protection in the contaminated scenarios while keeping the premium relatively small in the clean scenario. In particular, all the corresponding points are located above the line $y = x$, showing that the gain from robustness is larger than the price paid for using the robust models.

As the sample size increases, the same qualitative behavior is observed. The Normal model performs well in the clean setting, but its estimation error deteriorates in the presence of contaminated observations. The Student’s t model, especially for moderate degrees of freedom such as $\nu = 4$, is more stable across the three scenarios.

2.4.3. Inference on principal components under non-identifiability

In this subsection, we study the behaviour of $\mathbf{Z}_{\text{PC}}$, as defined in [subsection 2.2.3](#). We compare how well the posterior summary of $f_{\text{PC}}(\hat{\mathbf{X}})$ recovers the ground-truth quantity $f_{\text{PC}}(\mathbf{X}_k^0)$. From the variational posterior, we draw $(\mathbf{Z}^{(s)}, \mathbf{W}^{(s)})$ and reconstruct:

$$\mathbf{X}^{(s)} = \mathbf{Z}^{(s)}(\mathbf{W}^{(s)})^\top, \quad s = 1, \dots, S,$$

with $S = 200,000$, and compute

$$f_{\text{PC}}(\mathbf{X}^{(s)}) = \mathbf{X}^{(s)} \mathbf{R}_k(\mathbf{X}^{(s)}).$$

We then use the empirical posterior mean

$$\bar{\mathbf{Z}}_{\text{PC}} = \frac{1}{S} \sum_{s=1}^S f_{\text{PC}}(\mathbf{X}^{(s)})$$

as the point estimate of $\mathbf{Z}_{\text{PC}}$. All comparisons are made with respect to the ground truth $f_{\text{PC}}(\mathbf{X}_k^0)$. To compare two columns, we use the notion of acute angle between two non-zero vectors. For $a, b \in \mathbb{R}^n$, this angle is

$$\theta(a, b) = \arccos \left(\frac{|a^\top b|}{\|a\|_2 \|b\|_2} \right) \in [0, \pi/2].$$

We report all angles in degrees. For two matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times k}$, we compare them column by column:

$$\theta_j(\mathbf{A}, \mathbf{B}) = \theta(\mathbf{A}_{\cdot j}, \mathbf{B}_{\cdot j}), \quad j = 1, \dots, k.$$

We first consider a clean dataset with $n = 50$. The point-estimate comparison is reported in [Table 3](#). The Gaussian and Student’s t models behave similarly in this setting which suggests

that, in the absence of contamination, the robust Student’s t model does not pay a substantial price relative to the Gaussian model.

Table 3. Clean data: acute angles, in degrees, between $\bar{\mathbf{Z}}_{\text{PC}}$ and $f_{\text{PC}}(\mathbf{X}_k^0)$.

Model	PC1	PC2	PC3
Gaussian	13	35	36
Student’s t	14	36	35

To visualize posterior uncertainty, we focus on PC1 as similar results were observed for other PCs. For each posterior draw, we compute the acute angle between $f_{\text{PC}}(\mathbf{X}^{(s)})_{\cdot 1}$ and $(\bar{\mathbf{Z}}_{\text{PC}})_{\cdot 1}$. The histogram and kernel density estimate are shown in Figure 3. The two models have very similar posterior variability for PC1 in the clean setting, which is consistent with the point-estimate comparison above.

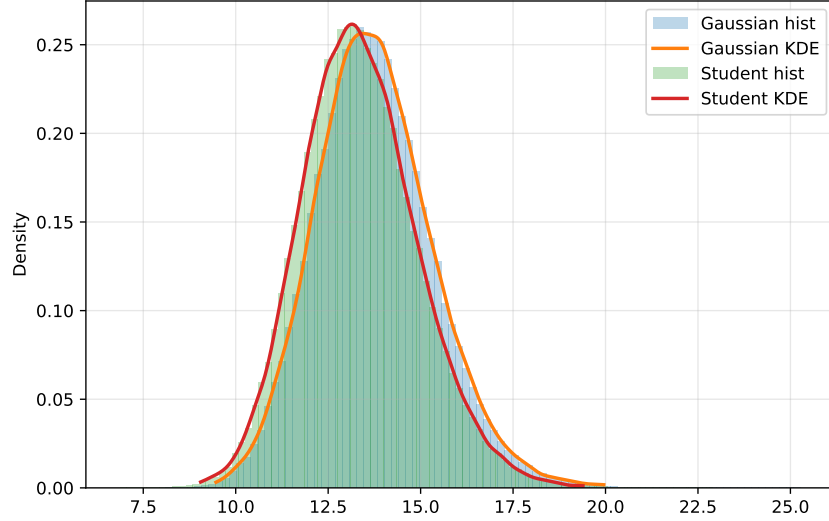


Figure 3. Clean data: histogram and kernel density estimate of the posterior acute angles between $f_{\text{PC}}(\mathbf{X}^{(s)})_{\cdot 1}$ and $(\bar{\mathbf{Z}}_{\text{PC}})_{\cdot 1}$, based on $S = 200,000$ variational posterior draws.

Contaminated data. We now consider a contaminated dataset. The contamination setting is $p = 0.05$ and $\omega = 20$, which is similar to the row-wise contamination scenario considered in the previous subsection. Although only a small proportion of rows is contaminated, the contamination magnitude is large enough to affect the Gaussian model.

The comparison is reported in Table 4. In this case, there is a clear difference between the two models. The Gaussian model gives large angles (compared to the clean case) for all three components, approximately 47° , 67° , and 60° . In contrast, the Student’s t model gives substantially smaller angles, approximately 19° , 27° , and 36° . Thus, under contamination, the Student’s t posterior average recovers $f_{\text{PC}}(\mathbf{X}_k^0)$ much better than the Gaussian posterior average.

The posterior variability for PC1 is shown in Figure 4.

Table 4. Contaminated data: acute angles, in degrees, between $\bar{\mathbf{Z}}_{\text{PC}}$ and $f_{\text{PC}}(\mathbf{X}_k^0)$.

Model	PC1	PC2	PC3
Gaussian	47	67	60
Student’s t	19	27	36

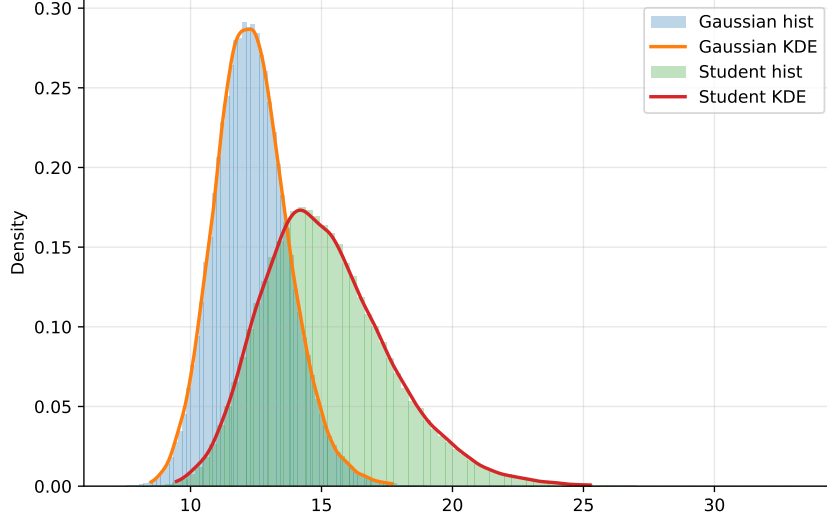


Figure 4. Contaminated data: histogram and kernel density estimate of the posterior acute angles between $f_{\text{PC}}(\mathbf{X}^{(s)})_{.1}$ and $(\bar{\mathbf{Z}}_{\text{PC}})_{.1}$, based on $S = 200,000$ variational posterior draws.

Overall, these results show that the Gaussian and Student’s t models behave similarly on clean data. Under contamination, however, the Gaussian model gives a much poorer recovery of $f_{\text{PC}}(\mathbf{X}_k^0)$, while the Student’s t model remains closer to the ground-truth.

2.5. Conclusion

In this paper, we proposed a robust Bayesian PCA by extending existing heavy-tailed formulations to a broader class of models with regularly varying or log-regularly varying error distributions. We established a theoretical result characterizing robustness, showing that the proposed model can limit the influence of outliers while preserving the main advantages of the Bayesian PCA with normal errors. We also showed that the implementation is tractable through latent variable augmentation, which leads naturally to practical inference procedures based on Gibbs sampling or CAVI. Apart from the theoretical analysis, we also provided a comprehensive case study comparing the statistical and computational performance of Gibbs sampling and CAVI in the context of robust Bayesian PCA models. Finally, by focusing on the identifiable quantity induced by the low-rank reconstruction, we developed a principled way of doing statistical inference on principal components in the presence of outliers.

2.6. Appendix of Chapter 2

2.6.1. Derivation of Gibbs and CAVI updates

2.6.1.1. Gibbs samplers for the normal and Student's t models. Throughout this appendix, we use the priors introduced in [subsection 2.2.2](#), namely

$$w_{\ell j} \sim \mathcal{N}(0,1), \quad z_{ij} \sim \mathcal{N}(0, \sigma_j^2), \quad \tau \sim \text{Gamma}(c_\tau, d_\tau),$$

where the gamma distribution is parameterized by shape and scale. In matrix form, this gives

$$\begin{aligned} \pi(\mathbf{W}) &\propto \exp\left(-\frac{1}{2}\text{tr}(\mathbf{W}^\top \mathbf{W})\right), \\ \pi(\mathbf{Z}) &\propto \exp\left(-\frac{1}{2}\text{tr}(\mathbf{Z}\Lambda\mathbf{Z}^\top)\right), \quad \Lambda := \text{diag}(\sigma_1^{-2}, \dots, \sigma_k^{-2}), \\ \pi(\tau) &= \frac{\tau^{c_\tau-1} \exp(-\tau/d_\tau)}{d_\tau^{c_\tau} \Gamma(c_\tau)}. \end{aligned}$$

For the derivations, it is convenient to write the model in matrix form as

$$\mathbf{X} = \mathbf{Z}\mathbf{W}^\top + \mathbf{E}, \quad \mathbf{X} \in \mathbb{R}^{n \times d}, \quad \mathbf{Z} \in \mathbb{R}^{n \times k}, \quad \mathbf{W} \in \mathbb{R}^{d \times k},$$

where the rows of $\mathbf{X}$ and $\mathbf{Z}$ are denoted by $x_1, \dots, x_n$ and $z_1, \dots, z_n$, respectively. We use $\|\cdot\|_F$ for the Frobenius norm. We also use the notation introduced in [subsection 2.6.4](#): for a random matrix A , the expression $A \sim \mathcal{N}(M, \Sigma^{(c)} \otimes \Sigma^{(r)})$ means that $\text{vec}(A) \sim \mathcal{N}(\text{vec}(M), \Sigma^{(c)} \otimes \Sigma^{(r)})$. In particular, the matrix-valued Gaussian conditionals below are written with covariance matrices of the form $\Sigma \otimes I_d$ or $\Sigma \otimes I_n$.

Algorithm 2 A Gibbs sampler for the normal-error BPCA model

(1) Initialization: choose starting values $\mathbf{W}^{(0)}$, $z_1^{(0)}, \dots, z_n^{(0)}$, and $\tau^{(0)}$, and let $\mathbf{Z}^{(0)} := (z_1^{(0)}, \dots, z_n^{(0)})^\top$.

(2) Update $\mathbf{W}$: set

$$\Sigma_{\mathbf{W}}^{(t+1)} = \left(\tau^{(t)} (\mathbf{Z}^{(t)})^\top \mathbf{Z}^{(t)} + I_k \right)^{-1}, \quad M_{\mathbf{W}}^{(t+1)} = \tau^{(t)} \mathbf{X}^\top \mathbf{Z}^{(t)} \Sigma_{\mathbf{W}}^{(t+1)},$$

and sample

$$\mathbf{W}^{(t+1)} \sim \mathcal{N}\left(M_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)} \otimes I_d\right).$$

(3) Update the latent matrix $\mathbf{Z}$: set

$$\Sigma_{\mathbf{Z}}^{(t+1)} = \left(\tau^{(t)} (\mathbf{W}^{(t+1)})^\top \mathbf{W}^{(t+1)} + \Lambda \right)^{-1}, \quad M_{\mathbf{Z}}^{(t+1)} = \tau^{(t)} \mathbf{X} \mathbf{W}^{(t+1)} \Sigma_{\mathbf{Z}}^{(t+1)},$$

and sample

$$\mathbf{Z}^{(t+1)} \sim \mathcal{N}\left(M_{\mathbf{Z}}^{(t+1)}, \Sigma_{\mathbf{Z}}^{(t+1)} \otimes I_n\right).$$

(4) Update τ : sample

$$\tau^{(t+1)} \sim \text{Gamma}\left(c_\tau + \frac{nd}{2}, \left[\frac{1}{d_\tau} + \frac{1}{2} \left\| \mathbf{X} - \mathbf{Z}^{(t+1)} (\mathbf{W}^{(t+1)})^\top \right\|_F^2 \right]^{-1}\right).$$

(5) Return to Step 2.

Algorithm 3 A Gibbs sampler for the Student's t BPCA model

(1) Initialization: choose starting values $\mathbf{W}^{(0)}$, $z_1^{(0)}, \dots, z_n^{(0)}$, $u_1^{(0)}, \dots, u_n^{(0)}$, and $\tau^{(0)}$, and let $\mathbf{Z}^{(0)} := (z_1^{(0)}, \dots, z_n^{(0)})^\top$.

(2) Update $\mathbf{W}$: let $D_u^{(t)} := \text{diag}(u_1^{(t)}, \dots, u_n^{(t)})$, set

$$\Sigma_{\mathbf{W}}^{(t+1)} = \left(\tau^{(t)} (\mathbf{Z}^{(t)})^\top D_u^{(t)} \mathbf{Z}^{(t)} + I_k \right)^{-1}, \quad M_{\mathbf{W}}^{(t+1)} = \tau^{(t)} \mathbf{X}^\top D_u^{(t)} \mathbf{Z}^{(t)} \Sigma_{\mathbf{W}}^{(t+1)},$$

and sample

$$\mathbf{W}^{(t+1)} \sim \mathcal{N}\left(M_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)} \otimes I_d\right).$$

(3) Update the latent variables $z_1, \dots, z_n$: because the weights $u_1^{(t)}, \dots, u_n^{(t)}$ need not be equal, this step does not reduce to a single matrix-normal update. For $1 \leq i \leq n$, set

$$\Sigma_{z_i}^{(t+1)} = \left(\tau^{(t)} u_i^{(t)} (\mathbf{W}^{(t+1)})^\top \mathbf{W}^{(t+1)} + \Lambda \right)^{-1},$$

$$\mu_{z_i}^{(t+1)} = \tau^{(t)} u_i^{(t)} \Sigma_{z_i}^{(t+1)} (\mathbf{W}^{(t+1)})^\top x_i, \quad z_i^{(t+1)} \sim \mathcal{N}\left(\mu_{z_i}^{(t+1)}, \Sigma_{z_i}^{(t+1)}\right).$$

Let $\mathbf{Z}^{(t+1)} := (z_1^{(t+1)}, \dots, z_n^{(t+1)})^\top$.

(4) Update the latent variables $u_1, \dots, u_n$: for $1 \leq i \leq n$,

$$u_i^{(t+1)} \sim \text{Gamma}\left(\frac{\nu + d}{2}, \frac{2}{\nu + \tau^{(t)} \|x_i - \mathbf{W}^{(t+1)} z_i^{(t+1)}\|^2}\right).$$

(5) Update τ : sample

$$\tau^{(t+1)} \sim \text{Gamma}\left(c_\tau + \frac{nd}{2}, \left[\frac{1}{d_\tau} + \frac{1}{2} \sum_{i=1}^n u_i^{(t+1)} \|x_i - \mathbf{W}^{(t+1)} z_i^{(t+1)}\|^2 \right]^{-1}\right).$$

(6) Return to Step 2.

Normal-error model. We have that, conditionally on $(z_i, \mathbf{W}, \tau)$,

$$x_i \mid z_i, \mathbf{W}, \tau \sim \mathcal{N}(\mathbf{W} z_i, \tau^{-1} I_d), \quad i = 1, \dots, n.$$

The likelihood is therefore

$$p(\mathbf{X} \mid \mathbf{Z}, \mathbf{W}, \tau) \propto \tau^{nd/2} \exp\left(-\frac{\tau}{2} \|\mathbf{X} - \mathbf{Z}\mathbf{W}^\top\|_F^2\right).$$

Conditional distribution of $\mathbf{W} \mid \mathbf{Z}, \tau, \mathbf{X}$. Collecting the terms that depend on $\mathbf{W}$ in the log-posterior gives

$$-\frac{\tau}{2} \|\mathbf{X} - \mathbf{Z}\mathbf{W}^\top\|_F^2 - \frac{1}{2} \text{tr}(\mathbf{W}^\top \mathbf{W}) = -\frac{1}{2} \text{tr}(\mathbf{W}^\top (\tau \mathbf{Z}^\top \mathbf{Z} + I_k) \mathbf{W} - 2\tau \mathbf{W}^\top \mathbf{Z}^\top \mathbf{X}) + \text{const.}$$

Completing the square with

$$\Sigma_{\mathbf{W}}^{(N)} := (\tau \mathbf{Z}^\top \mathbf{Z} + I_k)^{-1}, \quad M_{\mathbf{W}}^{(N)} := \tau \mathbf{X}^\top \mathbf{Z} \Sigma_{\mathbf{W}}^{(N)},$$

yields

$$\mathbf{W} \mid \mathbf{Z}, \tau, \mathbf{X} \sim \mathcal{N}(M_{\mathbf{W}}^{(N)}, \Sigma_{\mathbf{W}}^{(N)} \otimes I_d).$$

Conditional distribution of $\mathbf{Z} \mid \mathbf{W}, \tau, \mathbf{X}$. The terms that depend on $\mathbf{Z}$ are

$$-\frac{\tau}{2} \|\mathbf{X} - \mathbf{Z}\mathbf{W}^\top\|_F^2 - \frac{1}{2} \text{tr}(\mathbf{Z} \Lambda \mathbf{Z}^\top) = -\frac{1}{2} \text{tr}(\mathbf{Z} (\tau \mathbf{W}^\top \mathbf{W} + \Lambda) \mathbf{Z}^\top - 2\tau \mathbf{X} \mathbf{W} \mathbf{Z}^\top) + \text{const.}$$

Completing the square with

$$\Sigma_{\mathbf{Z}}^{(N)} := (\tau \mathbf{W}^\top \mathbf{W} + \Lambda)^{-1}, \quad M_{\mathbf{Z}}^{(N)} := \tau \mathbf{X} \mathbf{W} \Sigma_{\mathbf{Z}}^{(N)},$$

gives

$$\mathbf{Z} \mid \mathbf{W}, \tau, \mathbf{X} \sim \mathcal{N}(M_{\mathbf{Z}}^{(N)}, \Sigma_{\mathbf{Z}}^{(N)} \otimes I_n).$$

Equivalently, conditionally on $(\mathbf{W}, \tau, \mathbf{X})$, the rows $z_1, \dots, z_n$ are independent and satisfy

$$z_i \mid \mathbf{W}, \tau, x_i \sim \mathcal{N}(\tau \Sigma_{\mathbf{Z}}^{(N)} \mathbf{W}^\top x_i, \Sigma_{\mathbf{Z}}^{(N)}), \quad i = 1, \dots, n.$$

Conditional distribution of $\tau \mid \mathbf{Z}, \mathbf{W}, \mathbf{X}$. The likelihood contributes the factor $\tau^{nd/2} \exp\{-\frac{\tau}{2} \|\mathbf{X} - \mathbf{Z}\mathbf{W}^\top\|_F^2\}$; multiplying by the prior density of τ yields a gamma kernel with shape $c_\tau + nd/2$ and scale

$$\left[\frac{1}{d_\tau} + \frac{1}{2} \|\mathbf{X} - \mathbf{Z}\mathbf{W}^\top\|_F^2 \right]^{-1}.$$

Therefore,

$$\tau \mid \mathbf{Z}, \mathbf{W}, \mathbf{X} \sim \text{Gamma}\left(c_\tau + \frac{nd}{2}, \left[\frac{1}{d_\tau} + \frac{1}{2} \|\mathbf{X} - \mathbf{Z}\mathbf{W}^\top\|_F^2 \right]^{-1}\right).$$

Student's t error via Gaussian scale-mixture. For the Student's t model, we introduce the latent variables $u_1, \dots, u_n$ satisfying

$$x_i \mid z_i, \mathbf{W}, \tau, u_i \sim \mathcal{N}(\mathbf{W}z_i, (\tau u_i)^{-1} I_d), \quad u_i \sim \text{Gamma}(\nu/2, 2/\nu), \quad i = 1, \dots, n.$$

Let $D_u := \text{diag}(u_1, \dots, u_n)$, so that the notation of Section 2.2 corresponds to $\Xi = \tau D_u$. The augmented likelihood is

$$p(\mathbf{X} \mid \mathbf{Z}, \mathbf{W}, \tau, U) \propto \tau^{nd/2} \left(\prod_{i=1}^n u_i^{d/2} \right) \exp \left(-\frac{\tau}{2} \sum_{i=1}^n u_i \|x_i - \mathbf{W}z_i\|^2 \right).$$

Conditional distribution of $\mathbf{W} \mid \mathbf{Z}, U, \tau, \mathbf{X}$. Replacing the unweighted quadratic form by the weighted one yields

$$-\frac{1}{2} \text{tr}(\mathbf{W}^\top (\tau \mathbf{Z}^\top D_u \mathbf{Z} + I_k) \mathbf{W} - 2\tau \mathbf{W}^\top \mathbf{Z}^\top D_u \mathbf{X}) + \text{const.}$$

Completing the square with

$$\Sigma_{\mathbf{W}}^{(S)} := (\tau \mathbf{Z}^\top D_u \mathbf{Z} + I_k)^{-1}, \quad M_{\mathbf{W}}^{(S)} := \tau \mathbf{X}^\top D_u \mathbf{Z} \Sigma_{\mathbf{W}}^{(S)},$$

gives

$$\mathbf{W} \mid \mathbf{Z}, U, \tau, \mathbf{X} \sim \mathcal{N}(M_{\mathbf{W}}^{(S)}, \Sigma_{\mathbf{W}}^{(S)} \otimes I_d).$$

Conditional distribution of $z_i \mid \mathbf{W}, u_i, \tau, x_i$. For each $i \in \{1, \dots, n\}$, the terms involving z_i are

$$-\frac{1}{2} [z_i^\top (\tau u_i \mathbf{W}^\top \mathbf{W} + \Lambda) z_i - 2\tau u_i z_i^\top \mathbf{W}^\top x_i],$$

which is a Gaussian kernel in z_i . Completing the square with

$$\Sigma_{z_i}^{(S)} := (\tau u_i \mathbf{W}^\top \mathbf{W} + \Lambda)^{-1}, \quad \mu_{z_i}^{(S)} := \tau u_i \Sigma_{z_i}^{(S)} \mathbf{W}^\top x_i,$$

yields

$$z_i \mid \mathbf{W}, u_i, \tau, x_i \sim \mathcal{N}(\mu_{z_i}^{(S)}, \Sigma_{z_i}^{(S)}), \quad i = 1, \dots, n.$$

Conditional distribution of $\tau \mid \mathbf{Z}, \mathbf{W}, U, \mathbf{X}$. The augmented likelihood contributes $\tau^{nd/2} \exp\{-\frac{\tau}{2} \sum_i u_i \|x_i - \mathbf{W}z_i\|^2\}$; multiplying by $\pi(\tau)$ again yields a gamma kernel. Hence,

$$\tau \mid \mathbf{Z}, \mathbf{W}, U, \mathbf{X} \sim \text{Gamma} \left(c_\tau + \frac{nd}{2}, \left[\frac{1}{d_\tau} + \frac{1}{2} \sum_{i=1}^n u_i \|x_i - \mathbf{W}z_i\|^2 \right]^{-1} \right).$$

Conditional distribution of $u_i \mid \mathbf{W}, z_i, \tau, x_i$. From the likelihood, the terms depending on u_i are $u_i^{d/2} \exp\{-\frac{\tau}{2} u_i \|x_i - \mathbf{W}z_i\|^2\}$, and the prior contributes $u_i^{\nu/2-1} \exp\{-(\nu/2)u_i\}$. Therefore the posterior kernel is

$$u_i^{(\nu+d)/2-1} \exp \left(-\frac{\nu + \tau \|x_i - \mathbf{W}z_i\|^2}{2} u_i \right),$$

which is the kernel of a gamma distribution with shape $(\nu + d)/2$ and scale $2/(\nu + \tau\|x_i - \mathbf{W}z_i\|^2)$. Thus,

$$u_i \mid \mathbf{W}, z_i, \tau, x_i \sim \text{Gamma}\left(\frac{\nu + d}{2}, \frac{2}{\nu + \tau\|x_i - \mathbf{W}z_i\|^2}\right), \quad i = 1, \dots, n.$$

Summary of the full conditional distributions. Let $D_u := \text{diag}(u_1, \dots, u_n)$. Then

$$\begin{aligned} \text{Normal model:} \quad \mathbf{W} \mid \mathbf{Z}, \tau, \mathbf{X} &\sim \mathcal{N}\left(\tau \mathbf{X}^\top \mathbf{Z} \Sigma_{\mathbf{W}}^{(N)}, \Sigma_{\mathbf{W}}^{(N)} \otimes I_d\right), \quad \Sigma_{\mathbf{W}}^{(N)} = (\tau \mathbf{Z}^\top \mathbf{Z} + I_k)^{-1}, \\ \mathbf{Z} \mid \mathbf{W}, \tau, \mathbf{X} &\sim \mathcal{N}\left(\tau \mathbf{X} \mathbf{W} \Sigma_{\mathbf{Z}}^{(N)}, \Sigma_{\mathbf{Z}}^{(N)} \otimes I_n\right), \quad \Sigma_{\mathbf{Z}}^{(N)} = (\tau \mathbf{W}^\top \mathbf{W} + \Lambda)^{-1}, \\ \tau \mid \mathbf{Z}, \mathbf{W}, \mathbf{X} &\sim \text{Gamma}\left(c_\tau + \frac{nd}{2}, \left[\frac{1}{d_\tau} + \frac{1}{2}\|\mathbf{X} - \mathbf{Z} \mathbf{W}^\top\|_F^2\right]^{-1}\right). \end{aligned}$$

$$\begin{aligned} \text{Student's } t \text{ model:} \quad \mathbf{W} \mid \mathbf{Z}, U, \tau, \mathbf{X} &\sim \mathcal{N}\left(\tau \mathbf{X}^\top D_u \mathbf{Z} \Sigma_{\mathbf{W}}^{(S)}, \Sigma_{\mathbf{W}}^{(S)} \otimes I_d\right), \quad \Sigma_{\mathbf{W}}^{(S)} = (\tau \mathbf{Z}^\top D_u \mathbf{Z} + I_k)^{-1}, \\ z_i \mid \mathbf{W}, u_i, \tau, x_i &\sim \mathcal{N}(\mu_{z_i}^{(S)}, \Sigma_{z_i}^{(S)}), \quad \Sigma_{z_i}^{(S)} = (\tau u_i \mathbf{W}^\top \mathbf{W} + \Lambda)^{-1}, \\ \mu_{z_i}^{(S)} &= \tau u_i \Sigma_{z_i}^{(S)} \mathbf{W}^\top x_i, \quad i = 1, \dots, n, \\ \tau \mid \mathbf{Z}, \mathbf{W}, U, \mathbf{X} &\sim \text{Gamma}\left(c_\tau + \frac{nd}{2}, \left[\frac{1}{d_\tau} + \frac{1}{2} \sum_{i=1}^n u_i \|x_i - \mathbf{W}z_i\|^2\right]^{-1}\right), \\ u_i \mid \mathbf{W}, z_i, \tau, x_i &\sim \text{Gamma}\left(\frac{\nu + d}{2}, \frac{2}{\nu + \tau\|x_i - \mathbf{W}z_i\|^2}\right), \quad i = 1, \dots, n. \end{aligned}$$

2.6.1.2. Derivation of the CAVI updates in Algorithm 1. We now derive the updates used in Algorithm 1. At iteration t , the mean-field approximation is

$$q^{(t)}(\mathbf{W}, \mathbf{Z}, \tau, U) = q_{\mathbf{W}}^{(t)}(\mathbf{W}) q_\tau^{(t)}(\tau) \prod_{i=1}^n q_{z_i}^{(t)}(z_i) q_{u_i}^{(t)}(u_i),$$

where

$$\begin{aligned} q^{(t)}(\mathbf{W}) &= \mathcal{N}(\mathbf{W}; \mu_{\mathbf{W}}^{(t)}, \Sigma_{\mathbf{W}}^{(t)} \otimes I_d), \quad q^{(t)}(z_i) = \mathcal{N}(z_i; \mu_{z_i}^{(t)}, \Sigma_{z_i}^{(t)}), \\ q^{(t)}(u_i) &= \text{Gamma}(u_i; a_i^{(t)}, b_i^{(t)}), \quad q^{(t)}(\tau) = \text{Gamma}(\tau; c_\tau^{(t)}, d_\tau^{(t)}), \end{aligned}$$

with the gamma distributions parameterized by shape and scale and ignoring the subscripts to simplify the notation for this subsection since it is always clear from the context. We also define

$$\mu_{\mathbf{Z}}^{(t)} := (\mu_{z_1}^{(t)}, \dots, \mu_{z_n}^{(t)})', \quad \widehat{\Xi}^{(t)} := \text{diag}\left(c_\tau^{(t)} d_\tau^{(t)} a_1^{(t)} b_1^{(t)}, \dots, c_\tau^{(t)} d_\tau^{(t)} a_n^{(t)} b_n^{(t)}\right).$$

In what follows, $C_1, C_2, \dots$ denote generic constants that are independent of the variable currently being updated and may change from line to line.

We first record two equivalent expressions for the complete-data log-density. The first one isolates the dependence on $\mathbf{W}$:

$$\begin{aligned}
\log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau, U) &= \log p(\mathbf{X} \mid \mathbf{W}, \mathbf{Z}, \tau, U) + \log \pi(\mathbf{W}) + C_1 \\
&= -\frac{1}{2} \text{tr} [\Xi(\mathbf{X} - \mathbf{Z}\mathbf{W}')(\mathbf{X} - \mathbf{Z}\mathbf{W}')'] - \frac{1}{2} \text{tr}(\mathbf{W}\mathbf{W}') + C_1 \\
&= -\frac{1}{2} \text{tr} [-\Xi\mathbf{X}\mathbf{W}\mathbf{Z}' - \Xi\mathbf{Z}\mathbf{W}'\mathbf{X}' + \mathbf{W}\mathbf{Z}'\Xi\mathbf{Z}\mathbf{W}' + \mathbf{W}\mathbf{W}'] + C_1 \\
&= \text{tr}(\Xi\mathbf{X}\mathbf{W}\mathbf{Z}') - \frac{1}{2} \text{tr} [\mathbf{W}(\mathbf{Z}'\Xi\mathbf{Z} + I_k)\mathbf{W}'] + C_1, \tag{8}
\end{aligned}$$

where the last line uses standard properties of the trace. The second expression isolates the terms depending on $z_1, \dots, z_n, u_1, \dots, u_n$, and τ :

$$\begin{aligned}
\log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau, U) &= \sum_{i=1}^n [\log p(x_i \mid \mathbf{W}, z_i, \tau, u_i) + \log \pi(z_i) + \log \pi(u_i)] + \log \pi(\tau) + C_2 \\
&= \sum_{i=1}^n \left[\frac{d}{2} \log(\tau u_i) - \frac{\tau u_i}{2} (x_i - \mathbf{W}z_i)'(x_i - \mathbf{W}z_i) - \frac{1}{2} z_i' \Lambda z_i \right. \\
&\quad \left. + \left(\frac{\nu}{2} - 1 \right) \log u_i - \frac{\nu u_i}{2} \right] + (c_\tau - 1) \log \tau - \frac{\tau}{d_\tau} + C_2 \\
&= \sum_{i=1}^n \left[\frac{d}{2} \log \tau + \frac{d}{2} \log u_i - \frac{\tau u_i}{2} (x_i' x_i - x_i' \mathbf{W} z_i - z_i' \mathbf{W}' x_i) \right. \\
&\quad \left. - \frac{1}{2} z_i' (\tau u_i \mathbf{W}' \mathbf{W} + \Lambda) z_i + \left(\frac{\nu}{2} - 1 \right) \log u_i - \frac{\nu u_i}{2} \right] \tag{9} \\
&\quad + (c_\tau - 1) \log \tau - \frac{\tau}{d_\tau} + C_2.
\end{aligned}$$

Identification of $q^{(t+1)}(\mathbf{W})$. By the CAVI update formula,

$$\log q^{(t+1)}(\mathbf{W}) = \mathbb{E}_{q^{(t)}(\mathbf{Z})q^{(t)}(\tau)q^{(t)}(U)} [\log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau, U)] + C_3.$$

Using (8), together with $\mathbb{E}_t[\Xi] = \widehat{\Xi}^{(t)}$ and

$$\mathbb{E}_t[z_i z_i'] = \mu_{z_i}^{(t)} (\mu_{z_i}^{(t)})' + \Sigma_{z_i}^{(t)},$$

we obtain

$$\begin{aligned}
\log q^{(t+1)}(\mathbf{W}) &= \text{tr} \left(\widehat{\Xi}^{(t)} \mathbf{X} \mathbf{W} (\mu_{\mathbf{Z}}^{(t)})' \right) - \frac{1}{2} \text{tr} [\mathbf{W} (\mathbb{E}_t[\mathbf{Z}' \Xi \mathbf{Z}] + I_k) \mathbf{W}'] + C_3 \\
&= \text{tr} \left(\widehat{\Xi}^{(t)} \mathbf{X} \mathbf{W} (\mu_{\mathbf{Z}}^{(t)})' \right) - \frac{1}{2} \text{tr} \left[\mathbf{W} \left((\mu_{\mathbf{Z}}^{(t)})' \widehat{\Xi}^{(t)} \mu_{\mathbf{Z}}^{(t)} \right. \right. \\
&\quad \left. \left. + \sum_{i=1}^n c_\tau^{(t)} d_\tau^{(t)} a_i^{(t)} b_i^{(t)} \Sigma_{z_i}^{(t)} + I_k \right) \mathbf{W}' \right] + C_3.
\end{aligned}$$

Therefore the variational density $q^{(t+1)}(\mathbf{W})$ is, in the notation of [subsection 2.6.4](#), the matrix normal density

$$q^{(t+1)}(\mathbf{W}) = \mathcal{N}(\mathbf{W}; \mu_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)} \otimes I_d),$$

with

$$\begin{aligned} \Sigma_{\mathbf{W}}^{(t+1)} &= \left((\mu_{\mathbf{Z}}^{(t)})' \widehat{\Xi}^{(t)} \mu_{\mathbf{Z}}^{(t)} + \sum_{i=1}^n c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} \Sigma_{z_i}^{(t)} + I_k \right)^{-1}, \\ \mu_{\mathbf{W}}^{(t+1)} &= \mathbf{X}' \widehat{\Xi}^{(t)} \mu_{\mathbf{Z}}^{(t)} \Sigma_{\mathbf{W}}^{(t+1)}. \end{aligned}$$

Identification of $q^{(t+1)}(z_i)$. Since the variables $z_1, \dots, z_n$ appear in distinct summands in (9), each update can be derived separately. For any $i \in \{1, \dots, n\}$,

$$\log q^{(t+1)}(z_i) = \mathbb{E}_{q^{(t+1)}(\mathbf{W})q^{(t)}(\tau)q^{(t)}(u_i)} [\log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau, U)] + C_{4,i}.$$

By (9) and [Proposition 4](#),

$$\mathbb{E}_{q^{(t+1)}(\mathbf{W})}(\mathbf{W}'\mathbf{W}) = d\Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)},$$

which gives

$$\begin{aligned} \log q^{(t+1)}(z_i) &= -\frac{1}{2} z_i' \left(c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} \left(d\Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)} \right) + \Lambda \right) z_i \\ &\quad + c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} z_i' (\mu_{\mathbf{W}}^{(t+1)})' x_i + C_{4,i}. \end{aligned}$$

Hence the variational density $q^{(t+1)}(z_i)$ is Gaussian, that is,

$$q^{(t+1)}(z_i) = \mathcal{N}(z_i; \mu_{z_i}^{(t+1)}, \Sigma_{z_i}^{(t+1)}),$$

where

$$\begin{aligned} \Sigma_{z_i}^{(t+1)} &= \left(c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} \left(d\Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)} \right) + \Lambda \right)^{-1}, \\ \mu_{z_i}^{(t+1)} &= c_{\tau}^{(t)} d_{\tau}^{(t)} a_i^{(t)} b_i^{(t)} \Sigma_{z_i}^{(t+1)} (\mu_{\mathbf{W}}^{(t+1)})' x_i. \end{aligned}$$

Identification of $q^{(t+1)}(u_i)$. For each $i \in \{1, \dots, n\}$,

$$\log q^{(t+1)}(u_i) = \mathbb{E}_{q^{(t+1)}(\mathbf{W})q^{(t+1)}(z_i)q^{(t)}(\tau)} [\log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau, U)] + C_{5,i}.$$

From (9), we have

$$\log q^{(t+1)}(u_i) = \left(\frac{\nu}{2} + \frac{d}{2} - 1 \right) \log u_i - \left[\frac{c_{\tau}^{(t)} d_{\tau}^{(t)}}{2} R_i^{(t+1)} + \frac{\nu}{2} \right] u_i + C_{5,i},$$

where

$$\begin{aligned} R_i^{(t+1)} &:= x_i' x_i - x_i' \mu_{\mathbf{W}}^{(t+1)} \mu_{z_i}^{(t+1)} - (\mu_{z_i}^{(t+1)})' (\mu_{\mathbf{W}}^{(t+1)})' x_i \\ &\quad + \text{tr} \left[(\mu_{z_i}^{(t+1)} (\mu_{z_i}^{(t+1)})' + \Sigma_{z_i}^{(t+1)}) \left(d\Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)} \right) \right]. \end{aligned}$$

Thus the variational density $q^{(t+1)}(u_i)$ is gamma, namely,

$$q^{(t+1)}(u_i) = \text{Gamma}(u_i; a_i^{(t+1)}, b_i^{(t+1)}),$$

with parameters

$$a_i^{(t+1)} = \frac{\nu}{2} + \frac{d}{2}, \quad [b_i^{(t+1)}]^{-1} = \frac{c_\tau^{(t)} d_\tau^{(t)}}{2} R_i^{(t+1)} + \frac{\nu}{2}.$$

Identification of $q^{(t+1)}(\tau)$. Finally,

$$\log q^{(t+1)}(\tau) = \mathbb{E}_{q^{(t+1)}(\mathbf{W})q^{(t+1)}(\mathbf{Z})q^{(t+1)}(U)} [\log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau, U)] + C_6.$$

Again using (9), we obtain

$$\log q^{(t+1)}(\tau) = \left(\frac{dn}{2} + c_\tau - 1 \right) \log \tau - \left[\sum_{i=1}^n \frac{a_i^{(t+1)} b_i^{(t+1)}}{2} R_i^{(t+1)} + \frac{1}{d_\tau} \right] \tau + C_6.$$

Therefore the variational density $q^{(t+1)}(\tau)$ is gamma, namely,

$$q^{(t+1)}(\tau) = \text{Gamma}(\tau; c_\tau^{(t+1)}, d_\tau^{(t+1)}),$$

with parameters

$$c_\tau^{(t+1)} = \frac{dn}{2} + c_\tau, \quad [d_\tau^{(t+1)}]^{-1} = \sum_{i=1}^n \frac{a_i^{(t+1)} b_i^{(t+1)}}{2} R_i^{(t+1)} + \frac{1}{d_\tau}.$$

2.6.2. Proof of theoretical results

Proof of Proposition 1. The decomposition $f_\varepsilon(y) = C_\varepsilon f(\|y\|)$ follows directly from the definition of the multivariate Student's t density. Also, f is strictly positive and bounded, and it is monotone decreasing because $x \mapsto 1 + x^2/\nu$ is increasing on $\mathbb{R}_{\geq 0}$. To prove that f is regularly varying, let $m = (\nu + d)/2$. Then

$$\frac{f(x)}{x^{-(\nu+d)}} = \nu^{-m} \left(1 + \frac{x^2}{\nu} \right)^{-m} x^{2m} = \nu^{-m} \left(x^{-2} + \frac{1}{\nu} \right)^{-m} \longrightarrow \nu^{-m} \left(\frac{1}{\nu} \right)^{-m} = 1,$$

as $x \rightarrow \infty$. Therefore, Assumption 2 holds with $\alpha = \nu$.

In this work, $p(\cdot)$ denotes a generic PDF. For the mixture representation, let $U \sim \text{Gamma}(\nu/2, 2/\nu)$. Conditionally on $U = u$, we have

$$p(\varepsilon | u) = \frac{u^{d/2}}{(2\pi)^{d/2}} \exp \left(-\frac{u \|\varepsilon\|^2}{2} \right),$$

whereas the density of U is

$$p(u) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} u^{\nu/2-1} \exp \left(-\frac{\nu u}{2} \right), \quad u > 0.$$

Hence,

$$\begin{aligned}
p(\varepsilon) &= \int_0^\infty p(\varepsilon | u) p(u) du \\
&= \frac{(\nu/2)^{\nu/2}}{(2\pi)^{d/2} \Gamma(\nu/2)} \int_0^\infty u^{(\nu+d)/2-1} \exp\left(-\frac{u(\nu + \|\varepsilon\|^2)}{2}\right) du \\
&= \frac{(\nu/2)^{\nu/2}}{(2\pi)^{d/2} \Gamma(\nu/2)} \frac{\Gamma((\nu+d)/2)}{((\nu + \|\varepsilon\|^2)/2)^{(\nu+d)/2}} \\
&= \frac{\Gamma((\nu+d)/2)}{\Gamma(\nu/2) (\nu\pi)^{d/2}} \left(1 + \frac{\|\varepsilon\|^2}{\nu}\right)^{-(\nu+d)/2},
\end{aligned}$$

which is exactly $f_\varepsilon(\varepsilon)$. $\square$

Proof of Proposition 2. Let us first consider the case where f_ε is regularly varying. In this case, we recall $g(\tau) = \tau^{-\alpha/2}$. For all $(\mathbf{W}, \mathbf{Z}, \tau) \in \mathbb{R}^{d \times k} \times \mathbb{R}^{n \times k} \times \mathbb{R}^+$,

$$\lim_{\omega \rightarrow \infty} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau) f_\varepsilon(x_i)} = \lim_{\omega \rightarrow \infty} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{C_\varepsilon \|\sqrt{\tau}(x_i - \mathbf{W}z_i)\|^{-(\alpha+d)}} \frac{C_\varepsilon \|x_i\|^{-(\alpha+d)}}{f_\varepsilon(x_i)} \frac{\|x_i\|^{\alpha+d}}{\|x_i - \mathbf{W}z_i\|^{\alpha+d}} = 1.$$

Let us now consider the case where f_ε is log-regularly varying recall $g(\tau) = 1$. In this case, we have that

$$\begin{aligned}
\lim_{\omega \rightarrow \infty} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau) f_\varepsilon(x_i)} &= \lim_{\omega \rightarrow \infty} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{C_\varepsilon \|\sqrt{\tau}(x_i - \mathbf{W}z_i)\|^{-d} (\log \|\sqrt{\tau}(x_i - \mathbf{W}z_i)\|)^{-(\alpha+1)}} \\
&\quad \frac{C_\varepsilon \|x_i\|^{-d} (\log \|x_i\|)^{-(\alpha+1)}}{f_\varepsilon(x_i)} \\
&\quad \frac{\|x_i\|^d (\log \|x_i\|)^{\alpha+1}}{\|(x_i - \mathbf{W}z_i)\|^d (\log \|\sqrt{\tau}(x_i - \mathbf{W}z_i)\|)^{\alpha+1}} \\
&= 1.
\end{aligned}$$

$\square$

Proof of Proposition 3. We prove the result for the case where f_ε is regularly varying; the proof for the case where f_ε is log-regularly varying is similar. When f_ε is regularly varying,

$$\begin{aligned}
m(\mathbf{X}_{O^c}) &= \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau) g(\tau)^{|O|} \left[\prod_{i \in O^c} \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] d\tau d\mathbf{Z} d\mathbf{W} \\
&\leq (CC_\varepsilon)^{|O^c|} \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{(d|O^c| - \alpha|O|)/2} d\tau d\mathbf{Z} d\mathbf{W} \\
&= (CC_\varepsilon)^{|O^c|} \mathbb{E}[\tau^{(d|O^c| - \alpha|O|)/2}] < \infty,
\end{aligned}$$

using that $f_\varepsilon \leq CC_\varepsilon$ (Assumption 2) in the first inequality and, in the final inequality, that $\mathbb{E}[\tau^\kappa] < \infty$ for any $\kappa > 0$ (Assumption 1), given that $d|O^c| - \alpha|O| > 0$ (Assumption 3). Also,

$$\begin{aligned} m_\omega(\mathbf{X}) &= \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau) \left[\prod_{i=1}^n \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] d\tau d\mathbf{Z} d\mathbf{W} \\ &\leq (CC_\varepsilon)^n \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2} d\tau d\mathbf{Z} d\mathbf{W} \\ &= (CC_\varepsilon)^n \mathbb{E}[\tau^{nd/2}] < \infty, \end{aligned}$$

using, similarly, Assumption 2 in the first inequality and Assumption 1 in the final one. $\square$

Proof of Theorem 10. We first prove Result (a). We begin with the observation:

$$\begin{aligned} I(\omega) &:= \frac{m_\omega(\mathbf{X})}{m(\mathbf{X}_{O^c}) \prod_{i \in O} f_\varepsilon(x_i)} \\ &= \frac{m_\omega(\mathbf{X})}{m(\mathbf{X}_{O^c}) \prod_{i \in O} f_\varepsilon(x_i)} \iiint \pi_\omega(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}) d\tau d\mathbf{Z} d\mathbf{W} \\ &= \iiint \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau) \prod_{i=1}^n \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{m(\mathbf{X}_{O^c}) \prod_{i \in O} f_\varepsilon(x_i)} d\tau d\mathbf{Z} d\mathbf{W} \\ &= \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}_{O^c}) \prod_{i \in O} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau) f_\varepsilon(x_i)} d\tau d\mathbf{Z} d\mathbf{W}. \end{aligned} \quad (10)$$

We define the following sequence of functions indexed by ω :

$$F_\omega(\mathbf{W}, \mathbf{Z}, \tau) := \pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}_{O^c}) \prod_{i \in O} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau) f_\varepsilon(x_i)}. \quad (11)$$

We have that

$$\begin{aligned} F_\omega(\mathbf{W}, \mathbf{Z}, \tau) &= \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau) g(\tau)^{|O|}}{m(\mathbf{X}_{O^c})} \left[\prod_{i \in O^c} \tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] \prod_{i \in O} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau) f_\varepsilon(x_i)} \\ &= \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2}}{m(\mathbf{X}_{O^c})} \left[\prod_{i \in O^c} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] \prod_{i \in O} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)}. \end{aligned} \quad (12)$$

Our main strategy is the following: we show that there exists a “nice” set $\mathcal{S}(\omega) \subset \mathbb{R}^{dk} \times \mathbb{R}^{nk}$, such that, on $\mathcal{S}(\omega) \times (0, \infty)$, the function $F_\omega(\mathbf{W}, \mathbf{Z}, \tau)$ is essentially bounded by a polynomial in τ which is integrable and thus allows us to apply Lebesgue’s dominated convergence theorem. On the “bad” set, we cannot obtain such a bound, but we can prove that the integral vanishes because it is proportional to $\exp(-c_0\omega) \omega^\ell \rightarrow 0$ as $\omega \rightarrow \infty$ for some $c_0, \ell > 0$. We deduce this using a concentration inequality.

We now define our “nice” set:

$$\mathcal{S}(\omega) = \left\{ (\mathbf{W}, \mathbf{Z}) : \|\mathbf{W}z_i\| \leq \frac{\omega}{2} \text{ for all } i \right\}. \quad (13)$$

We split the integral in (10) as follows

$$I(\omega) = \iiint F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)} d\tau d\mathbf{Z} d\mathbf{W} + \iiint F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)^c} d\tau d\mathbf{Z} d\mathbf{W}. \quad (14)$$

We note that $\mathcal{S}(\omega) \uparrow \mathbb{R}^{dk} \times \mathbb{R}^{nk}$ or equivalently $\mathbb{1}_{\mathcal{S}(\omega)} \uparrow \mathbb{1}_{\mathbb{R}^{dk} \times \mathbb{R}^{nk}}$, as $\omega \uparrow \infty$, in the sense that for any fixed $(\mathbf{W}, \mathbf{Z})$, there exists ω large enough so that $\|\mathbf{W}z_i\| \leq \omega/2$ for all i .

Let us define

$$I_1(\omega) := \iiint F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)} d\tau d\mathbf{Z} d\mathbf{W} \quad \text{and} \quad I_2(\omega) := \iiint F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)^c} d\tau d\mathbf{Z} d\mathbf{W},$$

so that $I(\omega) = I_1(\omega) + I_2(\omega)$. We prove that $\lim_{\omega \rightarrow \infty} I_1(\omega) = 1$ and $\lim_{\omega \rightarrow \infty} I_2(\omega) = 0$, which concludes the proof of Result (a).

Part 1: proof that $\lim_{\omega \rightarrow \infty} I_1(\omega) = 1$.

By Lemma 8, the following upper bound holds on the set $\mathcal{S}(\omega) \times (0, \infty)$:

$$F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) (1 + \tau^\kappa),$$

where the constants $C_2, \kappa > 0$ are independent of ω and τ . The upper bound is integrable by Assumption 1. Therefore, by Lebesgue’s dominated convergence theorem,

$$\lim_{\omega \rightarrow \infty} I_1(\omega) = \iiint \lim_{\omega \rightarrow \infty} F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)} d\tau d\mathbf{Z} d\mathbf{W} = \iiint \pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}_{O^c}) d\tau d\mathbf{Z} d\mathbf{W} = 1,$$

using (11), (6), Proposition 3 and $\mathbb{1}_{\mathcal{S}(\omega)} \uparrow \mathbb{1}_{\mathbb{R}^{dk} \times \mathbb{R}^{nk}}$, as noted before.

Part 2: proof that $\lim_{\omega \rightarrow \infty} I_2(\omega) = 0$.

We have that

$$\begin{aligned} \iiint F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)^c} d\tau d\mathbf{Z} d\mathbf{W} &\stackrel{(a)}{\leq} \iiint C_1 \omega^{n(2\alpha+d+1)} \tau^{nd/2} \pi(\mathbf{W}, \mathbf{Z}, \tau) \mathbb{1}_{\mathcal{S}(\omega)^c} d\tau d\mathbf{Z} d\mathbf{W} \\ &\stackrel{(b)}{\leq} C_1 \mathbb{E}[\tau^{nd/2}] \omega^{n(2\alpha+d+1)} \mathbb{P} \left(\bigcup_{i=1}^n \left\{ (\mathbf{W}, \mathbf{Z}) : \|\mathbf{W}z_i\| > \frac{\omega}{2} \right\} \right) \\ &\stackrel{(c)}{\leq} C_1 \mathbb{E}[\tau^{nd/2}] \omega^{n(2\alpha+d+1)} \sum_{i=1}^n \mathbb{P} \left(\left\{ (\mathbf{W}, \mathbf{Z}) : \|\mathbf{W}z_i\| > \frac{\omega}{2} \right\} \right) \\ &\stackrel{(d)}{\leq} 2nd C_1 \mathbb{E}[\tau^{nd/2}] \omega^{n(2\alpha+d+1)} \exp(-L_{d,k}\omega), \end{aligned}$$

where $L_{d,k}$ is a positive constant only depending on d and k . At Step (a), we used [Lemma 10](#). At Step (b), using the prior independence of $\mathbf{W}, \mathbf{Z}$ and τ , we separated the terms that depend on τ to those that depend on $(\mathbf{W}, \mathbf{Z})$. Step (c) follows from the union bound. At Step (d), an upper bound on each $\mathbb{P}(\|\mathbf{W}z_i\| > \omega/2)$ was supplied by [Lemma 9](#). Thus, we have that

$$I_2(\omega) \leq 2ndC_1\mathbb{E}[\tau^{nd/2}]\omega^{n(2\alpha+d+1)}\exp(-L_{d,k}\omega) \rightarrow 0.$$

To establish Result (b), we first note the following algebraic identity: for any $(\mathbf{W}, \mathbf{Z}, \tau)$,

$$\pi_\omega(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}) = \pi(\mathbf{W}, \mathbf{Z}, \tau | \mathbf{X}_{O^c}) \left[\frac{m(\mathbf{X}_{O^c}) \prod_{i \in O} f_\varepsilon(x_i)}{m(\mathbf{X})} \right] \left[\prod_{i \in O} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau)f_\varepsilon(x_i)} \right],$$

but

$$\left[\frac{m(\mathbf{X}_{O^c}) \prod_{i \in O} f_\varepsilon(x_i)}{m(\mathbf{X})} \right] \left[\prod_{i \in O} \frac{\tau^{d/2} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{g(\tau)f_\varepsilon(x_i)} \right] \rightarrow 1,$$

following Result (a) and (2).

Result (c) follows from Scheffé's lemma (see [\[23\]](#)) and the pointwise convergence of density functions (Result (b)). $\square$

2.6.3. Three Lemmas

Lemma 8. *Suppose Assumptions 2 and 3 hold. Recall the definition of F_ω in (11) and $S(\omega)$ in (13). For ω large enough and $(\mathbf{W}, \mathbf{Z}, \tau) \in S(\omega) \times (0, \infty)$, there exist constants $C_2 > 0$ and $\kappa > 0$ such that*

$$F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) (1 + \tau^\kappa),$$

the constants C_2, κ being independent of ω and τ .

Proof. We observe that, on the set $S(\omega)$, if $i \in O$, then we have that $\|x_i - \mathbf{W}z_i\| \geq \|x_i\| - \|\mathbf{W}z_i\| \geq \omega/2$ by the reverse triangle inequality and using that $\|x_i\| = a_i + b_i\omega \geq \omega$ and $\|\mathbf{W}z_i\| \leq \omega/2$.

Since f is monotonically decreasing away from the origin and: 1) $\sqrt{\tau}\|x_i - \mathbf{W}z_i\| \geq \sqrt{\tau}\omega/2$, and 2) $\|x_i\| \leq |a_i| + \omega|b_i| \leq 2b_i\omega$, we have that

$$\frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)} = \frac{f(\sqrt{\tau}\|x_i - \mathbf{W}z_i\|)}{f(\|x_i\|)} \leq \frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)}. \quad (15)$$

To prove [Lemma 8](#), we consider two different cases. First, we prove the result when f is regularly varying. In this case, there exists $t_0 > 0$ such that for all $t > t_0$

$$\frac{1}{2}t^{-(\alpha+d)} \leq f(t) \leq \frac{3}{2}t^{-(\alpha+d)}. \quad (16)$$

For large ω , $2b_i\omega \geq t_0$, and therefore from (16),

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq 2(2b_i\omega)^{\alpha+d} f((\sqrt{\tau}\omega)/2) = C_4\omega^{\alpha+d} f((\sqrt{\tau}\omega)/2), \quad (17)$$

where $C_4 = 2(2\max b_i)^{\alpha+d}$. Next, we consider two situations. First, $\sqrt{\tau}\omega/2 > t_0$. In this situation from (17),

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq C_4\omega^{\alpha+d} \left(\frac{3}{2}(\sqrt{\tau}\omega/2)^{-(\alpha+d)}\right) = C_5\tau^{-(\alpha+d)/2},$$

where $C_5 = 3C_4/2^{\alpha+d+1}$.

Second, we consider that $\sqrt{\tau}\omega/2 \leq t_0 \iff \omega \leq 2t_0/\sqrt{\tau}$. In this situation from (17),

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq C_4 \left(\frac{2t_0}{\sqrt{\tau}}\right)^{\alpha+d} C = C_6\tau^{-(\alpha+d)/2},$$

where $C_6 = CC_4(2t_0)^{\alpha+d}$.

Thus, when f_ε is regularly varying from (15)

$$\frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)} = \frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq C_{11}\tau^{-(\alpha+d)/2},$$

for some constant $C_{11} > 0$. Consequently, letting $C_2 = (CC_\varepsilon)^{|O^c|}/m(\mathbf{X}_{O^c})$, we have that

$$\begin{aligned} F_\omega(\mathbf{W}, \mathbf{Z}, \tau) &= \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau)\tau^{nd/2}}{m(\mathbf{X}_{O^c})} \left[\prod_{i \in O^c} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] \prod_{i \in O} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)} \\ &\leq C_2\pi(\mathbf{W}, \mathbf{Z}, \tau)\tau^{nd/2}\tau^{-(\alpha+d)|O|/2} \\ &\leq C_2\pi(\mathbf{W}, \mathbf{Z}, \tau)\tau^{(d|O^c|-\alpha|O|)/2} \\ &\leq C_2\pi(\mathbf{W}, \mathbf{Z}, \tau)(1 + \tau^{(d|O^c|-\alpha|O|)/2}). \end{aligned}$$

Now, we deal with the case where f_ε is log-regularly varying. In this case, there exists $t_0 > 0$ such that for all $t > t_0$

$$\frac{1}{2}t^{-d}(\log t)^{-(\alpha+1)} \leq f(t) \leq \frac{3}{2}t^{-d}(\log t)^{-(\alpha+1)}. \quad (18)$$

In this case, taking ω large, and in particular, $2b_i\omega \geq t_0$, we get from (18)

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq 2(2b_i\omega)^d(\log(2b_i\omega))^{\alpha+1} f((\sqrt{\tau}\omega)/2) = C_7\omega^d(\log(2b_i\omega))^{\alpha+1} f((\sqrt{\tau}\omega)/2), \quad (19)$$

where $C_7 = 2(2\max b_i)^d$. Again, we consider two situations. First, let $\sqrt{\tau}\omega/2 \geq \omega^{1-\eta_d} > t_0$, where $\eta_d = (d+0.1)/(d+1)$ and clearly, $\eta_d \in (0,1)$. In this situation from (19),

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq C_7\omega^d(\log(2b_i\omega))^{\alpha+1} \left(\frac{3}{2}(\sqrt{\tau}\omega/2)^{-d}(\log(\sqrt{\tau}\omega/2))^{-(\alpha+1)}\right)$$

$$= C_8 \tau^{-d/2} \left(\frac{1 + \frac{\log(2b_i)}{\log \omega}}{1 + \frac{\log(\sqrt{\tau}/2)}{\log \omega}} \right)^{\alpha+1},$$

where $C_8 = 3C_7/2^{d+1}$. Note that, $\sqrt{\tau}\omega/2 \geq \omega^{1-\eta_d} \iff 1 + (\log(\sqrt{\tau}/2)/\log \omega) \geq 1 - \eta_d = 0.9/(d+1)$. Thus, all terms in the above bound, except $\tau^{-d/2}$, are constant with respect to τ and bounded with respect to ω .

Consequently, in this situation,

$$\begin{aligned} F_\omega(\mathbf{W}, \mathbf{Z}, \tau) &= \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2}}{m(\mathbf{X}_{O^c})} \left[\prod_{i \in O^c} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] \prod_{i \in O} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)} \\ &\leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2} \tau^{-d|O|/2} \\ &\leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) (1 + \tau^{d|O^c|/2}). \end{aligned}$$

Second, we consider that $\sqrt{\tau}\omega/2 < \omega^{1-\eta_d} \iff \tau < 4/\omega^{2\eta_d} \iff \tau^{(d+1)/2} < 2^{(d+1)}/\omega^{(d+1)\eta_d} = 2^{d+1}/\omega^{d+0.1}$. In this situation from (19),

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq \tau^{-(d+1)/2} \frac{2^{d+1}}{\omega^{d+0.1}} C_7 \omega^d (\log(2b_i\omega))^{\alpha+1} C \leq C_9 \tau^{-(d+1)/2} \frac{(\log(2b_i\omega))^{\alpha+1}}{\omega^{0.1}},$$

where $C_9 = CC_7 2^{d+1}$. The term $(\log(2b_i\omega))^{\alpha+1}/\omega^{0.1}$ tends to 0 as $\omega \rightarrow \infty$, and therefore, when ω is sufficiently large it is bounded by a constant which is independent of ω and τ . We thus have that

$$\frac{f((\sqrt{\tau}\omega)/2)}{f(2b_i\omega)} \leq C_{10} \tau^{-(d+1)/2},$$

for some absolute constant $C_{10} > 0$, independent of ω and τ . In this situation $\sqrt{\tau} < 2/\omega^{\eta_d} \leq 1$ (for ω sufficiently large), we thus have that

$$\begin{aligned} F_\omega(\mathbf{W}, \mathbf{Z}, \tau) &= \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2}}{m(\mathbf{X}_{O^c})} \left[\prod_{i \in O^c} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] \prod_{i \in O} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)} \\ &\leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2} \tau^{-(d+1)|O|/2} \\ &\leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{(d|O^c| - |O|)/2} \\ &\leq C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) \quad (\text{since } \tau \leq 1 \text{ and } d|O^c| - |O| > 0 \text{ by Assumption 3}) \\ &< C_2 \pi(\mathbf{W}, \mathbf{Z}, \tau) (1 + \tau^{d|O^c|/2}). \end{aligned}$$

□

Lemma 9. Let $\mathbf{A} \in \mathbb{R}^{d \times k}$ be a random matrix whose entries are sub-Gaussian. Let $v \in \mathbb{R}^k$ be a random vector which has a sub-Gaussian distribution. There exists a constant $L_{d,k} > 0$ which

depends only on d and k such that

$$\mathbb{P}\left(\|\mathbf{A}v\| > \frac{\omega}{2}\right) \leq 2d \exp(-L_{d,k}\omega), \quad \text{for every } \omega > 0.$$

Proof. We begin the proof with the definitions of sub-Gaussian and sub-exponential random variables, and present some of their properties which will be useful for the proof. A random variable U is called sub-Gaussian if there exists a constant $K_1 > 0$ such that $\mathbb{P}(|U| \geq t) \leq 2 \exp(-t^2/K_1)$ for all $t \geq 0$. A random variable U is called sub-exponential if there exists a constant $K_2 > 0$ such that $\mathbb{P}(|U| \geq t) \leq 2 \exp(-t/K_2)$ for all $t \geq 0$.

We now present definitions of ψ_2 - and ψ_1 -norms and explain how these norms are related to the notion of sub-Gaussian and sub-exponential random variables, respectively. The ψ_2 -norm of a random variable U , denoted by $\|U\|_{\psi_2}$, is defined as follows:

$$\|U\|_{\psi_2} = \inf \left\{ t > 0 : \mathbb{E} \left[\exp \left(\frac{U^2}{t^2} \right) \right] \leq 2 \right\}.$$

The ψ_1 -norm of a random variable U , denoted by $\|U\|_{\psi_1}$, is defined as follows:

$$\|U\|_{\psi_1} = \inf \left\{ t > 0 : \mathbb{E} \left[\exp \left(\frac{|U|}{t} \right) \right] \leq 2 \right\}.$$

The ψ_2 and ψ_1 -norms are inherently connected with the notion of sub-Gaussian and sub-exponential random variables, respectively. In particular, a random variable U is sub-Gaussian (sub-exponential) if and only if it has a finite ψ_2 - (ψ_1 -) norm [26, Sections 2.5 and 2.7]. Note that the presence of the factor 2 in the definitions of $\|U\|_{\psi_2}$ and $\|U\|_{\psi_1}$ is purely for convenience.

Let us denote the rows of $\mathbf{A}$ by $\mathbf{a}_1, \dots, \mathbf{a}_d \in \mathbb{R}^k$. By assumption of the lemma, all the coordinates of the random vectors $\mathbf{a}_i = (a_{i1}, \dots, a_{ik})$ and $v = (v_1, \dots, v_k)$ are sub-Gaussian, and therefore all have finite ψ_2 -norms. We first prove that the ψ_1 -norm of each inner product $\mathbf{a}_i'v$ is finite:

$$\|\mathbf{a}_i'v\|_{\psi_1} \leq \sum_{j=1}^k \|a_{ij}v_j\|_{\psi_1} \leq \sum_{j=1}^k \|a_{ij}\|_{\psi_2} \|v_j\|_{\psi_2} < \infty,$$

where we used that $\|\cdot\|_{\psi_1}$ is a norm at the first step, and then at the second step we used Lemma 2.7.7 of [26], which can be seen as an analogue of Cauchy–Schwarz inequality. Given that each $\mathbf{a}_i'v$ has finite ψ_1 -norm, then there exist positive constants $c_1, \dots, c_d$ such that for $i = 1, 2, \dots, d$,

$$\mathbb{P}(|\mathbf{a}_i'v| > T) \leq 2 \exp(-c_i T), \quad \text{for every } T > 0. \quad (20)$$

Consequently,

$$\begin{aligned} \mathbb{P}\left(\|\mathbf{A}v\| > \frac{\omega}{2}\right) &= \mathbb{P}\left(\|\mathbf{A}v\|^2 > \frac{\omega^2}{4}\right) \\ &= \mathbb{P}\left(\sum_{i=1}^d |\mathbf{a}_i'v|^2 > \frac{\omega^2}{4}\right) \end{aligned}$$

$$\begin{aligned}
&\leq \mathbb{P}\left(\exists i \in \{1, 2, \dots, d\} : |\mathbf{a}'_i v| > \frac{\omega}{2\sqrt{d}}\right) \\
&\leq \sum_{i=1}^d \mathbb{P}\left(|\mathbf{a}'_i v| > \frac{\omega}{2\sqrt{d}}\right) \quad (\text{union bound}) \\
&\leq \sum_{i=1}^d 2 \exp\left(-\frac{\omega c_i}{2\sqrt{d}}\right) \quad \left(\text{applying (20) with } T = \frac{\omega}{2\sqrt{d}}\right) \\
&\leq 2d \exp\left(-\frac{\omega \min\{c_1, \dots, c_d\}}{2\sqrt{d}}\right) \\
&= 2d \exp(-L_{d,k}\omega),
\end{aligned}$$

where $L_{d,k} := \min\{c_1, \dots, c_d\}/(2\sqrt{d})$, a constant which depends only on k and d (given that each c_i depends on k). $\square$

Lemma 10. Suppose Assumptions 2 and 3 hold. Recall the definition of F_ω in (11). For ω large enough there exists a constant $C_1 > 0$ such that

$$F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \leq C_1 \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2} \omega^{n(2\alpha+d+1)},$$

the constant C_1 being independent of ω and τ .

Proof. Recall (12):

$$F_\omega(\mathbf{W}, \mathbf{Z}, \tau) = \frac{\pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2}}{m(\mathbf{X}_{O^c})} \left[\prod_{i \in O^c} f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i)) \right] \prod_{i \in O} \frac{f_\varepsilon(\sqrt{\tau}(x_i - \mathbf{W}z_i))}{f_\varepsilon(x_i)}.$$

Furthermore, if $i \in O$, then for ω large enough, from (16), (18) and $\|x_i\| \leq 2b_i\omega$, we have that

$$\frac{1}{f_\varepsilon(x_i)} = \frac{1}{C_\varepsilon f(\|x_i\|)} \leq \frac{1}{2C_\varepsilon} \|x_i\|^{\alpha+d} (\log \|x_i\|)^{\alpha+1} \leq \frac{1}{2C_\varepsilon} \|x_i\|^{2\alpha+d+1} \leq C_3 \omega^{2\alpha+d+1},$$

for some sufficiently large positive constant $C_3 > 0$. Proposition 3 ensures that $m(\mathbf{X}_{O^c})$ is finite and recall that $f_\varepsilon \leq CC_\varepsilon$.

Combining everything, we obtain

$$F_\omega(\mathbf{W}, \mathbf{Z}, \tau) \leq \frac{(CC_\varepsilon)^n C_3^{|O|}}{m(\mathbf{X}_{O^c})} \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2} \omega^{(2\alpha+d+1)|O|} \leq C_1 \pi(\mathbf{W}, \mathbf{Z}, \tau) \tau^{nd/2} \omega^{n(2\alpha+d+1)},$$

for some sufficiently large positive constant C_1 . $\square$

2.6.4. The matrix normal distribution and its properties

We follow the exposition in [14]. Let $\mathbf{U} \in \mathbb{R}^{d \times n}$ and let $\mu \in \mathbb{R}^{d \times n}$. We say that $\mathbf{U}$ has a matrix normal distribution with mean parameter μ and covariance structure $\Sigma^{(c)} \otimes \Sigma^{(r)}$, where

$\Sigma^{(c)} \in \mathbb{R}^{n \times n}$ and $\Sigma^{(r)} \in \mathbb{R}^{d \times d}$ are symmetric positive-definite matrices, if its density is given by

$$f(\mathbf{U}; \mu, \Sigma^{(c)} \otimes \Sigma^{(r)}) = (2\pi)^{-dn/2} |\Sigma^{(c)}|^{-d/2} |\Sigma^{(r)}|^{-n/2} \times \exp \left(-\frac{1}{2} \text{tr} \left((\Sigma^{(r)})^{-1} (\mathbf{U} - \mu) (\Sigma^{(c)})^{-1} (\mathbf{U} - \mu)^\top \right) \right).$$

We use the notation

$$\mathbf{U} \sim \mathcal{N}(\mu, \Sigma^{(c)} \otimes \Sigma^{(r)}).$$

We now spend some words to clarify this notation. The vectorization operator vec transforms a matrix $\mathbf{U} \in \mathbb{R}^{d \times n}$ into the column vector obtained by stacking the columns of $\mathbf{U}$, that is,

$$\text{vec}(\mathbf{U}) = (u_{\cdot 1}^\top, \dots, u_{\cdot n}^\top)^\top,$$

where $u_{\cdot j}$ denotes the j th column of $\mathbf{U}$. Accordingly, the mean and covariance of $\mathbf{U}$ are defined as the mean and covariance of the random vector $\text{vec}(\mathbf{U})$, and the above notation is equivalent to

$$\text{vec}(\mathbf{U}) \sim \mathcal{N}_{dn}(\text{vec}(\mu), \Sigma^{(c)} \otimes \Sigma^{(r)}).$$

In the present work, the most common situation is the one where the rows of $\mathbf{U}$ are independent and identically distributed with common covariance matrix Σ . In that case, the covariance structure is $\Sigma \otimes I_d$, and this is precisely the form used in Appendix A.1 for the full conditionals of $\mathbf{W}$ and, in the normal model, of $\mathbf{Z}$.

A key result that we use repeatedly is the following proposition; see also [24].

Proposition 4. *Let $\mathbf{U}$ be a $d \times n$ matrix-valued random variable. If*

$$\mathbf{U} \sim \mathcal{N}(\mu, \Sigma \otimes I_d),$$

then

$$\mathbf{U}^\top \sim \mathcal{N}(\mu^\top, I_d \otimes \Sigma),$$

and the second non-central moments are given by

$$\mathbb{E}(\mathbf{U}^\top \Pi_1 \mathbf{U}) = \text{tr}(\Pi_1) \Sigma + \mu^\top \Pi_1 \mu,$$

$$\mathbb{E}(\mathbf{U} \Pi_2 \mathbf{U}^\top) = \text{tr}(\Pi_2 \Sigma) I_d + \mu \Pi_2 \mu^\top,$$

where Π_1 and Π_2 are $d \times d$ and $n \times n$ symmetric positive-definite matrices, respectively.

2.7. A mixing time experiment of the proposed Gibbs sampler

Let $\{X_t : t \geq 1\}$ be a π -ergodic Markov chain with transition kernel P , and suppose that a Central Limit Theorem (CLT) holds for a class of functions that includes a given function f which is square-integrable with respect to π . For simplicity, assume that $\mathbb{E}_\pi[f] = 0$ and that the

asymptotic variance of f is normalized to $\sigma_f^2 = 1$. Define

$$\delta_n = \sqrt{n} \sum_{t=1}^n f(X_t).$$

Then, under standard conditions, we have the convergence in distribution

$$\delta_n \Rightarrow \mathcal{N}(0,1).$$

We introduce a quantity τ_C , defined as the typical number of iterations required such that the null hypothesis $\delta_n \sim \mathcal{N}(0,1)$ cannot be rejected at a prescribed significance level. In other words, τ_C is the minimal time horizon at which the empirical behavior of the Markov chains has entered a CLT regime. Since τ_C is observable in practice via simulation, it provides a data-driven notion of convergence.

To understand τ_C , it is useful to relate it to two mixing time and integrated autocorrelation time. First, the *mixing time* τ_{mix} characterizes how quickly the distribution of X_t approaches the stationary distribution π , regardless of its initialization. Thus, τ_{mix} captures how long we must wait before the chain forgets its starting point and produces approximately unbiased samples.

Second, the *integrated autocorrelation time* τ_{int} quantifies the degree of dependence between successive samples. Even after the chain has mixed, consecutive draws are generally correlated. The quantity τ_{int} can be interpreted as the effective number of iterations required to obtain one approximately independent sample. In this sense, a thinned subsequence of the form $\{X_{\tau_{\text{mix}} + \ell \tau_{\text{int}}} : \ell \in \mathbb{N}\}$ behaves approximately like an i.i.d. sample from π .

To isolate the intrinsic difficulty of the estimation problem, consider another independent and identically distributed sequence $\{Y_t : t \geq 1\}$ with $Y_t \sim \pi$. Define τ^* as the analogue of τ_C for $\{Y_t\}$. The quantity τ^* depends only on the function f and the distribution π , and thus represents the fundamental statistical difficulty of estimating $\mathbb{E}_\pi[f]$.

By contrast, τ_C incorporates both the intrinsic difficulty of the problem and the additional cost induced by dependence in the Markov chain. Heuristically, one can decompose

$$\tau_C \approx \tau_{\text{mix}} + \tau^* \tau_{\text{int}}.$$

We interpret this as: one must first wait approximately τ_{mix} steps for the chain to reach stationarity, and then accumulate roughly τ^* approximately independent samples, each of which requires about τ_{int} iterations due to dependence. We design a simple numerical experiment to estimate τ_C .

2.7.1. Numerical simulation and the results

Fix a single contaminated dataset $\mathbf{X} \in \mathbb{R}^{n \times d}$ with known ground truth $\mathbf{X}_0$. In our experiments, $\mathbf{X}$ is generated from the row-wise contamination model in standardized errors as in [section 2.4](#).

For either the Gaussian or Student's t BPCA Gibbs sampler, iteration t produces the low-rank reconstruction

$$\widehat{\mathbf{X}}^{(t)} = \mathbf{Z}^{(t)}(\mathbf{W}^{(t)})^\top, \quad t = 1, 2, \dots, T.$$

We consider the scalar functional

$$u_t = f(\widehat{\mathbf{X}}^{(t)}), \quad \text{where} \quad f(\mathbf{Y}) = \|\mathbf{Y} - \mathbf{X}_0\|_F.$$

Under standard Markov chain CLT conditions, one expects that, for T sufficiently large,

$$\bar{u}_T := \frac{1}{T} \sum_{t=1}^T u_t \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{T}\right),$$

where μ is the posterior expectation of u_t and σ^2 is the Markov chain asymptotic variance.

To investigate at which finite horizon this Gaussian approximation becomes reasonable, we run $M = 100$ independent chains indexed by $j = 1, \dots, M$. For a fixed T , define the empirical mean from chain j by

$$\bar{u}_T^{(j)} = \frac{1}{T} \sum_{t=1}^T u_t^{(j)}.$$

We then examine the empirical distribution of

$$\{\bar{u}_T^{(j)}\}_{j=1}^M$$

across chains as T varies.

Let us consider the grid

$$\mathcal{T} = \{50, 100, 200, 400, 800, \dots, 25600, 51200, 102400, 204800, 409600\}.$$

For each model (Normal or Student's t), we first run the M chains up to 409600 iterations and discard the first 204800 draws. Using the retained draws, we compute estimators $\hat{\mu}$ and $\hat{\sigma}$ of the centering and scaling constants appearing in the CLT. We then form the standardized quantities

$$z_T^{(j)} = \sqrt{T} \frac{\bar{u}_T^{(j)} - \hat{\mu}}{\hat{\sigma}}, \quad j = 1, \dots, M.$$

The rationale is that *if* the CLT approximation is accurate at horizon T , then the collection

$$\{z_T^{(j)}\}_{j=1}^M$$

should be approximately distributed as a standard Gaussian sample.

To test normality, we use Shapiro–Wilk criterion. For each $T \in \mathcal{T}$, we apply the Shapiro–Wilk test to the sample

$$\{z_T^{(j)}\}_{j=1}^M.$$

Let $W_{n,m}(T)$ denote the corresponding Shapiro–Wilk statistic, where n is the sample size of the dataset and $m \in \mathcal{M} := \{\text{Gibbs Normal}, \text{Gibbs Student}\}$ indicates the Gaussian or Student’s t Gibbs sampler.

We define the onset of the Gaussian regime by

$$\tau_C(n,m) := \min\left\{T \in \mathcal{T} : W_{n,m}(T') \geq 0.98 \text{ for all } T' \in \mathcal{T} \text{ with } T' \geq T\right\}.$$

In words, $\tau_C(n,m)$ is the smallest horizon after which the Shapiro–Wilk statistic remains uniformly above the threshold 0.98 on the remainder of the grid.

The value 0.98 was chosen using a simple Monte Carlo calibration under the Gaussian null. More precisely, we generated 100 independent samples of size 100 from a standard normal distribution and computed the Shapiro–Wilk statistic W for each sample. In this experiment, the empirical 1% lower critical value was approximately 0.975, while the average W -value was approximately 0.98. We therefore adopt the cutoff $W \geq 0.98$ as a convenient and slightly conservative practical criterion for declaring that the across-chain standardized averages are close enough to Normality.

Table 5. Estimated CLT onset based on the Shapiro–Wilk W -criterion.

n	$\tau_C(n, \text{Gibbs Normal})$	$\tau_C(n, \text{Gibbs Student})$	Time (approx. in hours)
50	50	800	0.2
500	200	1600	1.5
5000	6400	204800	12

Table 5 reports the first horizon $\tau_C(n,m)$ after which the Shapiro–Wilk statistic remains above 0.98 for all larger values of T in the grid. This provides a simple empirical summary of how quickly the finite-time averages $\bar{u}_T$ enter a regime that is consistent with the Gaussian approximation predicted by the Markov chain CLT.

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Chapter 3

On the convergence of CAVI for Bayesian PCA

This chapter is based on joint work with my supervisors Philippe Gagnon and Florian Maire. This work is currently under resubmission for Statistics & Probability Letters.

Résumé

L'inférence variationnelle de type ascension coordonnée (IVAC) est un outil standard pour l'inférence bayésienne approximative à grande échelle; toutefois, ses propriétés théoriques demeurent encore mal comprises. Récemment, une ligne de recherche visant à caractériser ces propriétés théoriques dans des cas spécifiques a émergé. Nous contribuons à cette littérature en proposant une analyse théorique de l'IVAC pour l'analyse en composantes principales bayésienne (ACPB), un outil probabiliste populaire de réduction de dimension. Cette analyse est délicate, car le modèle est complexe, de grande dimension et non identifiable, avec une densité a posteriori multimodale et non-log-concave.

Abstract

Coordinate ascent variational inference (CAVI) is a standard tool for scalable approximate Bayesian inference, yet its theoretical properties are not well understood. Recently, a line of research characterizing its theoretical properties in specific cases has emerged. We contribute to this line of research by providing a theoretical analysis of CAVI for Bayesian principal component analysis, a popular probabilistic dimension reduction tool. The analysis is challenging as the model is complex, high-dimensional and non-identifiable, with a multimodal and non-log-concave posterior density.

Keywords: Coordinate ascent variational inference, Information-theoretic lower bounds, Mean-field variational inference, Power iteration, Principal component analysis

3.1. Introduction

Variational inference (VI) has become a central tool for approximate Bayesian inference in high-dimensional models, where sampling from the posterior distribution is often intractable. Given a posterior distribution π , VI approximates π by a tractable distribution q^* belonging to a prescribed family $\mathcal{Q}$ by solving an optimization problem:

$$q^* = \arg \min_{q \in \mathcal{Q}} \text{KL}(q \parallel \pi), \quad \text{KL}(q \parallel \pi) = \int q(\mathrm{d}x) \log \left(\frac{q(x)}{\pi(x)} \right), \quad (21)$$

where it is assumed to simplify that all $q \in \mathcal{Q}$ are absolutely continuous with respect to π and that π and all $q \in \mathcal{Q}$ admit densities with respect to Lebesgue measure.

The optimization problem in (21) is commonly solved via coordinate ascent variational inference (CAVI) when the Bayesian model exhibits a conjugate structure. In this situation, each CAVI update admits a closed-form expression and monotonically decreases the Kullback–Leibler (KL) divergence. Due to its simplicity and scalability, CAVI is widely used in practice [5, 6]. Despite its popularity, its theoretical convergence properties are not fully characterized.

As a first step towards complete characterization, the convergence properties of CAVI have recently been studied for specific statistical models, such as stochastic block models [22], Ising models [16] and generalized linear mixed models [10]. General results have also been recently introduced under a log-concave framework [13, 1] and under an abstract framework [2]. In this paper, we contribute to this growing literature by studying the convergence properties of CAVI for Bayesian principal component analysis (BPCA), a high-dimensional and complex model. The model complexity is due to, among others, its non-identifiability, with a posterior density which is multimodal and not log-concave.

BPCA is the Bayesian counterpart of probabilistic PCA [19, 3] which is a popular dimension reduction tool that gives access to uncertainty quantification. Despite the widespread use of PPCA and BPCA [12, 14, 21, 18], theoretical studies remain limited. Recent progress includes consistency of the maximum likelihood estimator in PPCA [8] and, on the Bayesian side, a selection procedure for the number of principal components (PCs) has been proposed and shown to be consistent in [7].

Given a data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with standardized columns and $n \geq d$, it is assumed that its lines $x_1, \dots, x_n \in \mathbb{R}^d$ are realizations of random variables $X_1, \dots, X_n$ modelled as:

$$X_i = \mathbf{W}Z_i + \tau^{-1/2}\epsilon_i, \quad i = 1, \dots, n, \quad (22)$$

where $\mathbf{W} \in \mathbb{R}^{d \times k}$ is an unknown loading matrix, $Z_i \in \mathbb{R}^k$ a latent vector, $\tau > 0$ a precision parameter, and $\epsilon_i \in \mathbb{R}^d$ a standardized normal error vector. The columns of the matrix $\mathbf{Z} \in \mathbb{R}^{n \times k}$, with lines given by Z_i , play an analogous role as the PCs in PCA. As in [3], Z_i is assigned a standard normal prior distribution and $\mathbf{W}$ is assigned a matrix normal prior distribution

$\mathcal{N}(\mathbf{0}, I_d \otimes \Lambda^{-1})$, where $\mathbf{0}$ is a null $d \times k$ matrix and Λ is a $k \times k$ diagonal matrix, implying that the rows of $\mathbf{W}$ are independent with shared covariance Λ^{-1} (see [11] for details about the matrix normal distribution). The prior distribution on τ is a gamma.

As in [4], $\mathcal{Q}$ is chosen to be the a mean-field family such that: for all $q \in \mathcal{Q}$, $q = q_{\mathbf{W}} q_{\mathbf{Z}} q_{\tau}$, with $q_{\mathbf{W}} = \mathcal{N}(\mu_{\mathbf{W}}, I_d \otimes \Sigma_{\mathbf{W}})$, $q_{\mathbf{Z}} = \mathcal{N}(\mu_{\mathbf{Z}}, I_n \otimes \Sigma_{\mathbf{Z}})$ and $q_{\tau} = \text{gamma}(\alpha_{\tau}, \beta_{\tau})$, where $\mu_{\mathbf{W}}$ and $\mu_{\mathbf{Z}}$ are matrices of size $n \times k$ and $d \times k$, respectively, $\Sigma_{\mathbf{W}}$ and $\Sigma_{\mathbf{Z}}$ belong to $\mathbb{S}_+^k$, the set of symmetric positive-definite matrices of size $k \times k$, and $\alpha_{\tau} > 0$ and $\beta_{\tau} > 0$ are shape and scale parameters, respectively.

To solve (21), CAVI is applied, where at each step a particular marginal distribution, say $q_{\mathbf{W}}$ (referred to as a *block*), is updated, while the other marginal distributions, in this case $q_{\mathbf{Z}}$ and q_{τ} , are held fixed. For instance, at iteration $t + 1$, we have that $q_{\mathbf{W}}^{(t+1)}$ corresponds to a matrix normal distribution with parameters $\mu_{\mathbf{W}}^{(t+1)}$ and $I_n \otimes \Sigma_{\mathbf{W}}^{(t+1)}$ given by

$$\Sigma_{\mathbf{W}}^{(t+1)} = (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} (n \Sigma_{\mathbf{Z}}^{(t)} + (\mu_{\mathbf{Z}}^{(t)})' \mu_{\mathbf{Z}}^{(t)}) + \Lambda)^{-1} \text{ and}$$

$$\mu_{\mathbf{W}}^{(t+1)} = (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} \mathbf{X}' \mu_{\mathbf{Z}}^{(t)}) \Sigma_{\mathbf{W}}^{(t+1)}.$$

Updates of $q_{\mathbf{Z}}$ and q_{τ} also have closed-form expressions (see subsection 3.4.1 for full derivations and the CAVI algorithm). We begin CAVI with an update of $q_{\mathbf{W}}$ and, therefore, only values for $\mu_{\mathbf{Z}}$, $\Sigma_{\mathbf{Z}}$, α_{τ} and β_{τ} are required for initialization. Note that when τ is assumed to be known, the optimization problem in (21) admits a global analytic solution [15].

Contributions. In section 3.2, for the case $k = 1$ (meaning that a single PC is used), we characterize the convergence of CAVI by using dynamical systems tools and by establishing a novel connection with the classical *power iteration* algorithm [20]. For $k \geq 2$, this dynamical systems approach appears unfeasible. In section 3.3, we explore the application of the recent abstract tools in [2] which are applicable in the non-log-concave case. In that paper, it is mentioned that studying such an application for complex statistical models is of interest. We show that it may be possible to obtain a theoretical result, but that the assumptions may be difficult or impossible to verify in practice. There is thus room for theoretical results covering the non-log-concave case. The proofs of all theoretical results are deferred to subsection 3.4.3.

3.2. Theoretical analysis of CAVI for $k = 1$

As mentioned, the case $k = 1$ reduces the model to one with a single PC. In this case, for all $q \in \mathcal{Q}$, we have that $\mu_{\mathbf{W}} \in \mathbb{R}^d$ and $\mu_{\mathbf{Z}} \in \mathbb{R}^n$ are vectors, and $\Sigma_{\mathbf{W}}$ and $\Sigma_{\mathbf{Z}}$ are positive real numbers. The CAVI updates thus simplify substantially: each step is governed by one of the three maps F, G or H that maps the current parameter values to the next ones. These maps are explicit and derived directly from the CAVI algorithm; for completeness, all expressions are

provided in [subsection 3.4.2](#). CAVI therefore corresponds to the following update equations:

$$\begin{aligned} \left(\mu_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)} \right) &= F \left(\mu_{\mathbf{Z}}^{(t)}, \Sigma_{\mathbf{Z}}^{(t)}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)} \right), \\ \left(\mu_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)}, \mu_{\mathbf{Z}}^{(t+1)}, \Sigma_{\mathbf{Z}}^{(t+1)} \right) &= G \left(\mu_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)} \right), \\ \left(\mu_{\mathbf{Z}}^{(t+1)}, \Sigma_{\mathbf{Z}}^{(t+1)}, \alpha_{\tau}^{(t+1)}, \beta_{\tau}^{(t+1)} \right) &= H \left(\mu_{\mathbf{W}}^{(t+1)}, \Sigma_{\mathbf{W}}^{(t+1)}, \mu_{\mathbf{Z}}^{(t+1)}, \Sigma_{\mathbf{Z}}^{(t+1)} \right), \quad t = 0, 1, \dots \end{aligned} \quad (23)$$

More precisely, a full CAVI iteration corresponds to the composition $H \circ G \circ F$. To analyze the convergence of CAVI, we leverage the representations above and the following *polar decomposition* of $\mu_{\mathbf{Z}}^{(t)}$ and $\mu_{\mathbf{W}}^{(t)}$:

$$\mu_{\mathbf{Z}}^{(t)} = \underbrace{\|\mu_{\mathbf{Z}}^{(t)}\|}_{\text{scaling}} \underbrace{\left(\frac{\mu_{\mathbf{Z}}^{(t)}}{\|\mu_{\mathbf{Z}}^{(t)}\|} \right)}_{\text{direction}} \quad \text{and} \quad \mu_{\mathbf{W}}^{(t)} = \underbrace{\|\mu_{\mathbf{W}}^{(t)}\|}_{\text{scaling}} \underbrace{\left(\frac{\mu_{\mathbf{W}}^{(t)}}{\|\mu_{\mathbf{W}}^{(t)}\|} \right)}_{\text{direction}},$$

where $\|\cdot\|$ is the Euclidean norm; when applied to a matrix, $\|\cdot\|$ represents the Frobenius norm. In the following two subsections, the convergence question is addressed into two parts: first, for the directional component, and next, for the scaling component, together with that of $\Sigma_{\mathbf{Z}}^{(t)}$, $\Sigma_{\mathbf{W}}^{(t)}$, $\alpha_{\tau}^{(t)}$ and $\beta_{\tau}^{(t)}$.

3.2.1. Exponential convergence of the directional component

Let us assume that the data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ has full rank. Let $\lambda_1 > \dots > \lambda_d > 0 = \lambda_{d+1} = \dots = \lambda_n$ denote the eigenvalues of $\mathbf{X}\mathbf{X}'$ with corresponding orthonormal eigenvectors $\mu_1, \dots, \mu_n$. Finally, let $\text{sgn}(x)$ denote the sign of a non-zero real number x . We now state the theoretical result describing the behaviour of the directional components.

Theorem 11. *Assume that the data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ has full rank. Given an initialization $(\mu_{\mathbf{Z}}^{(0)}, \Sigma_{\mathbf{Z}}^{(0)}, \alpha^{(0)}, \beta^{(0)}) \in \mathbb{R}^n \setminus \{0\} \times \mathbb{R}_+^3$, let $c_j = \mu_j' \mu_{\mathbf{Z}}^{(0)}$, $j = 1, \dots, n$, and $i = \min\{1 \leq j \leq n : c_j \neq 0\}$. If $i < d$, then there exist constants $c_0, c'_0 > 0$ such that for all $t > 1$,*

$$\left\| \frac{\mu_{\mathbf{Z}}^{(t)}}{\|\mu_{\mathbf{Z}}^{(t)}\|} - \text{sgn}(c_i) \mu_i \right\| \leq c_0 \left(1 - \frac{\rho_i}{\lambda_i} \right)^t, \quad \left\| \frac{\mu_{\mathbf{W}}^{(t)}}{\|\mu_{\mathbf{W}}^{(t)}\|} - \text{sgn}(c_i) \frac{\mathbf{X}' \mu_i}{\|\mathbf{X}' \mu_i\|} \right\| \leq c'_0 \left(1 - \frac{\rho_i}{\lambda_i} \right)^t,$$

where $\rho_i = \lambda_i - \lambda_{i+1}$.

[Theorem 11](#) states that, if $c_1 \neq 0$ (which holds with probability 1 under initialization with continuous random variables), then $\mu_{\mathbf{Z}}^{(t)} / \|\mu_{\mathbf{Z}}^{(t)}\|$ converges exponentially fast to the first PC. Also, $\mu_{\mathbf{W}}^{(t)} / \|\mu_{\mathbf{W}}^{(t)}\|$ converges exponentially to the first principal direction. In other words, VI of BPCA recovers classical PCA as point estimates. It is natural to obtain such a result, as in the case $k = 1$, the orthogonality underlying PCA is automatically obtained in BPCA. When $k > 1$,

unfortunately, this is not the case, and the point estimates obtained via VI of BPCA do not respect orthogonality (as observed in numerical experiments¹ in [subsection 3.4.4](#)).

In the proof of [Theorem 11](#), we use that the iterates $\mu_{\mathbf{Z}}^{(t)} / \|\mu_{\mathbf{Z}}^{(t)}\|$ can be written as

$$\mu_{\mathbf{Z}}^{(t)} / \|\mu_{\mathbf{Z}}^{(t)}\| = (\mathbf{X}\mathbf{X}')^t \mu_{\mathbf{Z}}^{(0)} / \|(\mathbf{X}\mathbf{X}')^t \mu_{\mathbf{Z}}^{(0)}\|. \quad (24)$$

We have an analogous expression for the iterates $\mu_{\mathbf{W}}^{(t)} / \|\mu_{\mathbf{W}}^{(t)}\|$. The expression in (24) corresponds to *power iteration* that can be used, in this case, for computing the leading normalized singular vectors of $\mathbf{X}\mathbf{X}'$. The expression for the iterates $\mu_{\mathbf{W}}^{(t)} / \|\mu_{\mathbf{W}}^{(t)}\|$ also corresponds to power iteration, but for computing the leading normalized singular vectors of $\mathbf{X}'\mathbf{X}$.

3.2.2. On the convergence of the scaling component and $(\Sigma_{\mathbf{Z}}^{(t)}, \Sigma_{\mathbf{W}}^{(t)}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)})$

We begin by analyzing the sequence $(\|\mu_{\mathbf{Z}}^{(t)}\|, \Sigma_{\mathbf{Z}}^{(t)}, \xi^{(t)})$, where $\xi^{(t)} = \alpha_{\tau}^{(t)} \beta_{\tau}^{(t)}$ (the analysis will have implications for the sequence $(\|\mu_{\mathbf{W}}^{(t)}\|, \Sigma_{\mathbf{W}}^{(t)})$). Note that in the CAVI updates, $\alpha_{\tau}^{(t)} = \alpha_{\tau}^{(0)} + (nd)/2$ and, therefore, $\alpha_{\tau}^{(t)}$ is fixed throughout and $\xi^{(t)}$ is a scaled version of $\beta_{\tau}^{(t)}$ which varies with t . In the rest of the section, we use the definitions $a^{(t)} := \|\mu_{\mathbf{Z}}^{(t)}\|$ and $b^{(t)} := \Sigma_{\mathbf{Z}}^{(t)}$ to simplify the notation. Given [Theorem 11](#), one can exploit that the normalized version of $\mu_{\mathbf{Z}}^{(t)}$ converges to μ_1 (assuming $c_1 \neq 0$) and initialize CAVI with $\mu_{\mathbf{Z}}^{(0)} = a^{(0)} \mu_1$ for some $a^{(0)} > 0$.

We now investigate whether the sequence $(a^{(t)}, b^{(t)}, \xi^{(t)})$ admits a limit. With the initialization $(a^{(0)} \mu_1, b^{(0)}, \xi^{(0)})$, we are in fact guaranteed that $c_1 \neq 0$ in [Theorem 11](#) and $\mu_{\mathbf{Z}}^{(t)} = \text{sgn}(c_1) \mu_1 \|\mu_{\mathbf{Z}}^{(t)}\|$ and $\mu_{\mathbf{W}}^{(t)} = \text{sgn}(c_1) \mathbf{X}' \mu_1 \|\mu_{\mathbf{W}}^{(t)}\|$ with only $\|\mu_{\mathbf{Z}}^{(t)}\|$ and $\|\mu_{\mathbf{W}}^{(t)}\|$ varying in $\mu_{\mathbf{Z}}^{(t)}$ and $\mu_{\mathbf{W}}^{(t)}$, respectively. Additionally, CAVI updates of the blocks $q_{\mathbf{Z}}$ and q_{τ} are governed by an explicit mapping $\Phi : \mathbb{R}_+^3 \rightarrow \mathbb{R}_+^3$ (see [subsubsection 3.4.2.1](#) for the expression) such that

$$(a^{(t+1)}, b^{(t+1)}, \xi^{(t+1)}) = \Phi(a^{(t)}, b^{(t)}, \xi^{(t)}), \quad t = 0, 1, \dots$$

If the sequence $(a^{(t)}, b^{(t)}, \xi^{(t)})$ converges to a limit (a^*, b^*, ξ^*) , then this limit must satisfy the fixed point condition $\Phi(a^*, b^*, \xi^*) = (a^*, b^*, \xi^*)$ because the map Φ is continuous. Consequently, characterizing all such fixed points provides insights into the possible limit points of the CAVI iterates. The next proposition is about this. Note that characterizing the fixed points of Φ implies the characterization of the limit of the sequence $(\|\mu_{\mathbf{W}}^{(t)}\|, \Sigma_{\mathbf{W}}^{(t)})$ through the mapping F given at the beginning of [section 3.2](#).

Proposition 5. *Assume that (a^*, b^*, ξ^*) is a fixed point of Φ . Then, $((a^*)^2, \xi^*)$ must be a common root of the polynomials P and Q (see the proof in [subsection 3.4.3](#) for their explicit expressions), which have at most 28 common roots. Moreover, assuming $((a^*)^2, \xi^*)$ is such a root, then $b^* = (\Lambda + \xi^*(a^*)^2) / ((\xi^*)^2 \lambda_1 - n\xi^*)$.*

¹The code to reproduce our numerical results is available online (see <https://github.com/curl1969/bpca-experiments>).

A user can exploit [Proposition 5](#) to conclude whether $(a^{(t)}, b^{(t)}, \xi^{(t)})$, and thus CAVI, converge. CAVI iterates may indeed not converge [\[1\]](#). The user can numerically solve the system of polynomial equations $P(t, w) = Q(t, w) = 0$. If there is no common root $(t, w) \in \mathbb{R}_+^2$ or $\lambda_1 < n/\xi^*$, then CAVI does not converge. Also, [Proposition 5](#) can be used to evaluate whether the convergence criterion in the CAVI algorithm should be refined (see [Algorithm 4](#) in [subsection 3.4.1](#)). Indeed, let us say that one stopped CAVI based on a given convergence criterion and obtain what would ideally be similar to (a^*, b^*, ξ^*) . This user can find the common roots and evaluate if what was obtained is close to one of these. If this is not the case, it may be because the convergence criterion should be refined (the other possibility is that CAVI does not actually converge, which is possible even if a common root is found).

3.3. Theoretical analysis of CAVI in the general case

In this section, we explore the application of the recent abstract tools in [\[2\]](#) for a convergence analysis of CAVI for BPCA in the general case, meaning with $k \geq 1$. In that paper, the focus is on the case where CAVI has two blocks, say $q_{\mathbf{Z}}$ and $q_{\mathbf{W}}$. Their tools can be extended to the case where CAVI has more blocks but the assumptions become more restrictive. In this section, we show that, even in the case with two blocks, the assumptions may be difficult or impossible to verify in practice, offering little hope for cases with more blocks. We thus treat τ as known and use the following mean-field family: for all $q \in \mathcal{Q}$, $q = q_{\mathbf{W}} q_{\mathbf{Z}}$, with $q_{\mathbf{W}} = \mathcal{N}(\mu_{\mathbf{W}}, I_d \otimes \Sigma_{\mathbf{W}})$ and $q_{\mathbf{Z}} = \mathcal{N}(\mu_{\mathbf{Z}}, I_n \otimes \Sigma_{\mathbf{Z}})$. CAVI in this case thus essentially corresponds to [Algorithm 4](#), but without the update for q_{τ} (see [subsection 3.4.1](#) for the details).

We start by presenting the abstract tools in [\[2\]](#), taking the form of a general theoretical result guaranteeing the exponential convergence of CAVI in KL divergence, and next we explore the application of this theoretical result in the special case of BPCA. Let $\mathcal{M}$ denote the set of stationary points of $\text{KL}(q \parallel \pi) = \text{KL}(q_{\mathbf{W}} q_{\mathbf{Z}} \parallel \pi)$. Theorem 5 in [\[15\]](#) indicates that $\mathcal{M} \neq \emptyset$. Let $q^* \in \mathcal{M}$ be a stationary point. For any $(q_{\mathbf{W}}, q_{\mathbf{Z}})$, Lemma 3.1 in [\[2\]](#) states that

$$\text{KL}(q_{\mathbf{W}} q_{\mathbf{Z}} \parallel \pi) - \text{KL}(q_{\mathbf{W}}^* q_{\mathbf{Z}}^* \parallel \pi) = \text{KL}(q_{\mathbf{W}} \parallel q_{\mathbf{W}}^*) + \text{KL}(q_{\mathbf{Z}} \parallel q_{\mathbf{Z}}^*) - \Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}}), \quad (25)$$

where

$$\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}}) = \iint (q_{\mathbf{W}}(\mathbf{W}) - q_{\mathbf{W}}^*(\mathbf{W})) (q_{\mathbf{Z}}(\mathbf{Z}) - q_{\mathbf{Z}}^*(\mathbf{Z})) \log \pi(\mathbf{W}, \mathbf{Z}) \, d\mathbf{W} \, d\mathbf{Z}. \quad (26)$$

The mathematical object in (26) was first introduced in [\[2\]](#). It controls the CAVI dynamics: it acts as a measure of “covariance” between the blocks $q_{\mathbf{W}}$ and $q_{\mathbf{Z}}$, and vanishes when the posterior density π factorizes. In this case, the optimization problem in (21) translates into two subproblems, each of which are solved in one CAVI step, where $q_{\mathbf{W}}^*$ and $q_{\mathbf{Z}}^*$ minimize the KL divergence with the marginal distributions in π .

We thus expect a fast convergence of CAVI when $\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})$ is small, relatively to $\text{KL}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*)$ and $\text{KL}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*)$ (recall (25)), along the CAVI trajectory. In [2], a sufficient condition, related to what is referred to as the *1/2-generalized correlation*, denoted by $\text{GCorr}_{1/2}$, is proposed:

$$\text{GCorr}_{1/2}(r_0) = \sup_{\substack{q_{\mathbf{W}} \in \mathcal{B}_{\mathbf{W}}^*(r_0) \setminus \{q_{\mathbf{W}}^*\}, \\ q_{\mathbf{Z}} \in \mathcal{B}_{\mathbf{Z}}^*(r_0) \setminus \{q_{\mathbf{Z}}^*\}}} \frac{|\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})|}{2\sqrt{D_{\text{KL},1/2}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*) D_{\text{KL},1/2}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*)}} \in (0,1), \quad (27)$$

where $\mathcal{B}_{\mathbf{Z}}^*(r_0) = \{q_{\mathbf{Z}} : \text{KL}(q_{\mathbf{Z}}^* \| q_{\mathbf{Z}}) \leq r_0\}$ is a KL ball of radius r_0 around $q_{\mathbf{Z}}^*$, with an analogous definition for $\mathcal{B}_{\mathbf{W}}^*(r_0)$, and $D_{\text{KL},1/2}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*) = (\text{KL}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*) + \text{KL}(q_{\mathbf{Z}}^* \| q_{\mathbf{Z}}))/2$ is the symmetric KL divergence between $q_{\mathbf{Z}}$ and $q_{\mathbf{Z}}^*$, with an analogous definition for $D_{\text{KL},1/2}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*)$. In Theorem 3.6 of [2], it is assumed that CAVI is initialized with $q_{\mathbf{Z}}^{(0)}$ within a symmetric KL ball, that is $q_{\mathbf{Z}}^{(0)} \in \{q_{\mathbf{Z}} : D_{\text{KL},1/2}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*) \leq r_0/2\}$, with $r_0 > 0$ such that $\text{GCorr}_{1/2}(r_0) \in (0,1)$, thus effectively ensuring that the generalized correlation is small on the whole CAVI trajectory and exponential convergence.

The denominator in $\text{GCorr}_{1/2}$ is there to make sure that $\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})$ is small relatively to the symmetric KL divergences $D_{\text{KL},1/2}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*)$ and $D_{\text{KL},1/2}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*)$, which is a stronger requirement than being small relatively to $\text{KL}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*)$ and $\text{KL}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*)$ (recall (25)). In our context of BPCA, by plugging the expression of $\log \pi(\mathbf{W}, \mathbf{Z})$ in (26) and after some simplifications, the condition $\text{GCorr}_{1/2}(r_0) \in (0,1)$ corresponds to

$$\sup_{\substack{q_{\mathbf{W}} \in \mathcal{B}_{\mathbf{W}}^*(r_0) \setminus \{q_{\mathbf{W}}^*\}, \\ q_{\mathbf{Z}} \in \mathcal{B}_{\mathbf{Z}}^*(r_0) \setminus \{q_{\mathbf{Z}}^*\}}} \frac{\left| \tau_0 \iint (q_{\mathbf{W}}(\mathbf{W}) - q_{\mathbf{W}}^*(\mathbf{W})) (q_{\mathbf{Z}}(\mathbf{Z}) - q_{\mathbf{Z}}^*(\mathbf{Z})) \|\mathbf{X} - \mathbf{Z}\mathbf{W}'\|^2 d\mathbf{W} d\mathbf{Z} \right|}{4\sqrt{D_{\text{KL},1/2}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*) D_{\text{KL},1/2}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*)}} \in (0,1). \quad (28)$$

The condition in (27), or equivalently (28) in our case, is not practical due to the presence of a supremum. As in the examples in [2], we thus derive a practical condition taking the form of an upper bound on $\text{GCorr}_{1/2}$. We explain how it is derived, highlighting its effectiveness (as assessed through a numerical experiment presented below) and providing a user guide for the application of the tools in [2]. The goal of the derivation of the upper bound is to get rid of $q_{\mathbf{W}}$ and $q_{\mathbf{Z}}$, which also allows to get rid of the supremum.

The first step uses the properties of the trace operator (see the proof of Lemma 12 in subsection 3.4.3.2) to rewrite the numerator in (28) as

$$\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}}) = \tau_0 \text{tr}((\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*)(\mu_{\mathbf{Z}}' - [\mu_{\mathbf{Z}}^*]') \mathbf{X}) - \frac{\tau_0}{2} \text{tr}((\Gamma_{\mathbf{W}} - \Gamma_{\mathbf{W}}^*)(\Gamma_{\mathbf{Z}}' - [\Gamma_{\mathbf{Z}}^*]')), \quad (29)$$

followed by the use of the sub-multiplicativity of the trace operator, yielding for instance $\|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\| \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\| \|\mathbf{X}\|$ for the first trace, where $\Gamma_{\mathbf{W}} = d\Sigma_{\mathbf{W}} + \mu_{\mathbf{W}}' \mu_{\mathbf{W}}$ and $\Gamma_{\mathbf{Z}} = n\Sigma_{\mathbf{Z}} + \mu_{\mathbf{Z}}' \mu_{\mathbf{Z}}$, $\Gamma_{\mathbf{W}}^*$ and $\Gamma_{\mathbf{Z}}^*$ being defined analogously.

The second step is a lower bound on the denominator in (28). For this, we developed a sharp lower bound on the symmetric KL between normal distributions (Lemma 13 in subsection 3.4.3.2). The result is of independent interest as traditional bounds based on Pinsker's inequality are often too loose to be useful in practice [9]. This bound yields a ratio of the form $(a_1 + a_2 + a_3 + a_4)/(b_1 + b_2 + b_3 + b_4)$ (see Lemma 21 in subsection 3.4.3.2), where a_i and b_i are all positive real numbers and with terms like $\|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\| \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|$ in the denominator.

The third step, inspired by the examples in [2], consists of a bound of the ratio $(a_1 + a_2 + a_3 + a_4)/(b_1 + b_2 + b_3 + b_4)$ by $\max a_i/b_i$. The form a_i/b_i yields useful cancellations of the aforementioned product terms like $\|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\| \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|$. After this step, we are left with terms of the form $(\|\mu_{\mathbf{Z}}\| + \|\mu_{\mathbf{Z}}^*\|)$ (coming from the second trace in (29)).

The final step is a triangle inequality which allows a bound of these remaining terms and to finally get rid of $q_{\mathbf{W}}$ and $q_{\mathbf{Z}}$. This inequality becomes an equality when the supremum in (28) is attained on the boundary (with other characteristics on the point q where the supremum is attained, see Proposition 6 in subsection 3.4.3.2). This final step appears in the examples in [2]. It allows for a bound which depends only on q^* , making the condition practical. The condition is as follows: the parameters of the tuple $(q_{\mathbf{Z}}^*, q_{\mathbf{W}}^*)$, meaning $(\mu_{\mathbf{Z}}^*, \Sigma_{\mathbf{Z}}^*, \mu_{\mathbf{W}}^*, \Sigma_{\mathbf{W}}^*) \in \mathbb{R}^{d \times k} \times \mathbb{S}_+^k \times \mathbb{R}^{n \times k} \times \mathbb{S}_+^k$, are such that

$$\max \left\{ \begin{aligned} &(\tau_0/\gamma_0) (\|\Sigma_{\mathbf{W}}^*\| \|\Sigma_{\mathbf{Z}}^*\|)^{1/2} \|\mathbf{X}\| + 2\|\mu_{\mathbf{W}}^*\| \|\mu_{\mathbf{Z}}^*\|, \\ &e\tau_0 (\|\Sigma_{\mathbf{W}}^*\| \|\Sigma_{\mathbf{Z}}^*\|) \sqrt{nd}, \\ &(\tau_0/\gamma_0^{1/2}) (\|\Sigma_{\mathbf{W}}^*\| \|\Sigma_{\mathbf{Z}}^*\|^{1/2}) \sqrt{2de} \|\mu_{\mathbf{Z}}^*\|, \\ &(\tau_0/\gamma_0^{1/2}) (\|\Sigma_{\mathbf{W}}^*\|^{1/2} \|\Sigma_{\mathbf{Z}}^*\|) \sqrt{2ne} \|\mu_{\mathbf{W}}^*\| \end{aligned} \right\} < 1. \quad (30)$$

where $\gamma_0 = (1 + e^{-1})/4$.

We are now ready to state the theoretical result derived from the general one in [2].

Theorem 12 (Local contraction of KL). *Assume that there exists at least one stationary point q^* that has parameters satisfying the condition in (30), and that for such a stationary point, the Hessian of $KL(q^* \|\pi)$ (with respect to the parameters) is non-singular. Then, there exists $r^* > 0$ such that, if the CAVI initialization satisfies $q_{\mathbf{Z}}^{(0)} \in \mathcal{B}_{\mathbf{Z}}^*(r^*)$, then for any $t \geq 1$,*

$$KL(q_{\mathbf{Z}}^{(t+1)} \| q_{\mathbf{Z}}^*) \leq C_{\mathbf{Z}} \kappa^t \text{ and } KL(q_{\mathbf{W}}^{(t+1)} \| q_{\mathbf{W}}^*) \leq C_{\mathbf{W}} \kappa^t,$$

where $C_{\mathbf{Z}}$ and $C_{\mathbf{W}}$ are two positive constants, independent of t , and $\kappa \in (0, 1)$ is a constant.

Empirically, we observed that the critical term in (30) is the first one, in the sense that the three others in (30) were observed to be smaller. The first term reflects the extent to which

$\tau_0 \|\mathbf{X} - \mathbf{Z}\mathbf{W}'\|$ is small over the integral in (28) relatively to $D_{\text{KL},1/2}(q_{\mathbf{Z}} \| q_{\mathbf{Z}}^*)$ and $D_{\text{KL},1/2}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*)$ (note that it is the only one depending explicitly on $\mathbf{X}$).

We performed a Monte Carlo experiment to observe in which cases the condition in (30) was verified, if any. The first step in the Monte Carlo experiment is as follows: we sample 100 synthetic data sets $\mathbf{X}_\rho \in \mathbb{R}^{n \times d}$ with independent rows, each of them having a normal distribution with a mean of zero and a equicorrelation covariance Σ_ρ (meaning unit variances and common pairwise correlation of $\rho \in (0, 1)$). The next step is to perform VI on each data set; a global minimizer q^* is thus obtained and we verify if the condition in (30) is satisfied. Theorem 1 in [15] states that all the global minimizers are equivalent, in the sense that they are the same up to a rotation, implying that the maximum in (30) is the same. For each data set, we set τ_0 to the PPCA maximum likelihood estimate [19].

We present results in Figure 5. They are for the case where $n = 20, d = 3$ and $k = 2$, with ρ ranging from 0.2 to 0.9. We observed similar results for different values of n, d and k . In Figure 5 (b), we observe that the condition in (30) is never met, yet exponential convergence of CAVI is always observed (see subsection 3.4.5).

To (tentatively) evaluate if the original condition in [2], meaning (27), is verified for those synthetic data sets, we derived a proxy of $\text{GCorr}_{1/2}$. This also allows to evaluate the tightness of our upper bound (yielding the condition in (30)). The proxy is conservative, in the sense that it is a lower bound on $\text{GCorr}_{1/2}$. For each $\mathbf{X}_\rho$, we obtain $(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*)$ and create a ball around it through many small random perturbations of the fixed point (see subsection 3.4.5 for the details). After initializing CAVI from each perturbation, we record the maximum value of the (exact) ratio in (27) along the CAVI trajectory to $(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*)$. Our proxy is the maximum of these maxima, mimicking the supremum over the ball in (27). The results are presented in Figure 5 (a). As our condition in (30), the condition in (27) using the proxy instead of $\text{GCorr}_{1/2}$ is never satisfied. The ratio of the mean of the left-hand side in (30) to that of the proxy is 2.3 when $\rho = 0.2$, and monotonically increases to 5.2 when $\rho = 0.9$. This shows that the bound derived in this section is relatively tight.

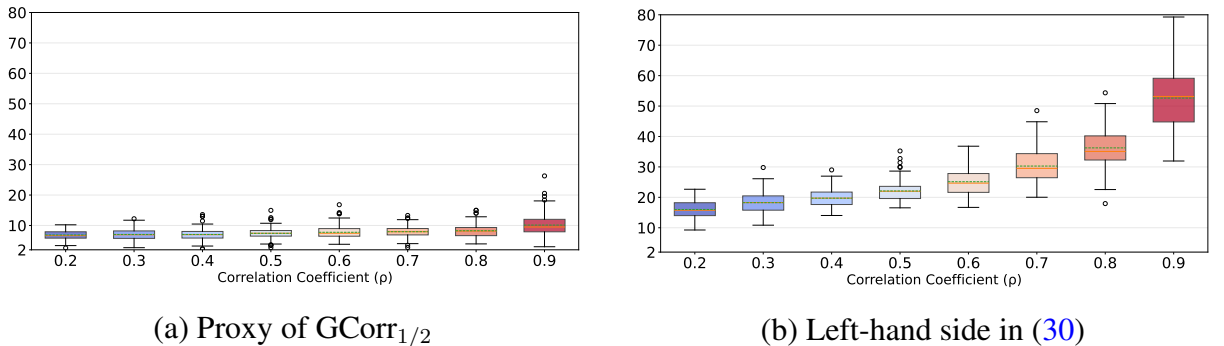


Figure 5. Proxy of $\text{GCorr}_{1/2}$ and left-hand side in (30) as a function of ρ for $(n, d, k) = (20, 3, 2)$.

3.4. Appendix of Chapter 3

3.4.1. Details of the derivation of [Algorithm 4](#)

In this appendix, we begin with the derivation of each marginal distribution used for the updates in [Algorithm 4](#) and end with [Algorithm 4](#).

3.4.1.1. The joint log-likelihood. In terms of matrix valued random variables the PPCA model can be rewritten as follows:

$$\mathbf{X} = \mathbf{Z}\mathbf{W}' + \mathbf{E},$$

where the rows of the $n \times d$ matrix $\mathbf{X}$ are given by the vectors $\{x_i\}_{1 \leq i \leq n}$, the rows of the $n \times k$ matrix $\mathbf{Z}$ are given by the latent variables $\{z_i\}_{1 \leq i \leq n}$ and the error matrix $\mathbf{E}$ has size $n \times d$ whose rows are given by the noise terms $\{\sigma \epsilon_i\}_{1 \leq i \leq n}$. We now rewrite the likelihood function and the prior distributions using the definition of matrix normal distribution presented in [section 3.4.3.2](#):

$$p(\mathbf{X}|\mathbf{W}, \mathbf{Z}, \tau) = \mathcal{N}(\mathbf{Z}\mathbf{W}', \tau^{-1} I_n \otimes I_d),$$

$$p(\mathbf{W}) = \mathcal{N}(0, I_d \otimes \Lambda^{-1}),$$

$$p(\mathbf{Z}) = \mathcal{N}(0, I_n \otimes I_k),$$

where we recall that Λ is a $k \times k$ diagonal matrix whose diagonal entries are given by $\sigma_1^{-2}, \dots, \sigma_k^{-2}$. We would like to mention that to keep the exposition simple, we did a little abuse of the notation above. For instance $p(\mathbf{X}|\mathbf{W}, \mathbf{Z}, \tau)$ is a probability density function while the right-hand side denotes a distribution $\mathcal{N}(\mathbf{Z}\mathbf{W}', \tau^{-1} I_n \otimes I_d)$, what we really meant there is this: $p(\mathbf{X}|\mathbf{W}, \mathbf{Z}, \tau) = f(\mathbf{X}; \mathbf{Z}\mathbf{W}', \tau^{-1} I_n \otimes I_d)$, using the notation introduced in [section 3.4.3.2](#). Therefore given the hyper-parameters $\sigma_1^2, \dots, \sigma_k^2, \alpha_\tau^{(0)}, \beta_\tau^{(0)}$, the logarithm of the joint PDF can be obtained as:

$$\begin{aligned} \log p(\mathbf{X}, \mathbf{W}, \mathbf{Z}, \tau) &= \log p(\mathbf{X}|\mathbf{W}, \mathbf{Z}, \tau) + \log p(\mathbf{W}) + \log p(\mathbf{Z}) + \log p(\tau) \\ &= \frac{dn}{2} \log \tau - \frac{1}{2} \tau \text{tr}((\mathbf{X} - \mathbf{Z}\mathbf{W}')(\mathbf{X} - \mathbf{Z}\mathbf{W}')') - \frac{1}{2} \text{tr}(\mathbf{W}\Lambda\mathbf{W}') \\ &\quad - \frac{1}{2} \text{tr}(\mathbf{Z}\mathbf{Z}') + (\alpha_\tau^{(0)} - 1) \log \tau - \frac{\tau}{\beta_\tau^{(0)}} + C \\ &= \frac{dn}{2} \log \tau - \frac{1}{2} \tau \text{tr}(\mathbf{X}\mathbf{X}' - \mathbf{X}\mathbf{W}\mathbf{Z}' - \mathbf{Z}\mathbf{W}'\mathbf{X}' + \mathbf{Z}\mathbf{W}'\mathbf{W}\mathbf{Z}') - \frac{1}{2} \text{tr}(\mathbf{W}\Lambda\mathbf{W}') \\ &\quad - \frac{1}{2} \text{tr}(\mathbf{Z}\mathbf{Z}') + (\alpha_\tau^{(0)} - 1) \log \tau - \frac{\tau}{\beta_\tau^{(0)}} + C \\ &= \frac{dn}{2} \log \tau - \frac{1}{2} \tau \text{tr}(\mathbf{X}'\mathbf{X}) - \frac{1}{2} \tau \text{tr}(-2\mathbf{W}\mathbf{Z}'\mathbf{X}) - \frac{1}{2} \tau \text{tr}(\mathbf{W}\mathbf{Z}'\mathbf{Z}\mathbf{W}') - \frac{1}{2} \text{tr}(\mathbf{W}\Lambda\mathbf{W}') \\ &\quad - \frac{1}{2} \text{tr}(\mathbf{Z}\mathbf{Z}') + (\alpha_\tau^{(0)} - 1) \log \tau - \frac{\tau}{\beta_\tau^{(0)}} + C, \end{aligned} \tag{31}$$

where at the final step we used properties of the trace operator and C is a constant that does not depend on the data $\mathbf{X}$ and the parameters $\mathbf{W}$, $\mathbf{Z}$, τ . We now derive the marginal distributions $q_{\mathbf{X}}^{(t+1)}(\mathbf{W})$, $q_{\mathbf{X}}^{(t+1)}(\mathbf{Z})$ and $q_{\mathbf{X}}^{(t+1)}(\tau)$ using (31) and analogous expressions.

3.4.1.2. Identification of $q_{\mathbf{X}}^{(t+1)}(\mathbf{W})$. We have that

$$\begin{aligned} q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) &\propto \exp \left(\int q_{\mathbf{X}}^{(t)}(\mathbf{Z}) q_{\mathbf{X}}^{(t)}(\tau) \log p(\mathbf{W}, \mathbf{Z}, \tau, \mathbf{X}) d\mathbf{Z} d\tau \right) \\ &\propto \exp \left[-\frac{1}{2} \text{tr} \left(-2\mathbf{W} \int q_{\mathbf{X}}^{(t)}(\mathbf{Z}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{Z}' \mathbf{X}) d\mathbf{Z} d\tau \right) \right. \\ &\quad \left. - \frac{1}{2} \text{tr} \left(\mathbf{W} \left(\int q_{\mathbf{X}}^{(t)}(\mathbf{Z}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{Z}' \mathbf{Z}) d\mathbf{Z} d\tau \right) \mathbf{W}' \right) - \frac{1}{2} \text{tr}(\mathbf{W} \Lambda \mathbf{W}') \right] \\ &= \exp \left[-\frac{1}{2} \text{tr} \left(-2\mathbf{W} \int q_{\mathbf{X}}^{(t)}(\mathbf{Z}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{Z}' \mathbf{X}) d\mathbf{Z} d\tau \right) \right. \\ &\quad \left. - \frac{1}{2} \text{tr} \left(\mathbf{W} \left(\int q_{\mathbf{X}}^{(t)}(\mathbf{Z}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{Z}' \mathbf{Z}) d\mathbf{Z} d\tau + \Lambda \right) \mathbf{W}' \right) \right]. \end{aligned}$$

Comparing the above exponentiated quadratic form in $\mathbf{W}$, we infer that $q_{\mathbf{X}}^{(t+1)}(\mathbf{W})$ is a matrix normal distribution of the form $\mathcal{N}(\mu_{\mathbf{W}}^{(t+1)}, I_d \otimes \Sigma_{\mathbf{W}}^{(t+1)})$, where the parameters are given by

$$\Sigma_{\mathbf{W}}^{(t+1)} = (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} (n \Sigma_{\mathbf{Z}}^{(t)} + (\mu_{\mathbf{Z}}^{(t)})' \mu_{\mathbf{Z}}^{(t)}) + \Lambda)^{-1},$$

using Proposition 7 and independence between $\mathbf{Z}$ and τ , and comparing the coefficient of $\mathbf{W}$ inside the trace factor, we obtain

$$\mu_{\mathbf{W}}^{(t+1)} = (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} \mathbf{X}' \mu_{\mathbf{Z}}^{(t)}) \Sigma_{\mathbf{W}}^{(t+1)}.$$

The parameters $\alpha_{\tau}^{(t)}$, $\beta_{\tau}^{(t)}$, $\mu_{\mathbf{Z}}^{(t)}$ and $\Sigma_{\mathbf{Z}}^{(t)}$ are defined below.

3.4.1.3. Identification of $q_{\mathbf{X}}^{(t+1)}(\mathbf{Z})$. We have that

$$\begin{aligned} q_{\mathbf{Z}}^{(t+1)}(\mathbf{Z}) &\propto \exp \left(\int q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) q_{\mathbf{X}}^{(t)}(\tau) \log p(\mathbf{W}, \mathbf{Z}, \tau, \mathbf{X}) d\mathbf{W} d\tau \right) \\ &\propto \exp \left[-\frac{1}{2} \text{tr} \left(-2 \int q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{W} \mathbf{Z}' \mathbf{X}) d\mathbf{W} d\tau \right) \right. \\ &\quad \left. - \frac{1}{2} \text{tr} \left(\mathbf{Z} \left(\int q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{W}' \mathbf{W}) d\mathbf{W} d\tau \right) \mathbf{Z}' \right) - \frac{1}{2} \text{tr}(\mathbf{Z} \mathbf{Z}') \right] \\ &= \exp \left[-\frac{1}{2} \text{tr} \left(-2 \int q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{W} \mathbf{Z}' \mathbf{X}) d\mathbf{W} d\tau \right) \right. \\ &\quad \left. - \frac{1}{2} \text{tr} \left(\mathbf{Z} \left(\int q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) q_{\mathbf{X}}^{(t)}(\tau) (\tau \mathbf{W}' \mathbf{W}) d\mathbf{W} d\tau + I_k \right) \mathbf{Z}' \right) \right], \end{aligned}$$

where we used $\text{tr}(\mathbf{MN}) = \text{tr}(\mathbf{NM})$ for matrices $\mathbf{M} \in \mathbb{R}^{m \times n}$ and $\mathbf{N} \in \mathbb{R}^{n \times m}$. As before, comparing the above exponentiated quadratic form in $\mathbf{Z}$ we infer that $q_{\mathbf{X}}^{(t+1)}(\mathbf{Z})$ is a matrix normal distribution of the form $\mathcal{N}(\mu_{\mathbf{Z}}^{(t+1)}, I_n \otimes \Sigma_{\mathbf{Z}}^{(t+1)})$, where the parameters are given by

$$\Sigma_{\mathbf{Z}}^{(t+1)} = (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} (d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)}) + I_k)^{-1},$$

using [Proposition 7](#) and independence between $\mathbf{W}$ and τ , and comparing the coefficient of $\mathbf{Z}$ inside the trace factor, we obtain

$$\mu_{\mathbf{Z}}^{(t+1)} = (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} \mathbf{X} \mu_{\mathbf{W}}^{(t+1)}) \Sigma_{\mathbf{Z}}^{(t+1)}.$$

3.4.1.4. Identification of $q_{\mathbf{X}}^{(t+1)}(\tau)$. We have that

$$\begin{aligned} q_{\mathbf{X}}^{(t+1)}(\tau) &\propto \exp \left(\int q_{\mathbf{X}}^{(t+1)}(\mathbf{W}) q_{\mathbf{X}}^{(t+1)}(\mathbf{Z}) \log p(\mathbf{W}, \mathbf{Z}, \tau, \mathbf{X}) d\mathbf{W} d\mathbf{Z} \right) \\ &\propto \exp \left[\left(\frac{dn}{2} + \alpha_{\tau}^{(0)} - 1 \right) \log \tau - \frac{\tau}{\beta_{\tau}^{(t+1)}} \right], \end{aligned}$$

where the updated scale parameter is given by

$$\begin{aligned} [\beta_{\tau}^{(t+1)}]^{-1} &= \frac{1}{\beta_{\tau}^{(0)}} + \frac{1}{2} \text{tr} \left(\mathbf{X}' \mathbf{X} - (\mu_{\mathbf{W}}^{(t+1)}) (\mu_{\mathbf{Z}}^{(t+1)})' \mathbf{X} - \mathbf{X}' (\mu_{\mathbf{Z}}^{(t+1)}) (\mu_{\mathbf{W}}^{(t+1)})' \right) \\ &\quad + \frac{1}{2} \text{tr} \left((d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)}) (n \Sigma_{\mathbf{Z}}^{(t+1)} + (\mu_{\mathbf{Z}}^{(t+1)})' \mu_{\mathbf{Z}}^{(t+1)}) \right), \end{aligned}$$

and

$$\alpha_{\tau}^{(t+1)} = \frac{dn}{2} + \alpha_{\tau}^{(0)}.$$

Algorithm 4 An explicit CAVI algorithm for BPCA

(1) Initialization: set $\mu_{\mathbf{Z}}^{(0)}, \Sigma_{\mathbf{Z}}^{(0)}, \alpha_{\tau}^{(0)}$ and $\beta_{\tau}^{(0)}$, and $q_{\mathbf{X}}^{(0)}(\mathbf{Z}) = \mathcal{N}(\mu_{\mathbf{Z}}^{(0)}, I_d \otimes \Sigma_{\mathbf{Z}}^{(0)})$ and $q_{\mathbf{X}}^{(0)}(\tau) = \Gamma(\alpha_{\tau}^{(0)}, \beta_{\tau}^{(0)})$. Fix $\varepsilon > 0$.

(2) Update the parameters of the distribution of $\mathbf{W}$:

$$\begin{aligned}\Sigma_{\mathbf{W}}^{(t+1)} &= (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} (n \Sigma_{\mathbf{Z}}^{(t)} + (\mu_{\mathbf{Z}}^{(t)})' \mu_{\mathbf{Z}}^{(t)}) + \Lambda)^{-1}, \\ \mu_{\mathbf{W}}^{(t+1)} &= (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} \mathbf{X}' \mu_{\mathbf{Z}}^{(t)}) \Sigma_{\mathbf{W}}^{(t+1)}.\end{aligned}$$

(3) Update the parameters of the distribution of the latent variable $\mathbf{Z}$:

$$\begin{aligned}\Sigma_{\mathbf{Z}}^{(t+1)} &= (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} (d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)}) + I_k)^{-1}, \\ \mu_{\mathbf{Z}}^{(t+1)} &= (\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)} \mathbf{X} \mu_{\mathbf{W}}^{(t+1)}) \Sigma_{\mathbf{Z}}^{(t+1)}.\end{aligned}$$

(4) Update the parameters of the distribution of τ :

$$\alpha_{\tau}^{(t+1)} = \frac{dn}{2} + c_{\tau},$$

$$\begin{aligned}\beta_{\tau}^{(t+1)} &= \left(\frac{1}{d_{\tau}} + \frac{1}{2} \text{tr} \left(\mathbf{X}' \mathbf{X} - (\mu_{\mathbf{W}}^{(t+1)}) (\mu_{\mathbf{Z}}^{(t+1)})' \mathbf{X} - \mathbf{X}' (\mu_{\mathbf{Z}}^{(t+1)}) (\mu_{\mathbf{W}}^{(t+1)})' \right) \right. \\ &\quad \left. + \frac{1}{2} \text{tr} \left((d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)}) (n \Sigma_{\mathbf{Z}}^{(t+1)} + (\mu_{\mathbf{Z}}^{(t+1)})' \mu_{\mathbf{Z}}^{(t+1)}) \right) \right)^{-1}.\end{aligned}$$

(5) Calculate the percentage of increase in ELBO:

$$\Delta^{(t)} \text{ELBO} := \frac{\text{ELBO}(q^{(t+1)}) - \text{ELBO}(q^{(t)})}{\text{ELBO}(q^{(t)})}.$$

(6) While $\Delta^{(t)} \text{ELBO} > \varepsilon$, go to Step 2.

In the case where τ is fixed to τ_0 , one can replace $\alpha_{\tau}^{(t)} \beta_{\tau}^{(t)}$ with τ_0 in [Algorithm 4](#).

3.4.2. Details of the analysis of CAVI dynamics for $k = 1$

In BPCA, as mentioned, the case $k = 1$ reduces the model to one with a single PC. In particular, $k = 1$ implies that the family $\mathcal{Q}$ is such that $\mu_{\mathbf{W}} \in \mathbb{R}^d$ and $\mu_{\mathbf{Z}} \in \mathbb{R}^n$ are vectors, and $\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{Z}}$ and $\alpha_{\tau}, \beta_{\tau}$ are positive real numbers. Also, CAVI takes a simpler form governed by the following mappings:

$$\begin{aligned}F: \mathbb{R}^n \times \mathbb{R}_+^3 &\rightarrow \mathbb{R}^d \times \mathbb{R}_+^3 \\ (\mu_{\mathbf{Z}}, \Sigma_{\mathbf{Z}}, \alpha_{\tau}, \beta_{\tau}) &\mapsto \left(\frac{\alpha_{\tau} \beta_{\tau} \mathbf{X}' \mu_{\mathbf{Z}}}{\alpha_{\tau} \beta_{\tau} (n \Sigma_{\mathbf{Z}} + \|\mu_{\mathbf{Z}}\|^2) + \Lambda}, \frac{1}{\alpha_{\tau} \beta_{\tau} (n \Sigma_{\mathbf{Z}} + \|\mu_{\mathbf{Z}}\|^2) + \Lambda}, \alpha_{\tau}, \beta_{\tau} \right), \\ G: \mathbb{R}^d \times \mathbb{R}_+^3 &\rightarrow \mathbb{R}^d \times \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}_+ \\ (\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}}, \alpha_{\tau}, \beta_{\tau}) &\mapsto \left(\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}}, \frac{\alpha_{\tau} \beta_{\tau} \mathbf{X} \mu_{\mathbf{W}}}{\alpha_{\tau} \beta_{\tau} (d \Sigma_{\mathbf{W}} + \|\mu_{\mathbf{W}}\|^2) + 1}, \frac{1}{\alpha_{\tau} \beta_{\tau} (d \Sigma_{\mathbf{W}} + \|\mu_{\mathbf{W}}\|^2) + 1} \right), \\ H: \mathbb{R}^d \times \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}_+ &\rightarrow \mathbb{R}^n \times \mathbb{R}_+^3\end{aligned}$$

$$(\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}}, \mu_{\mathbf{Z}}, \Sigma_{\mathbf{Z}}) \mapsto (\mu_{\mathbf{Z}}, \Sigma_{\mathbf{Z}}, H_1, H_2),$$

where the components of H are given by

$$H_1 = \frac{dn}{2} + \alpha_{\tau}^{(0)}, \text{ a constant function and}$$

$$H_2 = \left(\frac{1}{\beta_{\tau}^{(0)}} + \frac{1}{2} \text{tr} \left(\mathbf{X}' \mathbf{X} - (\mu_{\mathbf{W}}^{(t+1)})(\mu_{\mathbf{Z}}^{(t+1)})' \mathbf{X} - \mathbf{X}' (\mu_{\mathbf{Z}}^{(t+1)})(\mu_{\mathbf{W}}^{(t+1)})' \right) \right. \\ \left. + \frac{1}{2} \text{tr} \left((d\Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)})(n\Sigma_{\mathbf{Z}}^{(t+1)} + (\mu_{\mathbf{Z}}^{(t+1)})' \mu_{\mathbf{Z}}^{(t+1)}) \right) \right)^{-1}.$$

$\|\cdot\|$ is the Euclidean norm; when applied to a matrix, $\|\cdot\|$ represents the Frobenius norm, unless otherwise stated. CAVI indeed corresponds to the following update equations for the parameters of the matrix normal distributions of $\mathbf{W}$ and $\mathbf{Z}$ (which are in fact multivariate normal distributions in the case $k = 1$) and gamma distribution of τ .

3.4.2.1. On the convergence of the scaling component and $(\Sigma_{\mathbf{Z}}^{(t)}, \Sigma_{\mathbf{W}}^{(t)}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)})$. Given the initialization $(\mu_{\mathbf{Z}}^{(0)}, \Sigma_{\mathbf{Z}}^{(0)}, \alpha_{\tau}^{(0)}, \beta_{\tau}^{(0)}) = (a^{(0)}, b^{(0)}, \xi^{(0)})$, we are in fact guaranteed that $c_1 \neq 0$ in [Theorem 11](#) and CAVI updates are governed by the following mapping:

$$\Phi: \mathbb{R}_+^3 \rightarrow \mathbb{R}_+^3 \\ (a, b, \xi) \mapsto \left(\frac{\xi^2 a L(a, b, \xi) \lambda_1}{D(a, b, \xi)}, \frac{L(a, b, \xi)^2}{D(a, b, \xi)}, \frac{C}{R(a, b, \xi)} \right),$$

where $D(a, b, \xi) := d\xi L(a, b, \xi) + \xi^3 a^2 \lambda_1 + L(a, b, \xi)^2$, $C = \alpha_{\tau}^{(0)} + dn/2$, $L(a, b, \xi) = \xi \Gamma_2(a, b) + \Lambda$ and

$$R(a, b, \xi) = c_1 - \sqrt{\lambda_1} \Phi_1(a, b, \xi) T_1(a, b, \xi) \\ + \frac{1}{2} \Gamma_1(T_1(a, b, \xi), T_2(a, b, \xi)) \Gamma_2(\Phi_1(a, b, \xi), \Phi_2(a, b, \xi)). \quad (32)$$

with $c_1 = 1/\beta_{\tau}^{(0)} + \text{tr}(\mathbf{X}' \mathbf{X})/2$ and Φ_1, Φ_2 denoting the first two coordinates of Φ . The rational function $R(a, b, \xi)$ is complicated and we describe its each piece below:

$$T: \mathbb{R}_+^3 \rightarrow \mathbb{R}_+^2, \quad (a, b, \xi) \mapsto \left(\frac{\sqrt{\lambda_1} \xi a}{\xi \Gamma_2(a, b) + \Lambda}, \frac{1}{\xi \Gamma_2(a, b) + \Lambda} \right),$$

with $T(a, b, \xi) = (T_1(a, b, \xi), T_2(a, b, \xi))$, and finally $\Gamma_1: \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$, $\Gamma_2: \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ are mappings defined by $\Gamma_1(x, y) = dy + x^2$, $\Gamma_2(u, v) = nv + u^2$. In other words,

$$(a^{(t+1)}, b^{(t+1)}, \xi^{(t+1)}) = \Phi(a^{(t)}, b^{(t)}, \xi^{(t)}), \quad t = 0, 1, \dots$$

If the sequence $(a^{(t)}, b^{(t)}, \xi^{(t)})$ converges to a limit (a^*, b^*, ξ^*) , then this limit must satisfy the fixed point condition $\Phi(a^*, b^*, \xi^*) = (a^*, b^*, \xi^*)$ given that the map Φ is continuous and also $R(a^*, b^*, \xi^*) \neq 0$ (see [Lemma 24](#)). Consequently, characterizing all such fixed points provides insights into the possible limit points of the CAVI iterates. The next proposition is about this.

3.4.3. Proof of theoretical results

3.4.3.1. Proof of main results.

Proof of Theorem 11. Recall that the n eigenvalues of the matrix $\mathbf{X}\mathbf{X}'$ are sorted such that the first d eigen values are strictly positive $\lambda_1 > \lambda_2 > \lambda_3 > \dots > \lambda_d > 0$ and $\lambda_{d+1} = \dots = \lambda_n = 0$ with corresponding eigenvectors $\mu_1, \mu_2, \dots, \mu_n$, which forms an orthonormal basis of $\mathbb{R}^n$. Without loss of generality, we assume there exist constants $c_1, c_2, c_3, \dots, c_n$ such that $\tilde{\mu}_{\mathbf{Z}}^{(0)} := \mu_{\mathbf{Z}}^{(0)} / \|\mu_{\mathbf{Z}}^{(0)}\| = c_1\mu_1 + c_2\mu_2 + \dots + c_n\mu_n$ with $c_1 \neq 0$.

We set $\xi^{(t)} = \alpha_{\tau}^{(t)}\beta_{\tau}^{(t)}$. For $t \geq 1$, let us define the quantity $B_t = \xi^{(t)}(n\Sigma_{\mathbf{Z}}^{(t)} + \|\mu_{\mathbf{Z}}^{(t)}\|^2) + \Lambda$. From (23), we have that

$$\begin{aligned} (\mu_{\mathbf{Z}}^{(t+1)}, \Sigma_{\mathbf{Z}}^{(t+1)}, \alpha_{\tau}^{(t+1)}, \beta_{\tau}^{(t+1)}) &= (H \circ G \circ F)(\mu_{\mathbf{Z}}^{(t)}, \Sigma_{\mathbf{Z}}^{(t)}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)}) \\ &= (H \circ G)\left(\frac{\xi^{(t)}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}}{B_t}, \frac{1}{B_t}, \alpha_{\tau}^{(t)}, \beta_{\tau}^{(t)}\right), \end{aligned} \quad (33)$$

Thus, we have that

$$\begin{aligned} \mu_{\mathbf{Z}}^{(t+1)} &= \frac{\xi^{(t)}\mathbf{X} \left(\frac{\xi^{(t)}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}}{B_t} \right)}{\xi^{(t)} \left(\frac{d}{B_t} + \left\| \frac{\xi^{(t)}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}}{B_t} \right\|^2 \right) + 1} \\ &= \frac{(\xi^{(t)})^2 \mathbf{X}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)} / B_t}{\xi^{(t)} \left(\frac{d}{B_t} + \frac{\|\xi^{(t)}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}\|^2}{B_t^2} \right) + 1} \\ &= \frac{(\xi^{(t)})^2 B_t \mathbf{X}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}}{d\xi^{(t)}B_t + (\xi^{(t)})^3 \|\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}\|^2 + B_t^2}, \end{aligned} \quad (34)$$

and

$$\Sigma_{\mathbf{Z}}^{(t+1)} = \frac{B_t^2}{d\xi^{(t)}B_t + (\xi^{(t)})^3 \|\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}\|^2 + B_t^2},$$

which implies the normalized $(t+1)$ -th update of $\tilde{\mu}_{\mathbf{Z}}^{(t+1)} = \mathbf{X}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)} / \|\mathbf{X}\mathbf{X}'\mu_{\mathbf{Z}}^{(t)}\| = \mathbf{X}\mathbf{X}'\tilde{\mu}_{\mathbf{Z}}^{(t)} / \|\mathbf{X}\mathbf{X}'\tilde{\mu}_{\mathbf{Z}}^{(t)}\|$. This motivates us to look at the following iterative sequence on the unit circle:

$$\tilde{\mu}_{\mathbf{Z}}^{(0)} := \frac{\mu_{\mathbf{Z}}^{(0)}}{\|\mu_{\mathbf{Z}}^{(0)}\|}, \frac{\mathbf{A}\tilde{\mu}_{\mathbf{Z}}^{(0)}}{\|\mathbf{A}\tilde{\mu}_{\mathbf{Z}}^{(0)}\|}, \frac{\mathbf{A}^2\tilde{\mu}_{\mathbf{Z}}^{(0)}}{\|\mathbf{A}^2\tilde{\mu}_{\mathbf{Z}}^{(0)}\|}, \dots, \quad \text{where } \mathbf{A} = \mathbf{X}\mathbf{X}'.$$

We claim the above sequence converges to μ_1 . Note that for any $t \geq 1$,

$$\mathbf{A}^t \tilde{\mu}_{\mathbf{Z}}^{(0)} = c_1 \mathbf{A}^t \mu_1 + c_2 \mathbf{A}^t \mu_2 + \dots + c_n \mathbf{A}^t \mu_n$$

$$= c_1(\lambda_1)^t \left(\mu_1 + \frac{c_2}{c_1} \left(\frac{\lambda_2}{\lambda_1} \right)^t \mu_2 + \cdots + \frac{c_d}{c_1} \left(\frac{\lambda_d}{\lambda_1} \right)^t \mu_d \right).$$

Let us define the vector

$$v_t = \left(\mu_1 + \frac{c_2}{c_1} \left(\frac{\lambda_2}{\lambda_1} \right)^t \mu_2 + \cdots + \frac{c_d}{c_1} \left(\frac{\lambda_d}{\lambda_1} \right)^t \mu_d \right).$$

Thus we have that

$$\begin{aligned} \tilde{\mu}_{\mathbf{Z}}^{(t)} &= \frac{\mathbf{A}^t \tilde{\mu}_{\mathbf{Z}}^{(0)}}{\|\mathbf{A}^t \tilde{\mu}_{\mathbf{Z}}^{(0)}\|} \\ &= \frac{c_1}{|c_1|} \left(\frac{v_t}{\|v_t\|} \right) \\ &= \frac{c_1}{|c_1|} \frac{\mu_1 + \frac{c_2}{c_1} \left(\frac{\lambda_2}{\lambda_1} \right)^t \mu_2 + \cdots + \frac{c_d}{c_1} \left(\frac{\lambda_d}{\lambda_1} \right)^t \mu_d}{\left\| \mu_1 + \frac{c_2}{c_1} \left(\frac{\lambda_2}{\lambda_1} \right)^t \mu_2 + \cdots + \frac{c_d}{c_1} \left(\frac{\lambda_d}{\lambda_1} \right)^t \mu_d \right\|} \\ &\rightarrow \text{sgn}(c_1) \mu_1, \quad \text{as } t \rightarrow \infty. \end{aligned}$$

The second part of the proof follows from [Lemma 11](#) after noticing for all $t \geq 1$

$$\|v_t - \text{sgn}(c_1) \mu_1\| \leq \max\{|c_1|, |c_2|, \dots, |c_d|\} \left(\frac{d-1}{|c_1|} \right) \left(\frac{\lambda_2}{\lambda_1} \right)^t = \frac{c_0}{2} \left(1 - \frac{\rho_1}{\lambda_1} \right)^t,$$

for some $c_0 > 0$ which only depends on d , the initialization and the data $\mathbf{X}$, and consequently

$$\left\| \tilde{\mu}_{\mathbf{Z}}^{(t)} - \frac{c_1}{|c_1|} \mu_1 \right\| = \left\| \frac{c_1}{|c_1|} \left(\frac{v_t}{\|v_t\|} \right) - \frac{c_1}{|c_1|} \mu_1 \right\| = \left\| \frac{v_t}{\|v_t\|} - \mu_1 \right\| \leq c_0 \left(1 - \frac{\rho_1}{\lambda_1} \right)^t.$$

To obtain the upper bound on normalized iterates of $\mu_{\mathbf{W}}^{(t)}$, we pretend that our initialization is

$$\mu_{\mathbf{W}}^{(1)} = \frac{\xi^{(t)} \mathbf{X}' \mu_{\mathbf{Z}}^{(0)}}{B_0} = \frac{\xi^{(t)}}{B_0} (c_1 \mathbf{X}' \mu_1 + \cdots + c_n \mathbf{X}' \mu_n) = \frac{\xi^{(t)}}{B_0} \left(\sqrt{\lambda_1} c_1 \nu_1 + \cdots + \sqrt{\lambda_d} c_d \nu_d \right),$$

where $\nu_1, \dots, \nu_d$ are unit norm eigen vectors of $\mathbf{X}'\mathbf{X}$ corresponding to the eigen values $\lambda_1, \dots, \lambda_d$, forming an orthonormal basis of $\mathbb{R}^d$. Using similar analysis as before leads to:

$$\frac{\mu_{\mathbf{W}}^{(t+1)}}{\|\mu_{\mathbf{W}}^{(t+1)}\|} = \frac{\mathbf{X}' \mathbf{X} \mu_{\mathbf{W}}^{(t)}}{\|\mathbf{X}' \mathbf{X} \mu_{\mathbf{W}}^{(t)}\|} \quad t = 1, 2, \dots$$

In this case, using an analogous calculation as $\mu_{\mathbf{Z}}^{(t)}$, we obtain for $t \geq 2$,

$$\left\| \frac{\mu_{\mathbf{W}}^{(t+1)}}{\|\mu_{\mathbf{W}}^{(t+1)}\|} - \text{sgn}(c_1) \frac{\mathbf{X}' \mu_1}{\|\mathbf{X}' \mu_1\|} \right\| \leq c'_0 \left(1 - \frac{\rho_1}{\lambda_1} \right)^t,$$

where $c'_0 = C\beta^{(0)} c_0 \sqrt{\lambda_1} / \Lambda$. □

Proof of Proposition 5. We are given that (a^*, b^*, ξ^*) is a fixed point of Φ with $a^* > 0$, i.e., we have

$$(a^*, b^*, \xi^*) = \Phi(a^*, b^*, \xi^*),$$

equivalently, the following system of equations is satisfied

$$a^* = \frac{(\xi^*)^2 a^* L(a^*, b^*, \xi^*) \lambda_1}{d \xi^* L(a^*, b^*, \xi^*) + (\xi^*)^3 (a^*)^2 \lambda_1 + L(a^*, b^*, \xi^*)^2}$$

and

$$b^* = \frac{L(a^*, b^*, \xi^*)^2}{d \xi^* L(a^*, b^*, \xi^*) + (\xi^*)^3 (a^*)^2 \lambda_1 + L(a^*, b^*, \xi^*)^2}.$$

Plugging the value of b^* from the second equation, we have that

$$a^* = \frac{(\xi^*)^2 L(a^*, b^*, \xi^*) a^* \lambda_1 b^*}{L(a^*, b^*, \xi^*)^2} \implies b^* = \frac{L(a^*, b^*, \xi^*)}{(\xi^*)^2 \lambda_1}. \quad (35)$$

Equating two expressions for b^* , we obtain

$$\frac{L(a^*, b^*, \xi^*)}{(\xi^*)^2 \lambda_1} = \frac{L(a^*, b^*, \xi^*)^2}{d \xi^* L(a^*, b^*, \xi^*) + (\xi^*)^3 (a^*)^2 \lambda_1 + L(a^*, b^*, \xi^*)^2}.$$

Thus,

$$L(a^*, b^*, \xi^*)^2 + (d \xi^* - \lambda_1 (\xi^*)^2) L(a^*, b^*, \xi^*) + (\xi^*)^3 (a^*)^2 \lambda_1 = 0. \quad (36)$$

Note that also, $b^* = \frac{1}{n} \left(\frac{L(a^*, b^*, \xi^*) - \Lambda}{\xi^*} - (a^*)^2 \right)$ which implies,

$$\frac{L(a^*, b^*, \xi^*)}{(\xi^*)^2 \lambda_1} = \frac{1}{n} \left(\frac{L(a^*, b^*, \xi^*) - \Lambda}{\xi^*} - (a^*)^2 \right) \implies L(a^*, b^*, \xi^*) = \frac{\frac{\Lambda}{n \xi^*} + \frac{(a^*)^2}{n}}{\frac{1}{n \xi^*} - \frac{1}{(\xi^*)^2 \lambda_1}}.$$

Note that if we plug the value of $L(a^*, b^*, \xi^*)$ in (36), then we can obtain a quadratic equation, which is being satisfied by $(a^*)^2$. Let $t = (a^*)^2$. Then

$$L(a^*, b^*, \xi^*) = \xi^* \lambda_1 \left(\frac{\Lambda + \xi^* t}{\lambda_1 \xi^* - n} \right),$$

substituting this in (36), one obtains a quadratic equation in t , which means $(a^*)^2$ has at most 2 possibilities implying the conclusion. The details are as follows:

$$\left(\xi^* \lambda_1 \frac{\Lambda + \xi^* t}{\lambda_1 \xi^* - n} \right)^2 + (d \xi^* - \lambda_1 (\xi^*)^2) \left(\xi^* \lambda_1 \frac{\Lambda + \xi^* t}{\lambda_1 \xi^* - n} \right) + (\xi^*)^3 \lambda_1 t = 0,$$

and, after simplification,

$$\begin{aligned} & \lambda_1 (\xi^*)^2 t^2 + \xi^* t \left[2 \lambda_1 \Lambda + (d - \lambda_1 \xi^*) (\lambda_1 \xi^* - n) + (\lambda_1 \xi^* - n)^2 \right] \\ & + \left[\lambda_1 \Lambda^2 + (d - \lambda_1 \xi^*) (\lambda_1 \xi^* - n) \Lambda \right] = 0. \end{aligned}$$

Thus, $((a^*)^2, \xi^*)$ satisfies the equation $P(t, w) = 0$. Finally from (35) and the definition of $L(a^*, b^*, \xi^*)$,

$$b^* = \frac{L(a^*, b^*, \xi^*)}{(\xi^*)^2 \lambda_1} = \frac{\xi^*(nb^* + (a^*)^2) + \Lambda}{(\xi^*)^2 \lambda_1},$$

and solving for b^* , we obtain

$$b^* = \frac{\Lambda + \xi^*(a^*)^2}{(\xi^*)^2 \lambda_1 - n\xi^*}. \quad (37)$$

Using Lemma 24, at convergence, we have that

$$\begin{aligned} \frac{\xi^*}{C} &= \frac{1}{R(a^*, b^*, \xi^*)} = \left(c_1 - \sqrt{\lambda_1} \Phi_1(a^*, b^*, \xi^*) T_1(a^*, b^*, \xi^*) \right. \\ &\quad \left. + \frac{1}{2} \Gamma_1(T_1(a^*, b^*, \xi^*), T_2(a^*, b^*, \xi^*)) \right. \\ &\quad \left. \Gamma_2(\Phi_1(a^*, b^*, \xi^*), \Phi_2(a^*, b^*, \xi^*)) \right)^{-1} \\ &= \left(c_1 - \sqrt{\lambda_1} a^* T_1(a^*, b^*, \xi^*) \right. \\ &\quad \left. + \frac{1}{2} \Gamma_1(T_1(a^*, b^*, \xi^*), T_2(a^*, b^*, \xi^*)) \Gamma_2(a^*, b^*) \right)^{-1}. \end{aligned}$$

Thus, (a^*, b^*, ξ^*) is a root of the following equation $wR(u, v, w) - C = 0$, where using the definition of the mapping T_1 , one can rewrite $R(u, v, w)$ as

$$R(u, v, w) = c_1 - \frac{\lambda_1 w u^2}{w \Gamma_2(u, v) + \Lambda} + \frac{1}{2} \Gamma_1(T_1(u, v, w), T_2(u, v, w)) \Gamma_2(u, v).$$

Now let us compute,

$$\begin{aligned} \Gamma_1(T_1(u, v, w), T_2(u, v, w)) &= dT_2(u, v, w) + T_1^2(u, v, w) \\ &= \frac{d}{w \Gamma_2(u, v) + \Lambda} + \frac{\lambda_1 w^2 u^2}{(w \Gamma_2(u, v) + \Lambda)^2} \\ &= \frac{d(w \Gamma_2(u, v) + \Lambda) + \lambda_1 w^2 u^2}{(w \Gamma_2(u, v) + \Lambda)^2} \end{aligned}$$

Thus, we have that

$$\Gamma_1(T_1(u, v, w), T_2(u, v, w)) \Gamma_2(u, v) = \frac{[d(w \Gamma_2(u, v) + \Lambda) + \lambda_1 w^2 u^2] \Gamma_2(u, v)}{(w \Gamma_2(u, v) + \Lambda)^2}$$

To clear all denominators in the equation $wR(u, v, w) - C = 0$, we multiply by $(w \Gamma_2(u, v) + \Lambda)^2$:

$$(w \Gamma_2(u, v) + \Lambda)^2 [wR(u, v, w) - C] = 0$$

Expanding term by term:

$$\begin{aligned}
& w(w\Gamma_2(u,v) + \Lambda)^2 c_1 \\
& - w(w\Gamma_2(u,v) + \Lambda)^2 \cdot \frac{\lambda_1 w u^2}{w\Gamma_2(u,v) + \Lambda} \\
& + w(w\Gamma_2(u,v) + \Lambda)^2 \cdot \frac{1}{2} \cdot \frac{[d(w\Gamma_2(u,v) + \Lambda) + \lambda_1 w^2 u^2] \Gamma_2(u,v)}{(w\Gamma_2(u,v) + \Lambda)^2} \\
& - C(w\Gamma_2(u,v) + \Lambda)^2 = 0
\end{aligned}$$

Simplifying

$$\begin{aligned}
& w(w\Gamma_2(u,v) + \Lambda)^2 c_1 \\
& - \lambda_1 w^2 u^2 (w\Gamma_2(u,v) + \Lambda) \\
& + \frac{w}{2} \Gamma_2(u,v) [d(w\Gamma_2(u,v) + \Lambda) + \lambda_1 w^2 u^2] \\
& - C(w\Gamma_2(u,v) + \Lambda)^2 = 0
\end{aligned}$$

Let $G = w\Gamma_2(u,v) + \Lambda = w(nv + u^2) + \Lambda$. Then the above equation simplifies to

$$wG^2 c_1 - \lambda_1 w^2 u^2 G + \frac{w}{2} \Gamma_2(u,v) [dG + \lambda_1 w^2 u^2] - CG^2 = 0 \quad (38)$$

Expanding further,

$$\begin{aligned}
P_3(u,v,w) &= wG^2 c_1 - \lambda_1 w^2 u^2 G + \frac{dw}{2} \Gamma_2(u,v) G + \frac{\lambda_1 w^3 u^2}{2} \Gamma_2(u,v) - CG^2 \\
&= (wc_1 - C)G^2 - \lambda_1 w^2 u^2 G + \frac{dw}{2} \Gamma_2(u,v) G + \frac{\lambda_1 w^3 u^2}{2} \Gamma_2(u,v)
\end{aligned}$$

Finally, we deduce that (a^*, b^*, ξ^*) satisfies the coupled system of three polynomial equations:

- (1) $P_1(u,w) = \lambda_1 w^2 u^4 + wu^2 [2\lambda_1 \Lambda + (d - \lambda_1 w)(\lambda_1 w - n) + (\lambda_1 w - n)^2] + [\lambda_1 \Lambda^2 + (d - \lambda_1 w)(\lambda_1 w - n)\Lambda] = 0$
- (2) $P_2(u,v,w) = v(w^2 \lambda_1 - nw) - \Lambda - wu^2 = 0$
- (3) $P_3(u,v,w) = (wc_1 - C)G^2 - \lambda_1 w^2 u^2 G + \frac{dw}{2} \Gamma_2(u,v) G + \frac{\lambda_1 w^3 u^2}{2} \Gamma_2(u,v) = 0$

where $G = w(nv + u^2) + \Lambda$ and $\Gamma_2(u,v) = nv + u^2$. It is evident that P_2 is considerably simple and we can plug in $v = (\Lambda + wu^2)/(w^2 \lambda_1 - nw)$ in other two equations and obtain a simpler system of two polynomial equations. Furthermore, P_1, P_2, P_3 all depend on u only through u^2 , so it makes sense to substitute $t = u^2$, to simplify their degrees.

$$\begin{aligned}
G &= w(nv + t) + \Lambda \\
&= w \left(n \cdot \frac{\Lambda + wt}{w(w\lambda_1 - n)} + t \right) + \Lambda
\end{aligned}$$

$$\begin{aligned}
&= \frac{n(\Lambda + wt)}{w\lambda_1 - n} + wt + \Lambda \\
&= \frac{n\Lambda + nwt + (w\lambda_1 - n)(wt + \Lambda)}{w\lambda_1 - n} \\
&= \frac{w^2\lambda_1 t + w\lambda_1\Lambda}{w\lambda_1 - n} \\
&= \frac{w\lambda_1(\Lambda + wt)}{w\lambda_1 - n}.
\end{aligned}$$

Similarly, for Γ_2 :

$$\begin{aligned}
\Gamma_2(t, v) &= nv + t \\
&= \frac{n(\Lambda + wt)}{w(w\lambda_1 - n)} + t \\
&= \frac{n(\Lambda + wt) + wt(w\lambda_1 - n)}{w(w\lambda_1 - n)} \\
&= \frac{\Lambda n + w^2\lambda_1 t}{w(w\lambda_1 - n)}.
\end{aligned}$$

Substituting these into P_3 , we get:

$$\begin{aligned}
P_3(w, t) &= (wc_1 - C) \left(\frac{w\lambda_1(\Lambda + wt)}{w\lambda_1 - n} \right)^2 - \lambda_1 w^2 t \left(\frac{w\lambda_1(\Lambda + wt)}{w\lambda_1 - n} \right) \\
&\quad + \frac{dw}{2} \left(\frac{\Lambda n + w^2\lambda_1 t}{w(w\lambda_1 - n)} \right) \left(\frac{w\lambda_1(\Lambda + wt)}{w\lambda_1 - n} \right) \\
&\quad + \frac{\lambda_1 w^3 t}{2} \left(\frac{\Lambda n + w^2\lambda_1 t}{w(w\lambda_1 - n)} \right),
\end{aligned}$$

which implies

$$\begin{aligned}
P_3(w, t) &= (wc_1 - C) \cdot \frac{w^2\lambda_1^2(\Lambda + wt)^2}{(w\lambda_1 - n)^2} - \lambda_1 w^3 \lambda_1 t \cdot \frac{(\Lambda + wt)}{w\lambda_1 - n} \\
&\quad + \frac{d}{2} \cdot \lambda_1 \cdot \frac{(\Lambda n + w^2\lambda_1 t)(\Lambda + wt)}{(w\lambda_1 - n)^2} + \frac{\lambda_1 w^2 t}{2} \cdot \frac{(\Lambda n + w^2\lambda_1 t)}{(w\lambda_1 - n)}.
\end{aligned}$$

Multiplying through by the common denominator $(w\lambda_1 - n)^2$ (note that it is non-zero, otherwise b^* is undefined) to clear all fractions yields a polynomial in w and t . This is the final polynomial form of the equation:

$$\begin{aligned}
0 &= (wc_1 - C)w^2\lambda_1^2(\Lambda + wt)^2 - \lambda_1^2 w^3 t (w\lambda_1 - n)(\Lambda + wt) \\
&\quad + \frac{d}{2} \lambda_1 (\Lambda n + w^2\lambda_1 t)(\Lambda + wt) + \frac{\lambda_1 w^2 t}{2} (w\lambda_1 - n)(\Lambda n + w^2\lambda_1 t).
\end{aligned}$$

After substitution and simplification, we obtain the system:

$$\begin{aligned} P(t, w) &= \lambda_1 w^2 t^2 + wt[2\lambda_1 \Lambda + (d - \lambda_1 w)(\lambda_1 w - n) + (\lambda_1 w - n)^2] \\ &\quad + [\lambda_1 \Lambda^2 + (d - \lambda_1 w)(\lambda_1 w - n)\Lambda] = 0 \end{aligned} \quad (39)$$

$$\begin{aligned} Q(t, w) &= (wc_1 - C)w^2 \lambda_1^2 (\Lambda + wt)^2 - \lambda_1^2 w^3 t (w\lambda_1 - n)(\Lambda + wt) \\ &\quad + \frac{d}{2} \lambda_1 (\Lambda n + w^2 \lambda_1 t)(\Lambda + wt) + \frac{\lambda_1 w^2 t}{2} (w\lambda_1 - n)(\Lambda n + w^2 \lambda_1 t) = 0, \end{aligned}$$

which is satisfied by $((a^*)^2, \xi^*)$. We check using symbolic algebra in python that $\gcd(P, Q) = 1$. Thus, the polynomial system $P = 0, Q = 0$ has at most $7 \times 4 = 28$ solutions by Bézout's theorem. $\square$

Proof of Theorem 12. Let $r_0 > 0$ be fixed, and we pick some $(q_{\mathbf{W}}, q_{\mathbf{Z}}) \in \mathcal{B}_{\mathbf{W}}^*(r_0) \times \mathcal{B}_{\mathbf{Z}}^*(r_0)$. Let us denote the quantities: $A_1 = \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\|, A_2 = \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|, B_1 = \|\Sigma_{\mathbf{Z}} - \Sigma_{\mathbf{Z}}^*\|, B_2 = \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\|, C_1 = \|\mu_{\mathbf{Z}}\| + \|\mu_{\mathbf{Z}}^*\|, C_2 = \|\mu_{\mathbf{W}}\| + \|\mu_{\mathbf{W}}^*\|$. Recall the generalized correlation in (27) as:

$$\text{GCorr}_{1/2}(r_0) := \sup_{\substack{q_{\mathbf{W}} \in \mathcal{B}_{\mathbf{W}}^*(r_0) \setminus \{q_{\mathbf{W}}^*\}, \\ q_{\mathbf{Z}} \in \mathcal{B}_{\mathbf{Z}}^*(r_0) \setminus \{q_{\mathbf{Z}}^*\}}} \frac{|\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})|}{2\sqrt{D_{\text{KL},1/2}(q_{\mathbf{W}}\|q_{\mathbf{W}}^*) D_{\text{KL},1/2}(q_{\mathbf{Z}}\|q_{\mathbf{Z}}^*)}}.$$

Following [2], to establish Theorem 12, it suffices to show the existence of $r_0 > 0$ such that $\text{GCorr}_{1/2}(r_0) \in (0, 2)$. Using Lemma 12, we have that

$$\text{GCorr}_{1/2}(r_0) \leq \rho_0,$$

where

$$\rho_0 = \frac{\left(\tau_0 \|\mathbf{X}\| + \frac{C_1 C_2 \tau_0}{2}\right) A_1 A_2 + (nd\tau_0 B_1 B_2 + dC_1 \tau_0 B_2 A_1 + nC_2 \tau_0 A_2 B_1)/2}{\sqrt{Q_1 Q_2}}$$

with $Q_1 = D_{\text{KL},1/2}(q_{\mathbf{W}}\|q_{\mathbf{W}}^*)$ and $Q_2 = D_{\text{KL},1/2}(q_{\mathbf{Z}}\|q_{\mathbf{Z}}^*)$. From Lemma 13 we have that

$$\begin{aligned} Q_1 &\geq \frac{dc_3}{\|\Sigma_{\mathbf{W}}^*\|^2 e^{1+2(r_0/d)}} \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\|^2 + \frac{c_2}{\|\Sigma_{\mathbf{W}}^*\|} \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2 \\ &= L_{\mathbf{W}^*}^2 B_2^2 + M_{\mathbf{W}^*}^2 A_2^2, \end{aligned}$$

where

$$L_{\mathbf{W}^*}^2 = \frac{dc_3}{\|\Sigma_{\mathbf{W}}^*\|^2 e^{1+2(r_0/d)}} \quad \text{and} \quad M_{\mathbf{W}^*}^2 = \frac{c_2}{\|\Sigma_{\mathbf{W}}^*\|}.$$

Analogously using the same lemmas we obtain

$$Q_2 \geq L_{\mathbf{Z}^*}^2 B_1^2 + M_{\mathbf{Z}^*}^2 A_1^2,$$

where

$$L_{\mathbf{Z}^*}^2 = \frac{nc_3}{\|\Sigma_{\mathbf{Z}}^*\|^2 e^{1+2(r_0/n)}} \quad \text{and} \quad M_{\mathbf{Z}^*}^2 = \frac{c_2'}{\|\Sigma_{\mathbf{Z}}^*\|}.$$

Using Cauchy-Schwarz inequality, we therefore have that

$$\sqrt{Q_1} \geq \frac{1}{\sqrt{2}}(L_{\mathbf{W}^*}B_2 + M_{\mathbf{W}^*}A_2) \quad \text{and} \quad \sqrt{Q_2} \geq \frac{1}{\sqrt{2}}(L_{\mathbf{Z}^*}B_1 + M_{\mathbf{Z}^*}A_1),$$

Consequently,

$$\sqrt{Q_1 Q_2} \geq \frac{1}{2} (M_{\mathbf{W}^*} M_{\mathbf{Z}^*} A_1 A_2 + L_{\mathbf{W}^*} L_{\mathbf{Z}^*} B_1 B_2 + L_{\mathbf{W}^*} M_{\mathbf{Z}^*} B_2 A_1 + M_{\mathbf{W}^*} L_{\mathbf{Z}^*} A_2 B_1).$$

Thus we have that

$$\begin{aligned} \rho_0 &= \frac{\left(\tau_0 \|\mathbf{X}\| + \frac{C_1 C_2 \tau_0}{2} \right) A_1 A_2 + (nd\tau_0 B_1 B_2 + dC_1 \tau_0 B_2 A_1 + nC_2 \tau_0 A_2 B_1)/2}{\sqrt{Q_1 Q_2}} \\ &\leq 2 \frac{\left(\tau_0 \|\mathbf{X}\| + \frac{C_1 C_2 \tau_0}{2} \right) A_1 A_2 + (nd\tau_0 B_1 B_2 + dC_1 \tau_0 B_2 A_1 + nC_2 \tau_0 A_2 B_1)/2}{M_{\mathbf{W}^*} M_{\mathbf{Z}^*} A_1 A_2 + L_{\mathbf{W}^*} L_{\mathbf{Z}^*} B_1 B_2 + L_{\mathbf{W}^*} M_{\mathbf{Z}^*} B_2 A_1 + M_{\mathbf{W}^*} L_{\mathbf{Z}^*} A_2 B_1} \\ &\leq 2 \max \left\{ \frac{\tau_0 \|\mathbf{X}\| + \frac{\tau_0 C_1 C_2}{2}}{M_{\mathbf{W}^*} M_{\mathbf{Z}^*}}, \frac{nd\tau_0}{2L_{\mathbf{W}^*} L_{\mathbf{Z}^*}}, \frac{dC_1 \tau_0}{2L_{\mathbf{W}^*} M_{\mathbf{Z}^*}}, \frac{nC_2 \tau_0}{2M_{\mathbf{W}^*} L_{\mathbf{Z}^*}} \right\} \quad (\text{by Lemma 21}), \end{aligned}$$

where all the quantities $C_1, C_2, L_{\mathbf{W}^*}, L_{\mathbf{Z}^*}, M_{\mathbf{W}^*}, M_{\mathbf{Z}^*}$ depend on r_0 . For the condition as stated in [2] (see Theorem 4) to be satisfied we need to be able to find an $r_0 > 0$ such that the following

$$\max \left\{ \frac{\tau_0 \|\mathbf{X}\| + \frac{\tau_0 C_1 C_2}{2}}{M_{\mathbf{W}^*} M_{\mathbf{Z}^*}}, \frac{nd\tau_0}{2L_{\mathbf{W}^*} L_{\mathbf{Z}^*}}, \frac{dC_1 \tau_0}{2L_{\mathbf{W}^*} M_{\mathbf{Z}^*}}, \frac{nC_2 \tau_0}{2M_{\mathbf{W}^*} L_{\mathbf{Z}^*}} \right\} < 1$$

holds. Denoting $\gamma_0 = (1 + e^{-1})/4$, if r_0 is small enough, and in particular if $r_0 \rightarrow 0$, then we observe that each term in the max remains bounded above by its corresponding counterpart in the condition in (30), since $C_1 \leq 2\|\mu_{\mathbf{Z}}^*\|, C_2 \leq 2\|\mu_{\mathbf{W}}^*\|, L_{\mathbf{Z}^*}^2 \rightarrow nc_3/(e\|\Sigma_{\mathbf{Z}}^*\|^2), L_{\mathbf{W}^*}^2 \rightarrow dc_3/(e\|\Sigma_{\mathbf{W}}^*\|^2), M_{\mathbf{W}^*}^2 \rightarrow \gamma_0/\|\Sigma_{\mathbf{W}}^*\|$ and $M_{\mathbf{Z}^*}^2 \rightarrow \gamma_0/\|\Sigma_{\mathbf{Z}}^*\|$ by Lemma 15 (after plugging back the values c_2, c_2' and c_3 from Lemma 13). Thus, it is sufficient to require that $(\mu_{\mathbf{W}}^*, \Sigma_{\mathbf{W}}^*, \mu_{\mathbf{Z}}^*, \Sigma_{\mathbf{Z}}^*) \in \mathcal{I}$ since we can then select $r_0 > 0$ such that $\text{GCorr}_{1/2}(r_0) \in (0, 2)$. Finally, we choose $r^* > 0$ be sufficiently small such that $q_{\mathbf{Z}}^{(0)} \in \mathcal{B}_{\mathbf{Z}}^*(r^*)$ implies $D_{\text{KL}, 1/2}(q_{\mathbf{Z}}^{(0)} \| q_{\mathbf{Z}}^*) \leq r_0/2$ (this is possible because of Remark 1). $\square$

Remark 1. As shown in [2], above proof yields $D_{\text{KL}, 1/2}(q_{\mathbf{W}}^{(t)} \| q_{\mathbf{W}}^*) \leq C'_{\mathbf{W}} \kappa^t$ and $D_{\text{KL}, 1/2}(q_{\mathbf{Z}}^{(t)} \| q_{\mathbf{Z}}^*) \leq C'_{\mathbf{Z}} \kappa^t$ for some constants $C'_{\mathbf{W}}, C'_{\mathbf{Z}} > 0$, which is a stronger result than the exponential convergence of KL as stated in Theorem 12. The key reason why we can state Theorem 12 entirely in terms of KL divergence is due a recent work [23]. In particular, since $q_{\mathbf{Z}}^{(0)}$ and $q_{\mathbf{Z}}^*$ are matrix normal distributions with a special covariance structure that

imposes row independence, the KL divergence decomposes as a sum over independent rows each of which is a multivariate normal. This tensorization property of KL implies that controlling $\text{KL}(q_{\mathbf{Z}}^* \parallel q_{\mathbf{Z}}^{(0)})$ is sufficient to bound the reverse $\text{KL}(q_{\mathbf{Z}}^{(0)} \parallel q_{\mathbf{Z}}^*)$, and hence the symmetrized KL required in the hypothesis of Theorem 4 in [2].

Theorem 12 has the following corollary regarding the convergence of the parameters of the matrix normal distributions $q_{\mathbf{Z}}^{(t)} = \mathcal{N}(\mu_{\mathbf{Z}}^{(t)}, I_n \otimes \Sigma_{\mathbf{Z}}^{(t)})$ and $q_{\mathbf{W}}^{(t)} = \mathcal{N}(\mu_{\mathbf{W}}^{(t)}, I_d \otimes \Sigma_{\mathbf{W}}^{(t)})$.

Corollary 3 (Convergence of parameters). *Under the same assumptions as Theorem 12, one has, for any $t \geq 1$,*

$$\begin{aligned} \max\{\|\mu_{\mathbf{Z}}^{(t+1)} - \mu_{\mathbf{Z}}^*\|, \|\Sigma_{\mathbf{Z}}^{(t+1)} - \Sigma_{\mathbf{Z}}^*\|\} &\leq A_{\mathbf{Z}}\kappa^{t/2}, \\ \max\{\|\mu_{\mathbf{W}}^{(t+1)} - \mu_{\mathbf{W}}^*\|, \|\Sigma_{\mathbf{W}}^{(t+1)} - \Sigma_{\mathbf{W}}^*\|\} &\leq A_{\mathbf{W}}\kappa^{t/2}, \end{aligned}$$

where $A_{\mathbf{Z}}, A_{\mathbf{W}}$, are positive constants, independent of t . The constant κ is the same as in Theorem 12.

Proof of Corollary 3. Let $r_0 > 0$ such that $\text{GCorr}_{1/2}(r_0) \in (0, 2)$ which exists. Suppose, r^* be sufficiently small such that $q_{\mathbf{Z}}^{(0)} \in \mathcal{B}_{\mathbf{Z}}^*(r^*)$ implies $D_{\text{KL}, 1/2}(q_{\mathbf{Z}}^{(0)} \parallel q_{\mathbf{Z}}^*) \leq r_0/2$ (this is possible because of Remark 1). Following [2], it can be shown that $q_{\mathbf{Z}}^{(t)} \in \mathcal{B}_{\mathbf{Z}}^*(r_0)$ for all $t \geq 1$. Recall Lemma 13 and set

$$m_{\mathbf{Z}} = \min \left\{ \frac{nc_3}{\|\Sigma_{\mathbf{Z}}^*\|^2 e^{1+(2r_0/n)}}, \frac{c'_2}{\|\Sigma_{\mathbf{Z}}^*\|} \right\}.$$

Using the lower bounds in Lemma 13 and Remark 1, we have that

$$m_{\mathbf{Z}} \|\Sigma_{\mathbf{Z}}^{(t+1)} - \Sigma_{\mathbf{Z}}^*\|^2 \leq D_{\text{KL}, 1/2}(q_{\mathbf{Z}}^{(t+1)} \parallel q_{\mathbf{Z}}^*) \leq C'_{\mathbf{Z}}\kappa^t,$$

and

$$m_{\mathbf{Z}} \|\mu_{\mathbf{Z}}^{(t+1)} - \mu_{\mathbf{Z}}^*\|^2 \leq D_{\text{KL}, 1/2}(q_{\mathbf{Z}}^{(t+1)} \parallel q_{\mathbf{Z}}^*) \leq C'_{\mathbf{Z}}\kappa^t.$$

Therefore,

$$\max\{\|\mu_{\mathbf{Z}}^{(t+1)} - \mu_{\mathbf{Z}}^*\|, \|\Sigma_{\mathbf{Z}}^{(t+1)} - \Sigma_{\mathbf{Z}}^*\|\} \leq A_{\mathbf{Z}}\kappa^{t/2},$$

where $A_{\mathbf{Z}} = \sqrt{C'_{\mathbf{Z}}/m_{\mathbf{Z}}}$, a positive constant. The proof of the other inequality is similar. $\square$

We now state a theoretical result which guarantees existence of at least one solution to the optimization problem in (21) for BPCA when τ is fixed as in section 3.3.

3.4.3.2. Proof of auxiliary results. Suppose $q_{\mathbf{W}}^* = \mathcal{N}(\mu_{\mathbf{W}}^*, I \otimes \Sigma_{\mathbf{W}}^*)$ and $q_{\mathbf{Z}}^* = \mathcal{N}(\mu_{\mathbf{Z}}^*, I \otimes \Sigma_{\mathbf{Z}}^*)$. The goal of next proposition is to show that the application of triangle inequality in Lemma 15 cannot be avoided and as a consequence of the supremum in (28) could be attained on the boundary of the product of the KL balls $\mathcal{B}_{\mathbf{Z}}^*(r_0) \times \mathcal{B}_{\mathbf{W}}^*(r_0)$.

Proposition 6. *If $(q_{\mathbf{Z}}, q_{\mathbf{W}}) = (\mathcal{N}(\mu_{\mathbf{Z}}, I_d \otimes \Sigma_{\mathbf{Z}}^*), \mathcal{N}(\mu_{\mathbf{W}}, I_d \otimes \Sigma_{\mathbf{W}}^*))$, with $(q_{\mathbf{Z}}, q_{\mathbf{W}}) \in \mathcal{B}_{\mathbf{Z}}^*(r_0) \times \mathcal{B}_{\mathbf{W}}^*(r_0)$, then we have $\|\mu_{\mathbf{Z}}\| \leq \|\mu_{\mathbf{Z}}^*\| + \sqrt{2r_0/L}$ and $\|\mu_{\mathbf{Z}}\| \leq \|\mu_{\mathbf{Z}}^*\| + \sqrt{2r_0/M}$, where $(L, M) =$*

$(1/\lambda_{\max}((\Sigma_{\mathbf{Z}}^*)^{-1}), 1/\lambda_{\max}((\Sigma_{\mathbf{W}}^*)^{-1}))$. Furthermore, the equality is achieved, that is, there exists $(q_{\mathbf{Z}}^0, q_{\mathbf{W}}^0) = (\mathcal{N}(\mu_{\mathbf{Z}}^0, I_d \otimes \Sigma_{\mathbf{Z}}^*), \mathcal{N}(\mu_{\mathbf{W}}^0, I_d \otimes \Sigma_{\mathbf{W}}^*))$ such that $(q_{\mathbf{Z}}^0, q_{\mathbf{W}}^0) \in \mathcal{B}_{\mathbf{Z}}^*(r_0) \times \mathcal{B}_{\mathbf{W}}^*(r_0)$ with $\|\mu_{\mathbf{Z}}^0\| = \|\mu_{\mathbf{Z}}^*\| + \sqrt{2r_0/L}$ and $\|\mu_{\mathbf{W}}^0\| = \|\mu_{\mathbf{W}}^*\| + \sqrt{2r_0/M}$.

Proof. Note that,

$$\text{KL}(q_{\mathbf{Z}}^* \| q_{\mathbf{Z}}^0) = \frac{1}{2} (\text{vec}(\mu_{\mathbf{Z}}^0) - \text{vec}(\mu_{\mathbf{Z}}^*))' (\Sigma_{\mathbf{Z}}^*)^{-1} (\text{vec}(\mu_{\mathbf{Z}}^0) - \text{vec}(\mu_{\mathbf{Z}}^*)) \leq \frac{\lambda_{\max}((\Sigma_{\mathbf{Z}}^*)^{-1})}{2} \|\mu_{\mathbf{Z}}^0 - \mu_{\mathbf{Z}}^*\|^2.$$

$q_{\mathbf{Z}}^0 \in \mathcal{B}_{\mathbf{Z}}^*(r_0)$ if and only if $\|\mu_{\mathbf{Z}}^0 - \mu_{\mathbf{Z}}^*\| \leq \sqrt{2r_0/L}$. It is easy to construct a point $\mu_{\mathbf{Z}}^0$ on the boundary of the Euclidean ball centred at $\mu_{\mathbf{Z}}^*$ for which the equality holds, by choosing $\mu_{\mathbf{Z}}^0$ to be one of the intersections of the line containing 0 and $\mu_{\mathbf{Z}}^*$ and ball. For this point, $\|\mu_{\mathbf{Z}}^0\| = \|\mu_{\mathbf{Z}}^*\| + \sqrt{2r_0/L}$, since they are all collinear. The construction of $\mu_{\mathbf{W}}^0$ is analogous. $\square$

The Matrix Normal Distribution and its properties. We follow the exposition in [11]. A matrix normal distribution with mean parameter $\mu \in \mathbb{R}^{d \times n}$ and covariance structure $\Sigma^{(c)} \otimes \Sigma^{(r)}$ is defined through the following probability density function (PDF):

$$f(\mathbf{U}; \mu, \Sigma^{(c)} \otimes \Sigma^{(r)}) = (2\pi)^{-dn/2} |\Sigma^{(c)}|^{-n/2} |\Sigma^{(r)}|^{-d/2} \exp \left(-\frac{1}{2} \text{tr} \left(\Sigma^{(c)-1} (\mathbf{U} - \mu) (\Sigma^{(r)-1})' (\mathbf{U} - \mu)' \right) \right),$$

where $\Sigma^{(c)}$ and $\Sigma^{(r)}$ are two symmetric positive-definite matrices that denote the covariance across columns and rows of $\mathbf{U}$, respectively. We use the notation $\mathbf{U} \sim \mathcal{N}(\mu, \Sigma^{(c)} \otimes \Sigma^{(r)})$.

We now spend some words to elucidate the density function above. The vectorization operator ‘vec’ transforms a matrix $\mathbf{U} \in \mathbb{R}^{d \times n}$ into the column consisting of the column vectors of $\mathbf{U}$, i.e. $\text{vec}(\mathbf{U}) = (u'_{.1} \dots, u'_{.n})'$, where $u_{.j}$ is the j -th column of the matrix $\mathbf{U}$. The mean and covariance of the matrix $\mathbf{U}$ is defined to be the mean and covariance of this random vector $\text{vec}(\mathbf{U})$, which is a multivariate normal in our case and we write $\text{vec}(\mathbf{U}) \sim \mathcal{N}_{dn}(\text{vec}(\mu), \Sigma^{(c)} \otimes \Sigma^{(r)})$. The notation significantly simplifies if the columns of $\{u_{.j}\}_{j=1}^n$ are n i.i.d. observations with a shared covariance matrix Σ_U , and in that case $\text{cov}(\mathbf{U}) = \Sigma_U \otimes I_n$. In such cases, a key result that is used repeatedly is provided by the following proposition (see [17]).

Proposition 7. *Let $\mathbf{U}$ be a given $d \times n$ matrix valued random variable. If $\mathbf{U} \sim \mathcal{N}(\mu, I_d \otimes \Sigma)$, then $\mathbf{U}' \sim \mathcal{N}(\mu', \Sigma \otimes I_d)$ and the second non-central moments are given by*

$$\mathbb{E}(\mathbf{U}' \Pi_1 \mathbf{U}) = \text{tr}(\Pi_1) \Sigma + \mu' \Pi_1 \mu,$$

$$\mathbb{E}(\mathbf{U} \Pi_2 \mathbf{U}') = \text{tr}(\Pi_2 \Sigma) I_d + \mu \Pi_2 \mu',$$

where Π_1, Π_2 are $d \times d$ and $n \times n$ symmetric positive definite matrices, respectively.

For two $d \times k$ matrix normal distributions $q_{\mathbf{W}} = \mathcal{N}(\mu_{\mathbf{W}}, I_d \otimes \Sigma_{\mathbf{W}})$ and $q_{\mathbf{W}}^* = \mathcal{N}(\mu_{\mathbf{W}}^*, I_d \otimes \Sigma_{\mathbf{W}}^*)$, we define the quantity:

$$\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) = d \left(\text{tr}([\Sigma_{\mathbf{W}}^*]^{-1} \Sigma_{\mathbf{W}}) - k - \log \det([\Sigma_{\mathbf{W}}^*]^{-1} \Sigma_{\mathbf{W}}) \right).$$

We now record the following expression of KL-divergence between $q_{\mathbf{W}}$ and $q_{\mathbf{W}}^*$:

$$\begin{aligned} \text{KL}(q_{\mathbf{W}} \| q_{\mathbf{W}}^*) &= \frac{1}{2} \left(\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + (\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*))' \right. \\ &\quad \left. \cdot (I_d \otimes \Sigma_{\mathbf{W}}^*)^{-1} (\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*)) \right). \end{aligned} \quad (40)$$

Details of the derivation of log-posterior. In this appendix, our goal is to derive the log-posterior $\log \pi(\mathbf{W}, \mathbf{Z})$. We recall the BPCA model with normal error terms in terms of matrix-valued random variables as:

$$\mathbf{X} = \mathbf{Z}\mathbf{W}' + \mathbf{E}, \quad (41)$$

where the rows of the $n \times d$ data matrix $\mathbf{X}$ are observations x_i , the $n \times k$ matrix $\mathbf{Z}$ contains latent variables z_i , and $\mathbf{E}$ is an $n \times d$ noise matrix whose rows are scaled Gaussian noise vectors $\tau_0^{-1/2} \epsilon_i$ for $i = 1, 2, \dots, n$. The loading matrix $\mathbf{W}$ has size $d \times k$, and we fix the noise precision $\tau_0 > 0$ for simplicity. The likelihood and priors are given by (see [3]):

$$\begin{aligned} p(\mathbf{X} | \mathbf{W}, \mathbf{Z}) &= \mathcal{N}(\mathbf{X}; \mathbf{Z}\mathbf{W}', \tau_0^{-1} I_n \otimes I_d), \\ p(\mathbf{W}) &= \mathcal{N}(\mathbf{W}; \mathbf{0}, I_d \otimes \Lambda^{-1}), \quad p(\mathbf{Z}) = \mathcal{N}(\mathbf{Z}; \mathbf{0}, I_n \otimes I_k). \end{aligned}$$

where $\Lambda = \text{diag}(\sigma_1^{-2}, \dots, \sigma_k^{-2})$ is a diagonal matrix of precision hyperparameters. Assume that, $\Lambda \not\propto I_k$. In the above equation $\mathcal{N}(\mathbf{X}; \mathbf{Z}\mathbf{W}', \tau_0^{-1} I_n \otimes I_d)$ denotes a matrix normal density and an interpretation follows in [section 3.4.3.2](#). The posterior density is thus:

$$\pi(\mathbf{W}, \mathbf{Z}) \propto p(\mathbf{X} | \mathbf{W}, \mathbf{Z}) p(\mathbf{W}) p(\mathbf{Z}). \quad (42)$$

Therefore, given the hyper-parameter Λ , the logarithm of the posterior can be obtained as:

$$\begin{aligned} \log \pi(\mathbf{W}, \mathbf{Z}) &= \log p(\mathbf{X} | \mathbf{W}, \mathbf{Z}) + \log p(\mathbf{W}) + \log p(\mathbf{Z}) + \text{cst} \\ &= \tau_0 \text{tr}(\mathbf{W}\mathbf{Z}'\mathbf{X}) - \frac{1}{2} \tau_0 \text{tr}(\mathbf{W}\mathbf{Z}'\mathbf{Z}\mathbf{W}') - \frac{1}{2} \text{tr}(\mathbf{W}\Lambda\mathbf{W}') - \frac{1}{2} \text{tr}(\mathbf{Z}\mathbf{Z}') + \text{cst}, \end{aligned}$$

where at the final step we used properties of the trace operator and the constant term does not depend on the parameters $\mathbf{W}, \mathbf{Z}$.

Details of the derivation of ELBO. Let $q = (\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}}, \mu_{\mathbf{Z}}, \Sigma_{\mathbf{Z}}) \in \mathbb{R}^d \times \mathbb{S}_k^+ \times \mathbb{R}^n \times \mathbb{S}_k^+$ be given. To derive an explicit expression of $\text{ELBO}(q)$, we first recall the definition

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(\mathbf{W}, \mathbf{Z}, \mathbf{X})] - \mathbb{E}_q[\log q], \quad (43)$$

where $q_{\mathbf{W}} = \mathcal{N}(\mu_{\mathbf{W}}, I_d \otimes \Sigma_{\mathbf{W}})$, $q_{\mathbf{Z}} = \mathcal{N}(\mu_{\mathbf{Z}}, I_n \otimes \Sigma_{\mathbf{Z}})$ and $q = q_{\mathbf{W}} q_{\mathbf{Z}}$. Expanding we have that

$$\text{ELBO}(q) = \mathbb{E}_q[\log \pi(\mathbf{W}, \mathbf{Z})] + \mathbb{E}_q[-\log q_{\mathbf{W}}] + \mathbb{E}_q[-\log q_{\mathbf{Z}}] + \text{cst}.$$

The entropy terms in the above expressions are $d/2 \log \det(\Sigma_{\mathbf{W}})$ and $n/2 \log \det(\Sigma_{\mathbf{Z}})$, respectively. Using (45) in the proof of Lemma 12, we conclude that

$$\begin{aligned} \text{ELBO}(q) &= \tau_0 \text{tr}(\mu_{\mathbf{W}} \mu'_{\mathbf{Z}} \mathbf{X}) - \frac{\tau_0}{2} \text{tr}(\Gamma_{\mathbf{W}} \Gamma_{\mathbf{Z}}) - \frac{d}{2} \text{tr}(\Lambda \Sigma_{\mathbf{W}}) - \frac{1}{2} \text{tr}(\mu_{\mathbf{W}} \Lambda \mu'_{\mathbf{W}}) \\ &\quad - \frac{1}{2} \text{tr}(\Gamma_{\mathbf{Z}}) + \frac{d}{2} \log \det(\Sigma_{\mathbf{W}}) + \frac{n}{2} \log \det(\Sigma_{\mathbf{Z}}) + \text{cst}, \end{aligned}$$

where

$$\Gamma_{\mathbf{W}} = d\Sigma_{\mathbf{W}} + \mu'_{\mathbf{W}} \mu_{\mathbf{W}} \text{ and } \Gamma_{\mathbf{Z}} = n\Sigma_{\mathbf{Z}} + \mu'_{\mathbf{Z}} \mu_{\mathbf{Z}}.$$

Lemma 11. *Let $\{x_n\} \subset \mathbb{R}^\ell$ be a sequence of nonzero vectors such that for all $n \in \mathbb{N}$, we have $\|x_n - x\| \leq cr^n$ for some positive constant c and $r \in (0,1)$ fixed. Then we have that*

$$\left\| \frac{x_n}{\|x_n\|} - \frac{x}{\|x\|} \right\| \leq \frac{2cr^n}{\|x\|},$$

where $x \neq 0$.

Proof.

$$\begin{aligned} \left\| \frac{x_n}{\|x_n\|} - \frac{x}{\|x\|} \right\| &= \left\| \frac{x_n(\|x\| - \|x_n\|) + \|x_n\|(x_n - x)}{\|x_n\| \cdot \|x\|} \right\| \\ &\leq \frac{1}{\|x\|} \|\|x_n\| - \|x\|\| + \frac{1}{\|x\|} \|x_n - x\| \\ &\leq \frac{2}{\|x\|} \|x_n - x\|. \end{aligned}$$

□

Lemma 12. *Let $q_{\mathbf{W}}, q_{\mathbf{Z}}, q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*$ denote the matrix normal distributions $\mathcal{N}(\mu_{\mathbf{W}}, I_d \otimes \Sigma_{\mathbf{W}}), \mathcal{N}(\mu_{\mathbf{Z}}, I_n \otimes \Sigma_{\mathbf{Z}})$, and $\mathcal{N}(\mu_{\mathbf{W}}^*, I_d \otimes \Sigma_{\mathbf{W}}^*), \mathcal{N}(\mu_{\mathbf{Z}}^*, I_n \otimes \Sigma_{\mathbf{Z}}^*)$. Let us denote the quantities $A_1 = \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\|, A_2 = \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|, B_1 = \|\Sigma_{\mathbf{Z}} - \Sigma_{\mathbf{Z}}^*\|, B_2 = \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\|, C_1 = \|\mu_{\mathbf{Z}}\| + \|\mu_{\mathbf{Z}}^*\|, C_2 = \|\mu_{\mathbf{W}}\| + \|\mu_{\mathbf{W}}^*\|$. We have that*

$$\begin{aligned} |\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})| &\leq \left(\tau_0 \|\mathbf{X}\| + \frac{C_1 C_2 \tau_0}{2} \right) A_1 A_2 \\ &\quad + \frac{1}{2} \left(nd \tau_0 B_1 B_2 + d C_1 \tau_0 B_2 A_1 + n C_2 \tau_0 A_2 B_1 \right). \end{aligned}$$

where the quantity $\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})$ is defined in (26).

Proof. For any two non-negative integrable functions $f(\mathbf{W}), g(\mathbf{Z})$, we define the following functional

$$\delta(f, g) = \iint f(\mathbf{W}) g(\mathbf{Z}) \log \pi(\mathbf{W}, \mathbf{Z}) \, d\mathbf{W} \, d\mathbf{Z},$$

provided the quantity is finite. By definition, we have that

$$\begin{aligned}\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}}) &= \iint (q_{\mathbf{W}}(\mathbf{W}) - q_{\mathbf{W}}^*(\mathbf{W})) (q_{\mathbf{Z}}(\mathbf{Z}) - q_{\mathbf{Z}}^*(\mathbf{Z})) \log \pi(\mathbf{W}, \mathbf{Z}) \, d\mathbf{W} \, d\mathbf{Z} \\ &= \delta(q_{\mathbf{W}}, q_{\mathbf{Z}}) - \delta(q_{\mathbf{W}}^*, q_{\mathbf{Z}}) - \delta(q_{\mathbf{W}}, q_{\mathbf{Z}}^*) + \delta(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*).\end{aligned}\quad (44)$$

Let us now focus on

$$\begin{aligned}\delta(q_{\mathbf{W}}, q_{\mathbf{Z}}) &= \mathbb{E}[\log p(\mathbf{X}|\mathbf{W}, \mathbf{Z})] + \mathbb{E}[\log p(\mathbf{W})] + \mathbb{E}[\log p(\mathbf{Z})] + \text{cst} \\ &= \tau_0 \text{tr}(\mu_{\mathbf{W}} \mu_{\mathbf{Z}}' \mathbf{X}) - \frac{\tau_0}{2} \mathbb{E}[\text{tr}(\mathbf{W} \mathbf{Z}' \mathbf{Z} \mathbf{W})] - \frac{1}{2} \mathbb{E}[\text{tr}(\mathbf{W} \Lambda \mathbf{W}')] - \frac{1}{2} \mathbb{E}[\text{tr}(\mathbf{Z} \mathbf{Z}')] + \text{cst}.\end{aligned}$$

Detailed derivation of the log-posterior can be found in [section 3.4.3.2](#). To compute the expectations of the above quadratic forms, let us define the following quantities for convenience

$$\Gamma_{\mathbf{W}} = d\Sigma_{\mathbf{W}} + \mu_{\mathbf{W}}' \mu_{\mathbf{W}} \text{ and } \Gamma_{\mathbf{Z}} = n\Sigma_{\mathbf{Z}} + \mu_{\mathbf{Z}}' \mu_{\mathbf{Z}}.$$

Now from [Lemma 20](#) it follows that

$$\begin{aligned}\mathbb{E}[\text{tr}(\mathbf{W} \mathbf{Z}' \mathbf{Z} \mathbf{W}')] &= \text{tr}(\Gamma_{\mathbf{W}} \Gamma_{\mathbf{Z}}), \\ \mathbb{E}[\text{tr}(\mathbf{W} \Lambda \mathbf{W}')] &= d \text{tr}(\Lambda \Sigma_{\mathbf{W}}) + \text{tr}(\mu_{\mathbf{W}} \Lambda \mu_{\mathbf{W}}').\end{aligned}$$

Therefore, we have that

$$\mathbb{E}[\log \pi(\mathbf{W}, \mathbf{Z})] = \tau_0 \text{tr}(\mu_{\mathbf{W}} \mu_{\mathbf{Z}}' \mathbf{X}) - \frac{\tau_0}{2} \text{tr}(\Gamma_{\mathbf{W}} \Gamma_{\mathbf{Z}}) - \frac{d}{2} \text{tr}(\Lambda \Sigma_{\mathbf{W}}) - \frac{1}{2} \text{tr}(\mu_{\mathbf{W}} \Lambda \mu_{\mathbf{W}}') - \frac{1}{2} \text{tr}(\Gamma_{\mathbf{Z}}) + \text{cst}. \quad (45)$$

Combining everything in (44) (note that only the cross-terms survive),

$$\begin{aligned}\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}}) &= \delta(q_{\mathbf{W}}, q_{\mathbf{Z}}) - \delta(q_{\mathbf{W}}^*, q_{\mathbf{Z}}) - \delta(q_{\mathbf{W}}, q_{\mathbf{Z}}^*) + \delta(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*) \\ &= \tau_0 \text{tr}((\mu_{\mathbf{W}} \mu_{\mathbf{Z}}' - \mu_{\mathbf{W}}^* \mu_{\mathbf{Z}}' - \mu_{\mathbf{W}} [\mu_{\mathbf{Z}}^*]' + \mu_{\mathbf{W}}^* [\mu_{\mathbf{Z}}^*]') \mathbf{X}) \\ &\quad - \frac{\tau_0}{2} \text{tr}((\Gamma_{\mathbf{W}} \Gamma_{\mathbf{Z}}' - \Gamma_{\mathbf{W}}^* \Gamma_{\mathbf{Z}}' - \Gamma_{\mathbf{W}} [\Gamma_{\mathbf{Z}}^*]' + \Gamma_{\mathbf{W}}^* [\Gamma_{\mathbf{Z}}^*]')) \\ &= \tau_0 \text{tr}((\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*) (\mu_{\mathbf{Z}}' - [\mu_{\mathbf{Z}}^*]') \mathbf{X}) \\ &\quad - \frac{\tau_0}{2} \text{tr}((\Gamma_{\mathbf{W}} - \Gamma_{\mathbf{W}}^*) (\Gamma_{\mathbf{Z}}' - [\Gamma_{\mathbf{Z}}^*]'))\end{aligned}$$

Using [Lemma 19](#) and the submultiplicativity of the Frobenius norm, we get that

$$|\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})| \leq \tau_0 \|\mathbf{X}\| \cdot \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\| \cdot \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\| + (\tau_0/2) \|\Gamma_{\mathbf{W}} - \Gamma_{\mathbf{W}}^*\| \cdot \|\Gamma_{\mathbf{Z}} - \Gamma_{\mathbf{Z}}^*\|. \quad (46)$$

We can simplify the right side further. Note that by triangle inequality

$$\begin{aligned}\|\Gamma_{\mathbf{W}} - \Gamma_{\mathbf{W}}^*\| &\leq d \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\| + \|\mu_{\mathbf{W}}' \mu_{\mathbf{W}} - [\mu_{\mathbf{W}}^*]' \mu_{\mathbf{W}}^*\| \\ &\leq d \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\| + (\|\mu_{\mathbf{W}}\| + \|\mu_{\mathbf{W}}^*\|) \cdot \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|,\end{aligned}$$

since $\mu'_{\mathbf{W}}\mu_{\mathbf{W}} - [\mu_{\mathbf{W}}^*]'\mu_{\mathbf{W}}^* = \mu'_{\mathbf{W}}(\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*) + (\mu'_{\mathbf{W}} - [\mu_{\mathbf{W}}^*]')\mu_{\mathbf{W}}^*$, and analogously,

$$\begin{aligned}\|\Gamma_{\mathbf{Z}} - \Gamma_{\mathbf{Z}}^*\| &\leq n\|\Sigma_{\mathbf{Z}} - \Sigma_{\mathbf{Z}}^*\| + \|\mu'_{\mathbf{Z}}\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\mu_{\mathbf{Z}}^*\| \\ &\leq n\|\Sigma_{\mathbf{Z}} - \Sigma_{\mathbf{Z}}^*\| + (\|\mu_{\mathbf{Z}}\| + \|\mu_{\mathbf{Z}}^*\|) \cdot \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\|.\end{aligned}$$

Combining everything in (46) we obtain

$$\begin{aligned}|\Delta_{q^*}(q_{\mathbf{W}}, q_{\mathbf{Z}})| &\leq \tau_0\|\mathbf{X}\|A_1A_2 + \frac{\tau_0}{2}(dB_2 + C_2A_2)(nB_1 + C_1A_1) \\ &= \tau_0\|\mathbf{X}\|A_1A_2 + \frac{1}{2}\left(C_1C_2\tau_0A_1A_2 + nd\tau_0B_1B_2\right. \\ &\quad \left.+ dC_1\tau_0B_2A_1 + nC_2\tau_0A_2B_1\right) \\ &= \left(\tau_0\|\mathbf{X}\| + \frac{C_1C_2\tau_0}{2}\right)A_1A_2 \\ &\quad + \frac{1}{2}\left(nd\tau_0B_1B_2 + dC_1\tau_0B_2A_1 + nC_2\tau_0A_2B_1\right).\end{aligned}$$

□

Lemma 13. *Let $r_0 > 0$ be given. For any $(q_{\mathbf{W}}, q_{\mathbf{Z}}) \in \mathcal{B}_{\mathbf{W}}^*(r_0) \times \mathcal{B}_{\mathbf{Z}}^*(r_0)$, we have the following lower bounds*

$$\begin{aligned}D_{KL,1/2}(q_{\mathbf{W}}\|q_{\mathbf{W}}^*) &\geq \frac{dc_3}{\|\Sigma_{\mathbf{W}}^*\|^2 e^{1+2(r_0/d)}}\|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\|^2 + \frac{c_2}{\|\Sigma_{\mathbf{W}}^*\|}\|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2, \\ D_{KL,1/2}(q_{\mathbf{Z}}\|q_{\mathbf{Z}}^*) &\geq \frac{nc_3}{\|\Sigma_{\mathbf{Z}}^*\|^2 e^{1+2(r_0/n)}}\|\Sigma_{\mathbf{Z}} - \Sigma_{\mathbf{Z}}^*\|^2 + \frac{c'_2}{\|\Sigma_{\mathbf{Z}}^*\|}\|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\|^2,\end{aligned}$$

where $c_2 = 1/4(1 + \exp(-(1 + (2r_0/d))))$, $c'_2 = 1/4(1 + \exp(-(1 + (2r_0/n))))$ and $c_3 = 1/2$.

Proof. We only prove the first lower bound as the other follows analogously. Let us write $\mathcal{P}_{\mathbf{W}^*} = I_d \otimes \Sigma_{\mathbf{W}}^*$. Therefore, by definition of KL-divergence between two matrix normal variables in (40), we have that

$$\begin{aligned}\text{KL}(q_{\mathbf{W}}\|q_{\mathbf{W}}^*) &= \frac{1}{2}\left(\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + (\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*))'\mathcal{P}_{\mathbf{W}^*}^{-1}(\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*))\right) \\ &\geq \frac{1}{2}\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + \frac{1}{2}\lambda_{\min}(\mathcal{P}_{\mathbf{W}^*}^{-1})\|\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*)\|_2^2 \\ &= \frac{1}{2}\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + \frac{1}{2\|\Sigma_{\mathbf{W}}^*\|_{\text{op}}}\|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2.\end{aligned}$$

Analogous inequality also holds for $\text{KL}(q_{\mathbf{W}}^*\|q_{\mathbf{W}})$. Thus we arrive at the following follower bound

$$\begin{aligned}D_{KL,1/2}(q_{\mathbf{W}}\|q_{\mathbf{W}}^*) &= \frac{1}{2}[\text{KL}(q_{\mathbf{W}}\|q_{\mathbf{W}}^*) + \text{KL}(q_{\mathbf{W}}^*\|q_{\mathbf{W}})] \\ &\geq \frac{1}{2}(\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + \mathcal{J}(\Sigma_{\mathbf{W}}^*, \Sigma_{\mathbf{W}})) + \frac{1}{4}\left(\frac{1}{\|\Sigma_{\mathbf{W}}^*\|_{\text{op}}} + \frac{1}{\|\Sigma_{\mathbf{W}}\|_{\text{op}}}\right)\|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2\end{aligned}$$

$$\begin{aligned}
&\geq \frac{1}{2} (\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + \mathcal{J}(\Sigma_{\mathbf{W}}^*, \Sigma_{\mathbf{W}})) \\
&\quad + \frac{1}{4} \left(\frac{1}{\|\Sigma_{\mathbf{W}}^*\|} + \frac{1}{\|\Sigma_{\mathbf{W}}^*\| e^{1+2(r_0/d)}} \right) \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2 \quad (\text{by Lemmas 23 and 14}) \\
&= \frac{1}{2} (\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + \mathcal{J}(\Sigma_{\mathbf{W}}^*, \Sigma_{\mathbf{W}})) + \frac{c_2}{\|\Sigma_{\mathbf{W}}^*\|} \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2.
\end{aligned}$$

To finish the proof we focus on the term $\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*)$. We have that

$$\begin{aligned}
\frac{1}{2} (\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*) + \mathcal{J}(\Sigma_{\mathbf{W}}^*, \Sigma_{\mathbf{W}})) &= \frac{d}{2} (\text{tr}([\Sigma_{\mathbf{W}}^*]^{-1} \Sigma_{\mathbf{W}}) + \text{tr}(\Sigma_{\mathbf{W}}^{-1} \Sigma_{\mathbf{W}}^*) - 2k) \\
&\geq \frac{d}{2 \|\Sigma_{\mathbf{W}}^*\|_{\text{op}} \|\Sigma_{\mathbf{W}}\|_{\text{op}}} \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\|^2 \quad (\text{by Lemma 17}) \\
&\geq \frac{dc_3}{\|\Sigma_{\mathbf{W}}^*\|^2 e^{1+2(r_0/d)}} \|\Sigma_{\mathbf{W}} - \Sigma_{\mathbf{W}}^*\|^2 \quad (\text{by Lemmas 23 and 15})
\end{aligned}$$

The second part of the Lemma follows analogously. $\square$

Lemma 14. *For every $r_0 > 0$, the following inclusions hold*

$$\begin{aligned}
\mathcal{B}_{\mathbf{W}}^*(r_0) &\subseteq \{(\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}}) : \|\Sigma_{\mathbf{W}}\|_{\text{op}} \leq \|\Sigma_{\mathbf{W}}^*\| e^{1+2(r_0/d)}, \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2 \leq 2r_0 e^{(1+2(r_0/d))} \|\Sigma_{\mathbf{W}}^*\|_{\text{op}}\}, \\
\mathcal{B}_{\mathbf{Z}}^*(r_0) &\subseteq \{(\mu_{\mathbf{Z}}, \Sigma_{\mathbf{Z}}) : \|\Sigma_{\mathbf{Z}}\|_{\text{op}} \leq \|\Sigma_{\mathbf{Z}}^*\| e^{1+2(r_0/n)}, \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\|^2 \leq 2r_0 e^{(1+2(r_0/n))} \|\Sigma_{\mathbf{Z}}^*\|_{\text{op}}\}.
\end{aligned}$$

Proof. We only prove the first inclusion. Recall the expression for KL divergence for matrix normal distributions in (40). The term $\mathcal{J}(\Sigma_{\mathbf{W}}, \Sigma_{\mathbf{W}}^*)$ can be re-written in the following way (see [2]):

$$\text{KL}(q_{\mathbf{W}}^* \| q_{\mathbf{W}}) = \frac{d}{2} \sum_{j=1}^k \psi(\lambda_j(\Delta)) + \frac{1}{2} (\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*))' (I_d \otimes \Sigma_{\mathbf{W}})^{-1} (\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*)),$$

where $\psi(x) = x - \log x - 1$, $\Delta = \Sigma_{\mathbf{W}}^{-1/2} \Sigma_{\mathbf{W}}^* \Sigma_{\mathbf{W}}^{-1/2}$ and $\{\lambda_j(\Delta)\}_{j=1}^d$ denote the eigenvalues in the sorted order. Therefore, $\text{KL}(q_{\mathbf{W}}^* \| q_{\mathbf{W}}) \leq r_0$ implies

$$\psi(\lambda_j(\Delta)) \leq 2r_0/d \quad \text{for } j = 1, \dots, d,$$

$$(\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*))' (I_d \otimes \Sigma_{\mathbf{W}})^{-1} (\text{vec}(\mu_{\mathbf{W}}) - \text{vec}(\mu_{\mathbf{W}}^*)) \leq 2r_0.$$

We have $\psi(x) \geq 0$ for every $x \in (0, \infty)$ with equality if and only if $x = 1$. Since $\lim_{x \rightarrow \infty} \psi(x) = \lim_{x \rightarrow 0} \psi(x) = \infty$, from Lemma 22 it follows that for any $c > 0$, there exists an interval $[\ell_c, u_c]$ containing 1 such that $\psi(x) \leq c$ if and only if $x \in [\ell_c, u_c]$. Using the same lemma, we also have that $\ell_c \in (e^{-(1+c)}, e^{-c})$. Thus $\psi(\lambda_j(\Delta)) \leq 2r_0/d$ implies $\lambda_j(\Delta) \geq e^{-(1+2(r_0/d))}$ for all j . Note that $\lambda_j(\Delta) = \lambda_j(\Sigma_{\mathbf{W}}^{-1} \Sigma_{\mathbf{W}}^*)$ holds as **AB** and **BA** have same set of positive eigen values. Now by Lemma 16 we have that

$$e^{-(1+2(r_0/d))} \leq \lambda_{\min}(\Sigma_{\mathbf{W}}^{-1} \Sigma_{\mathbf{W}}^*) \leq \lambda_{\min}(\Sigma_{\mathbf{W}}^{-1}) \lambda_{\max}(\Sigma_{\mathbf{W}}^*).$$

Also, the other condition implies

$$\lambda_{\min}(\Sigma_{\mathbf{W}}^{-1}) \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2 \leq 2r_0 \implies \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\|^2 \leq 2r_0 e^{(1+2(r_0/d))} \lambda_{\max}(\Sigma_{\mathbf{W}}^*).$$

$$\begin{aligned} \|\Sigma_{\mathbf{W}}\|_{\text{op}} &= \lambda_{\max}(\Sigma_{\mathbf{W}}) \\ &= \frac{1}{\lambda_{\min}(\Sigma_{\mathbf{W}}^{-1})} \\ &\leq \|\Sigma_{\mathbf{W}}^*\|_{\text{op}} e^{1+2(r_0/d)} \\ &\leq \|\Sigma_{\mathbf{W}}^*\| e^{1+2(r_0/d)} \quad (\text{using Lemma 23}). \end{aligned}$$

The upper bounds on $\|\Sigma_{\mathbf{Z}}\|_{\text{op}}$ and $\|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\|^2$ follow analogously. $\square$

Lemma 15. *Let $r_0 > 0$ be given. For any $(q_{\mathbf{W}}, q_{\mathbf{Z}}) \in \mathcal{B}_{\mathbf{W}}^*(r_0) \times \mathcal{B}_{\mathbf{Z}}^*(r_0)$, we have the following inequalities*

$$\|\mu_{\mathbf{W}}\| \leq \|\mu_{\mathbf{W}}^*\| + \|\Sigma_{\mathbf{W}}\|_{\text{op}}^{1/2} G_1(r_0) \text{ and } \|\mu_{\mathbf{Z}}\| \leq \|\mu_{\mathbf{Z}}^*\| + \|\Sigma_{\mathbf{Z}}\|_{\text{op}}^{1/2} G_2(r_0),$$

where G_1, G_2 are strictly positive functions such that $\lim_{r \rightarrow 0+} G_1(r) = \lim_{r \rightarrow 0+} G_2(r) = 0$.

Proof. The claims follow from Lemma 14 and triangle inequality:

$$\|\mu_{\mathbf{W}}\| \leq \|\mu_{\mathbf{W}}^*\| + \|\mu_{\mathbf{W}} - \mu_{\mathbf{W}}^*\| \leq \|\mu_{\mathbf{W}}^*\| + \|\Sigma_{\mathbf{W}}\|_{\text{op}}^{1/2} G_1(r_0),$$

$$\|\mu_{\mathbf{Z}}\| \leq \|\mu_{\mathbf{Z}}^*\| + \|\mu_{\mathbf{Z}} - \mu_{\mathbf{Z}}^*\| \leq \|\mu_{\mathbf{Z}}^*\| + \|\Sigma_{\mathbf{Z}}\|_{\text{op}}^{1/2} G_2(r_0),$$

with $G_1(r) = \sqrt{2re^{(1+2(r/d))}}$ and $G_2(r) = \sqrt{2re^{(1+2(r/n))}}$. $\square$

Lemma 16. *Let $\mathbf{A}$ and $\mathbf{B}$ are two $k \times k$ symmetric positive definite matrices. We have that*

$$\lambda_{\min}(\mathbf{AB}) \leq \lambda_{\min}(\mathbf{A}) \lambda_{\max}(\mathbf{B}).$$

Proof. Note that the matrices $\mathbf{B}^{1/2}, \mathbf{B}^{1/2} \mathbf{A} \mathbf{B}^{1/2}$ are symmetric. By Courant-Fischer-Weyl mini-max principle we have that

$$\begin{aligned} \lambda_{\min}(\mathbf{AB}) &= \lambda_{\min}(\mathbf{B}^{1/2} \mathbf{A} \mathbf{B}^{1/2}) \\ &= \min_{\substack{\mathcal{F} \subset \mathbb{R}^k \\ \dim(\mathcal{F})=1}} \left(\max_{x \in \mathcal{F} \setminus \{0\}} \frac{\langle \mathbf{B}^{1/2} \mathbf{A} \mathbf{B}^{1/2} x, x \rangle}{\langle x, x \rangle} \right) \\ &= \min_{\substack{\mathcal{F} \subset \mathbb{R}^k \\ \dim(\mathcal{F})=1}} \left(\max_{x \in \mathcal{F} \setminus \{0\}} \frac{\langle \mathbf{B}^{1/2} \mathbf{A} \mathbf{B}^{1/2} x, x \rangle}{\langle \mathbf{B}^{1/2} x, \mathbf{B}^{1/2} x \rangle} \frac{\langle \mathbf{B} x, x \rangle}{\langle x, x \rangle} \right). \end{aligned}$$

From above we readily obtain

$$\lambda_{\min}(\mathbf{AB}) \leq \lambda_{\max}(\mathbf{B}) \min_{\substack{\mathcal{F} \subseteq \mathbb{R}^k \\ \dim(\mathcal{F})=1}} \left(\max_{x \in \mathcal{F} \setminus \{0\}} \frac{\langle \mathbf{B}^{1/2} \mathbf{A} \mathbf{B}^{1/2} x, x \rangle}{\langle \mathbf{B}^{1/2} x, \mathbf{B}^{1/2} x \rangle} \right).$$

Let $\mathcal{F} = \text{span}(\mathbf{B}^{-1/2} v_1)$ where v_1 is an unit norm eigen vector of $\mathbf{A}$ which corresponds to $\lambda_{\min}(\mathbf{A})$. Evaluating on this subspace yields

$$\begin{aligned} \lambda_{\min}(\mathbf{AB}) &\leq \lambda_{\max}(\mathbf{B}) \left(\max_{\ell \in \mathbb{R} \setminus \{0\}} \frac{\langle \mathbf{B}^{1/2} \mathbf{A} \ell v_1, \mathbf{B}^{-1/2} \ell v_1 \rangle}{\langle \ell v_1, \ell v_1 \rangle} \right) \\ &= \lambda_{\max}(\mathbf{B}) \left(\frac{\langle \mathbf{B}^{1/2} \mathbf{A} v_1, \mathbf{B}^{-1/2} v_1 \rangle}{\langle v_1, v_1 \rangle} \right) \\ &= \lambda_{\max}(\mathbf{B}) \lambda_{\min}(\mathbf{A}) \left(\frac{\langle \mathbf{B}^{1/2} v_1, \mathbf{B}^{-1/2} v_1 \rangle}{\langle v_1, v_1 \rangle} \right) \\ &= \lambda_{\min}(\mathbf{A}) \lambda_{\max}(\mathbf{B}). \end{aligned}$$

□

Lemma 17. For two $k \times k$ symmetric positive definite matrices $\mathbf{A}$ and $\mathbf{B}$ we have that

$$\text{tr}(\mathbf{A}^{-1} \mathbf{B}) + \text{tr}(\mathbf{B}^{-1} \mathbf{A}) - 2k \geq \frac{1}{\lambda_{\max}(\mathbf{A}) \lambda_{\max}(\mathbf{B})} \|\mathbf{A} - \mathbf{B}\|^2.$$

Proof. Let us denote

$$\phi(\mathbf{A}, \mathbf{B}) = \text{tr}(\mathbf{A}^{-1} \mathbf{B}) + \text{tr}(\mathbf{B}^{-1} \mathbf{A}) - 2k.$$

We have that

$$\begin{aligned} \phi(\mathbf{A}, \mathbf{B}) &= \text{tr}(\mathbf{A}^{-1} \mathbf{B} - I_k) + \text{tr}(\mathbf{B}^{-1} \mathbf{A} - I_k) = \text{tr}(\mathbf{A}^{-1}(\mathbf{B} - \mathbf{A})) + \text{tr}(\mathbf{B}^{-1}(\mathbf{A} - \mathbf{B})) \\ &= \text{tr}((\mathbf{A}^{-1} - \mathbf{B}^{-1})(\mathbf{B} - \mathbf{A})), \end{aligned}$$

but note that since

$$\mathbf{A}^{-1} - \mathbf{B}^{-1} = \mathbf{A}^{-1} \mathbf{B} \mathbf{B}^{-1} - \mathbf{A}^{-1} \mathbf{A} \mathbf{B}^{-1} = \mathbf{A}^{-1}(\mathbf{B} - \mathbf{A}) \mathbf{B}^{-1},$$

we have that

$$\phi(\mathbf{A}, \mathbf{B}) = \text{tr}(\mathbf{A}^{-1}(\mathbf{B} - \mathbf{A}) \mathbf{B}^{-1}(\mathbf{B} - \mathbf{A})).$$

Observing that $\mathbf{M} = (\mathbf{B} - \mathbf{A}) \mathbf{B}^{-1}(\mathbf{B} - \mathbf{A}) = (\mathbf{B} - \mathbf{A})' \mathbf{B}^{-1}(\mathbf{B} - \mathbf{A})$, we clearly have that $\mathbf{M}$ is a positive definite (and symmetric) matrix. Thus, we have that

$$\begin{aligned} \phi(\mathbf{A}, \mathbf{B}) &\geq \frac{1}{\lambda_{\max}(\mathbf{A})} \text{tr}((\mathbf{B} - \mathbf{A}) \mathbf{B}^{-1}(\mathbf{B} - \mathbf{A})) \\ &= \frac{1}{\lambda_{\max}(\mathbf{A})} \text{tr}(\mathbf{B}^{-1}(\mathbf{B} - \mathbf{A})^2) \end{aligned}$$

$$\geq \frac{1}{\lambda_{\max}(\mathbf{A})\lambda_{\max}(\mathbf{B})} \text{tr}((\mathbf{B} - \mathbf{A})^2) \quad (\text{using Lemma 18}).$$

The conclusion follows since $\mathbf{B} - \mathbf{A}$ is symmetric, and therefore $\text{tr}((\mathbf{B} - \mathbf{A})^2) = \text{tr}((\mathbf{B} - \mathbf{A})(\mathbf{B} - \mathbf{A})') = \|\mathbf{B} - \mathbf{A}\|^2$. $\square$

Lemma 18. For two $k \times k$ symmetric positive definite matrices $\mathbf{A}$ and $\mathbf{B}$ we have that

$$\text{tr}(\mathbf{AB}) \geq \lambda_{\min}(\mathbf{A}) \text{tr}(\mathbf{B}).$$

Proof. The matrix $\mathbf{C} = \mathbf{A} - \lambda_{\min}(\mathbf{A})\mathbf{I}_k$ is positive semi definite. Thus, there exist matrices $\mathbf{U}, \mathbf{V}$ such that $\mathbf{C} = \mathbf{UU}'$ and $\mathbf{B} = \mathbf{VV}'$, and consequently,

$$\text{tr}(\mathbf{CB}) = \text{tr}(\mathbf{UU}'\mathbf{VV}') = \text{tr}(\mathbf{U}'\mathbf{VV}'\mathbf{U}) = \text{tr}((\mathbf{U}'\mathbf{V})(\mathbf{U}'\mathbf{V})') \geq 0.$$

Therefore,

$$\text{tr}((\mathbf{A} - \lambda_{\min}(\mathbf{A})\mathbf{I}_k)\mathbf{B}) \geq 0 \implies \text{tr}(\mathbf{AB}) \geq \lambda_{\min}(\mathbf{A}) \text{tr}(\mathbf{B}).$$

$\square$

Lemma 19. For any two matrices $\mathbf{A}$ and $\mathbf{B}$ compatible for multiplication such that $\mathbf{AB}$ is a $d \times d$ square matrix, we have that

$$\text{tr}(\mathbf{AB}) \leq \|\mathbf{A}\| \cdot \|\mathbf{B}\|.$$

Proof. Note that if $\mathbf{A}$ and $\mathbf{B}$ were square matrices the result would have been a simple application of Cauchy-Schwarz inequality. Suppose the sizes of the matrices $\mathbf{A}$ and $\mathbf{B}$ are $d \times n$ and $n \times d$, respectively. Let $\mathbf{A} = (A_{ij})$ and $\mathbf{B} = (B_{ij})$. We have that

$$\begin{aligned} |\text{tr}(\mathbf{AB})| &= |\langle \mathbf{A}, \mathbf{B}' \rangle_F| \quad (\langle \cdot, \cdot \rangle_F \text{ is the Frobenius inner product}) \\ &\leq \|\mathbf{A}\| \cdot \|\mathbf{B}'\| \quad (\text{Cauchy-Schwarz for inner product spaces}) \\ &= \|\mathbf{A}\| \cdot \|\mathbf{B}\|. \end{aligned}$$

$\square$

Lemma 20. We have that

$$\mathbb{E}[\text{tr}(\mathbf{WZ}'\mathbf{ZW}')] = \text{tr}(\Gamma_{\mathbf{W}}\Gamma_{\mathbf{Z}}) \quad \text{and} \quad \mathbb{E}[\mathbf{W}\Lambda\mathbf{W}'] = d\text{tr}(\Lambda\Sigma_{\mathbf{W}}) + \text{tr}(\mu_{\mathbf{W}}\Lambda\mu_{\mathbf{W}}'),$$

where the independent random variables $\mathbf{W}$ and $\mathbf{Z}$ are such that $\mathbf{W} \sim \mathcal{N}(\mu_{\mathbf{W}}, \mathbf{I}_d \otimes \Sigma_{\mathbf{W}})$ and $\mathbf{Z} \sim \mathcal{N}(\mu_{\mathbf{Z}}, \mathbf{I}_n \otimes \Sigma_{\mathbf{Z}})$.

Proof. Recall the following quantities for convenience

$$\Gamma_{\mathbf{W}} = d\Sigma_{\mathbf{W}} + \mu_{\mathbf{W}}'\mu_{\mathbf{W}} \quad \text{and} \quad \Gamma_{\mathbf{Z}} = n\Sigma_{\mathbf{Z}} + \mu_{\mathbf{Z}}'\mu_{\mathbf{Z}}.$$

We compute the above expectations using [Proposition 7](#) as follows:

$$\begin{aligned}
\mathbb{E}[\text{tr}(\mathbf{W}\mathbf{Z}'\mathbf{Z}\mathbf{W}')] &= \mathbb{E}[\text{tr}(\mathbf{Z}\mathbf{W}'\mathbf{W}\mathbf{Z}')] \\
&= \mathbb{E}[\text{tr}(\mathbb{E}[\mathbf{Z}\mathbf{W}'\mathbf{W}\mathbf{Z}' | \mathbf{W}])] \\
&= \mathbb{E}[\text{tr}(\text{tr}(\mathbf{W}'\mathbf{W}\Sigma_{\mathbf{Z}}) I_n + \mu_{\mathbf{Z}}(\mathbf{W}'\mathbf{W}) \mu'_{\mathbf{Z}})] \\
&= \mathbb{E}[n\text{tr}(\mathbf{W}'\mathbf{W}\Sigma_{\mathbf{Z}}) + \text{tr}(\mu_{\mathbf{Z}}(\mathbf{W}'\mathbf{W}) \mu'_{\mathbf{Z}})] \\
&= n\text{tr}(\mathbb{E}[\mathbf{W}'\mathbf{W}] \Sigma_{\mathbf{Z}}) + \text{tr}(\mu_{\mathbf{Z}} \mathbb{E}[\mathbf{W}'\mathbf{W}] \mu'_{\mathbf{Z}}) \\
&= n\text{tr}(\Gamma_{\mathbf{W}} \Sigma_{\mathbf{Z}}) + \text{tr}(\mu_{\mathbf{Z}} \Gamma_{\mathbf{W}} \mu'_{\mathbf{Z}}) \\
&= n\text{tr}(\Gamma_{\mathbf{W}} \Sigma_{\mathbf{Z}}) + \text{tr}(\Gamma_{\mathbf{W}} \mu'_{\mathbf{Z}} \mu_{\mathbf{Z}}) \\
&= \text{tr}(\Gamma_{\mathbf{W}} (n\Sigma_{\mathbf{Z}} + \mu'_{\mathbf{Z}} \mu_{\mathbf{Z}})) \\
&= \text{tr}(\Gamma_{\mathbf{W}} \Gamma_{\mathbf{Z}}).
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}[\text{tr}(\mathbf{W}\Lambda\mathbf{W}')] &= \text{tr}(\mathbb{E}[\mathbf{W}\Lambda\mathbf{W}']) \\
&= \text{tr}[\text{tr}(\Lambda\Sigma_{\mathbf{W}}) I_d + \mu_{\mathbf{W}}\Lambda\mu'_{\mathbf{W}}] \\
&= d\text{tr}(\Lambda\Sigma_{\mathbf{W}}) + \text{tr}(\mu_{\mathbf{W}}\Lambda\mu'_{\mathbf{W}}).
\end{aligned}$$

□

Lemma 21. For positive real numbers a_1, a_2, a_3, a_4 and b_1, b_2, b_3, b_4 , we have the following inequality

$$\frac{a_1 + a_2 + a_3 + a_4}{b_1 + b_2 + b_3 + b_4} \leq \max \left\{ \frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \frac{a_4}{b_4} \right\}.$$

Proof. We first prove the following inequality for positive real numbers a'_1, a'_2, b'_1, b'_2

$$\frac{a'_1 + a'_2}{b'_1 + b'_2} \leq \max \left\{ \frac{a'_1}{b'_1}, \frac{a'_2}{b'_2} \right\}.$$

There are two possible scenarios: either 1) $b'_1/b'_2 \leq a'_1/a'_2$, or 2) $b'_1/b'_2 > a'_1/a'_2$. It can be checked for the first case $(a'_1 + a'_2)/(b'_1 + b'_2) \leq a'_1/b'_1$, and for the second case $(a'_1 + a'_2)/(b'_1 + b'_2) < a'_2/b'_2$ hold. Let, $a'_1 = a_1 + a_2, a'_2 = a_3 + a_4$ and $b'_1 = b_1 + b_2, b'_2 = b_3 + b_4$. We have the left hand side is equal to

$$\begin{aligned}
\frac{a'_1 + a'_2}{b'_1 + b'_2} &\leq \max \left\{ \frac{a'_1}{b'_1}, \frac{a'_2}{b'_2} \right\} \\
&= \max \left\{ \frac{a_1 + a_2}{b_1 + b_2}, \frac{a_3 + a_4}{b_3 + b_4} \right\} \\
&\leq \max \left\{ \max \left\{ \frac{a_1}{b_1}, \frac{a_2}{b_2} \right\}, \max \left\{ \frac{a_3}{b_3}, \frac{a_4}{b_4} \right\} \right\}
\end{aligned}$$

$$= \max \left\{ \frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \frac{a_4}{b_4} \right\}.$$

□

Lemma 22. Let $\psi : (0, \infty) \rightarrow (0, \infty)$ be the function defined by $\psi(x) = x - 1 - \log x$. Let $c > 0$ be given, there exist real numbers ℓ_c, u_c such that $\ell_c < 1 < u_c$ and $[\ell_c, u_c] = \{x : \psi(x) \leq c\}$. Additionally, $e^{-(1+c)} \leq \ell_c \leq e^{-c}$.

Proof. The function ψ decreases monotonically on $(0, 1)$ and then increases monotonically on $(1, \infty)$. $\psi(t) = 0$ if and only if $t = 1$. Therefore, the first part of the claim follows. Since it must be the case that $\psi(\ell_c) = c$, we have that

$$\log(\ell_c) - \ell_c = -(1+c) \implies \ell_c e^{-\ell_c} = e^{-(c+1)} \implies \ell_c = e^{\ell_c} e^{-(1+c)} \in (e^{-(1+c)}, e^{-c}).$$

□

Lemma 23. For a $d \times d$ positive definite symmetric matrix $\mathbf{A}$ we have the following inequality

$$\frac{1}{\sqrt{d}} \|\mathbf{A}\| \leq \|\mathbf{A}\|_{op} \leq \|\mathbf{A}\|,$$

and the lower and upper bounds cannot be improved.

Proof. Skipped. Elementary. □

Lemma 24. Let $a, b, \alpha, \beta > 0$ be given and $t \geq 0$ be an integer. If $(\mu_{\mathbf{Z}}^{(t)}, \Sigma_{\mathbf{Z}}^{(t)} \alpha^{(t)}, \beta^{(t)}) = (a\mu_1, b, \alpha, \beta)$, then the next update $\beta^{(t+1)}$ in [Algorithm 4](#) is given by

$$\beta^{(t+1)} = \frac{1}{R(a, b, \alpha\beta)},$$

where R is defined in [\(32\)](#). Moreover, $R(u, v, w) \geq 1/\beta^{(0)}$ for any $(u, v, w) \in \mathbb{R}_+^3$. Consequently, the sequence $\beta^{(t)}$ is bounded above by $\beta^{(0)}$ for all $t \geq 1$.

Proof. We recall the mappings:

$$T : \mathbb{R}_+^3 \rightarrow \mathbb{R}_+^2, \quad (a, b, \xi) \mapsto \left(\frac{\sqrt{\lambda_1} \xi a}{\xi \Gamma_2(a, b) + \Lambda}, \frac{1}{\xi \Gamma_2(a, b) + \Lambda} \right),$$

with $T(a, b, \xi) = (T_1(a, b, \xi), T_2(a, b, \xi))$, and finally $\Gamma_1 : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$, $\Gamma_2 : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ are mappings defined by $\Gamma_1(x, y) = dy + x^2$, $\Gamma_2(u, v) = nv + u^2$. From the proof of [Theorem 11](#) in [subsection 3.4.3](#) (after applying F in the composition $H \circ G \circ F$), we have that

$$\begin{aligned} (\|\mu_{\mathbf{W}}^{(t+1)}\|, \Sigma_{\mathbf{W}}^{(t+1)}) &= (T_1(a, b, \alpha\beta), T_2(a, b, \alpha\beta)), \\ (\|\mu_{\mathbf{Z}}^{(t+1)}\|, \Sigma_{\mathbf{Z}}^{(t+1)}) &= (\Phi_1(a, b, \alpha\beta), \Phi_2(a, b, \alpha\beta)). \end{aligned}$$

Note that, $\mu_{\mathbf{Z}}^{(t+1)} = \|\mu_{\mathbf{Z}}^{(t+1)}\| \mu_1 = \Phi_1(a, b, \alpha\beta) \mu_1$ and $\mu_{\mathbf{W}}^{(t+1)} = \|\mu_{\mathbf{W}}^{(t+1)}\| \nu_1 = T_1(a, b, \alpha\beta) \nu_1$, where ν_1 is the unit norm right singular vector of $\mathbf{X}$ corresponding to μ_1 . From the update equation of $\beta^{(t+1)}$ in [Algorithm 4](#) we have that

$$\begin{aligned} \beta_{\tau}^{(t+1)} &= \left(\frac{1}{\beta_{\tau}^{(0)}} + \frac{1}{2} \text{tr} \left(\mathbf{X}' \mathbf{X} - (\mu_{\mathbf{W}}^{(t+1)}) (\mu_{\mathbf{Z}}^{(t+1)})' \mathbf{X} - \mathbf{X}' (\mu_{\mathbf{Z}}^{(t+1)}) (\mu_{\mathbf{W}}^{(t+1)})' \right) \right. \\ &\quad \left. + \frac{1}{2} \text{tr} \left((d \Sigma_{\mathbf{W}}^{(t+1)} + (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)}) (n \Sigma_{\mathbf{Z}}^{(t+1)} + (\mu_{\mathbf{Z}}^{(t+1)})' \mu_{\mathbf{Z}}^{(t+1)}) \right) \right)^{-1}. \end{aligned}$$

Upto the first trace term, we can simplify as

$$\begin{aligned} &\frac{1}{\beta_{\tau}^{(0)}} + \frac{1}{2} \text{tr} \left(\mathbf{X}' \mathbf{X} - \mu_{\mathbf{W}}^{(t+1)} (\mu_{\mathbf{Z}}^{(t+1)})' \mathbf{X} - \mathbf{X}' \mu_{\mathbf{Z}}^{(t+1)} (\mu_{\mathbf{W}}^{(t+1)})' \right) \\ &= c_1 - \text{tr} \left(\mathbf{X}' \mu_{\mathbf{Z}}^{(t+1)} (\mu_{\mathbf{W}}^{(t+1)})' \right) \\ &= c_1 - \sqrt{\lambda_1} \Phi_1(a, b, \alpha\beta) T_1(a, b, \alpha\beta) \text{tr}(\nu_1 \nu_1') \\ &= c_1 - \sqrt{\lambda_1} \Phi_1(a, b, \alpha\beta) T_1(a, b, \alpha\beta) \end{aligned}$$

by definition of c_1 . The claim for the expression of R follows after rewriting the final trace term using the mappings: $\Phi_1, \Phi_2, T_1, T_2, \Gamma_1, \Gamma_2$. The lower bound on R follows after completing square and rewriting R in the following way

$$\begin{aligned} &\frac{1}{\beta_{\tau}^{(0)}} + \frac{1}{2} \left\| \mathbf{X} - \mu_{\mathbf{W}}^{(t+1)} (\mu_{\mathbf{Z}}^{(t+1)})' \right\|^2 + \frac{1}{2} \text{tr} \left(d \Sigma_{\mathbf{W}}^{(t+1)} n \Sigma_{\mathbf{Z}}^{(t+1)} \right) \\ &+ \frac{1}{2} \text{tr} \left(d \Sigma_{\mathbf{W}}^{(t+1)} (\mu_{\mathbf{Z}}^{(t+1)})' \mu_{\mathbf{Z}}^{(t+1)} + n \Sigma_{\mathbf{Z}}^{(t+1)} (\mu_{\mathbf{W}}^{(t+1)})' \mu_{\mathbf{W}}^{(t+1)} \right) \geq \frac{1}{\beta_{\tau}^{(0)}}. \end{aligned}$$

□

3.4.4. Rotational ambiguity and flatness of loss function at stationary point

In this appendix, we show two things. First, if the prior hyperparameter Λ is poorly chosen, then Hessian of $\text{KL}(q^* \|\pi)$ is singular. Second, through numerical experiments, we show that when $k \geq 2$, the columns of $\mu_{\mathbf{Z}}^*$ are not orthogonal. For the purpose of this appendix, we treat τ as fixed: the same conclusions hold when τ is allowed to vary, since the arguments below do not depend on τ .

Proposition 8 (Rotational ambiguity). *Suppose the hyper-parameter $\Lambda = \lambda I_k$ for some $\lambda > 0$ and $q^* = (\mu_{\mathbf{W}}^*, \Sigma_{\mathbf{W}}^*, \mu_{\mathbf{Z}}^*, \Sigma_{\mathbf{Z}}^*) \in \mathcal{M}$ be a solution to the minimization problem in (21). Let $\mathbf{R}$ be any $k \times k$ orthogonal matrix. If we set $q^* \mathbf{R} = (\mu_{\mathbf{W}}^* \mathbf{R}, \mathbf{R}' \Sigma_{\mathbf{W}}^* \mathbf{R}, \mu_{\mathbf{Z}}^* \mathbf{R}, \mathbf{R}' \Sigma_{\mathbf{Z}}^* \mathbf{R})$, then $q^* \mathbf{R}$ is also a solution to the minimization problem in (21). Consequently, the Hessian of the loss function Ψ is singular if $k \geq 2$.*

Proof. We have that $\Lambda = \lambda I_k$ for some $\lambda > 0$. We note the loss function

$$\begin{aligned}\Psi(\mu_{\mathbf{W}}, \Sigma_{\mathbf{W}}, \mu_{\mathbf{Z}}, \Sigma_{\mathbf{Z}}) &= \frac{\tau_0}{2} \|\mathbf{X} - \mu_{\mathbf{Z}} \mu'_{\mathbf{W}}\|^2 + \frac{1}{2} \text{tr}(\mu'_{\mathbf{W}} \Lambda \mu_{\mathbf{W}}) + \frac{1}{2} \text{tr}(\mu'_{\mathbf{Z}} \mu_{\mathbf{Z}}) + \frac{d}{2} \text{tr}(\Lambda \Sigma_{\mathbf{W}}) \\ &\quad + \frac{n}{2} \text{tr}(\Sigma_{\mathbf{Z}}) + \frac{\tau_0 d n}{2} \text{tr}(\Sigma_{\mathbf{W}} \Sigma_{\mathbf{Z}}) + \frac{\tau_0 d}{2} \text{tr}(\Sigma_{\mathbf{W}} \mu'_{\mathbf{Z}} \mu_{\mathbf{Z}}) \\ &\quad + \frac{\tau_0 n}{2} \text{tr}(\mu'_{\mathbf{W}} \mu_{\mathbf{W}} \Sigma_{\mathbf{Z}}) - \frac{k}{2} \log \det(\Sigma_{\mathbf{W}}) - \frac{k}{2} \log \det(\Sigma_{\mathbf{Z}}) + \text{cst}.\end{aligned}$$

Therefore, $\Psi(q^*) = \Psi(q^* \mathbf{R})$, as all the trace terms and Frobenius norms remain invariant under the proposed transformations. $\square$

When Λ is a multiple of the identity matrix, [Proposition 8](#) presents an immediate roadblock to proving results such as [Theorem 12](#) whenever $k \geq 2$. This arises because arbitrarily close minimizers exist. More formally, for any $\eta > 0$, there exists a rotation $\mathbf{R}_\eta$ such that $\text{KL}(q^* \mathbf{R}_\eta \| q^*) \leq \eta$. In such cases, the singularity of the Hessian of the loss function Ψ leads to flatness at q^* , a phenomenon that is numerically demonstrated in the next subsection.

3.4.4.1. Numerical Experiment. In this subsection, we numerically illustrate that the Hessian of Ψ is flat at q^* when the prior hyperparameter Λ is proportional to I_k for $k \geq 2$. Conversely, when Λ is not proportional to I_k , the Hessian is non-singular, ensuring that q^* is an isolated local minimizer. Additionally, It is shown that the Hessian is always non-singular when $k = 1$, as there is no rotational ambiguity in this case. We focus on the simplest case, $k = 2$, although similar claims hold for higher rank models and can be easily verified numerically with our code. Note that the numerical experiments presented here are independent of CAVI, as our focus is on an analytical property of the loss function Ψ . Suppose $q^* = (\mu_{\mathbf{W}}^*, \Sigma_{\mathbf{W}}^*, \mu_{\mathbf{Z}}^*, \Sigma_{\mathbf{Z}}^*)$ is a local minimum (or a saddle point) of Ψ . We recall the loss function $\Psi(q^*)$ from above, and observe that the choice of model hyper-parameter Λ influences Ψ through the trace terms.

Let $\Lambda_1 = I_2$, $\Lambda_2 = \text{diag}(1, 2)$. We begin with the set-up: $n = 4$, $d = 3$, $k = 2$, $\tau_0 = 100$ and we set $\Lambda = \Lambda_2$, so that $\Lambda \not\propto I_2$. Each dataset, denoted as $\mathbf{X}_n$, is generated independently from the model in (41) with

$$\mathbf{X}_n \sim \mathcal{N}(\mathbf{Z}_0 \mathbf{W}'_0, \tau_0^{-1} I_n \otimes I_d), \quad (47)$$

where $\mathbf{W}_0 = (1, \dots, 1)'$ and $\mathbf{Z}_0 \sim \mathcal{N}(0, I_n \otimes I_k)$, with $\tau_0 = 100$. We run [Algorithm 4](#) with the initialization $\mu_{\mathbf{Z}}^{(0)}$, an $n \times k$ matrix with all entries set to 0.1, and $\Sigma_{\mathbf{Z}}^{(0)} = I_k$. To record the value of q^* , we first prepare a dataset $\mathbf{X}_4$ as described in (47) and then run CAVI ([Algorithm 4](#)). We also observed that, in this case, the two columns of $\mu_{\mathbf{Z}}^*$ form an angle of 56.6° and hence are not orthogonal.

However, this alone was insufficient to locate a stationary point of Ψ , as the gradient norm of Ψ at the converged point q_{CAVI}^* remained relatively large ($\approx 10^{-4}$). To reduce this value, we performed an additional round of Newton's descent on Ψ , initializing from q_{CAVI}^* and iterating until we reached the nearest local minimizer (or a saddle point) q^* , where the gradient norm

reduced to $\approx 10^{-14}$. Next, we repeat the same steps with the important difference being that $\Lambda = \Lambda_1$.

It is important to note that Newton's descent alone was ineffective in locating q^* , as it converged extremely slowly. In contrast, the combination of CAVI and Newton's descent yielded convergence in just a few seconds.

We compute the Hessian $H(\Psi)$ at q^* in both cases using Newton's method, and observe that $H(\Psi)$ becomes singular when $\Lambda = \Lambda_1$, exhibiting a zero eigenvalue with a corresponding eigenvector v_0 . This was empirically confirmed by the observation that, when $\Lambda \propto I_k$, the smallest eigenvalue (in absolute value) was on the order of machine precision ($\approx 10^{-15}$), whereas in the alternative case it was significantly larger ($\approx 10^{-2}$). This implies that q^* is not isolated when $\Lambda \propto I_k$, as Ψ is flat along the direction of v_0 , meaning $\Psi(q^* + \alpha v_0)$ remains constant for small perturbations α . However, when $\Lambda = \Lambda_2$, i.e., Λ is not proportional to I_2 , $H(\Psi)$ is non-singular, ensuring local convexity of Ψ at q^* . Consequently, q^* is an isolated local minimizer.

The local behavior of Ψ differs fundamentally when $k = 1$ as Λ is a positive real number in this case, eliminating rotational ambiguity in the model. In this case, $H(\Psi)$ is always non-singular, ensuring that Ψ is always locally convex regardless of the choice Λ . The situation is highlighted through the plots in [Figure 6](#).

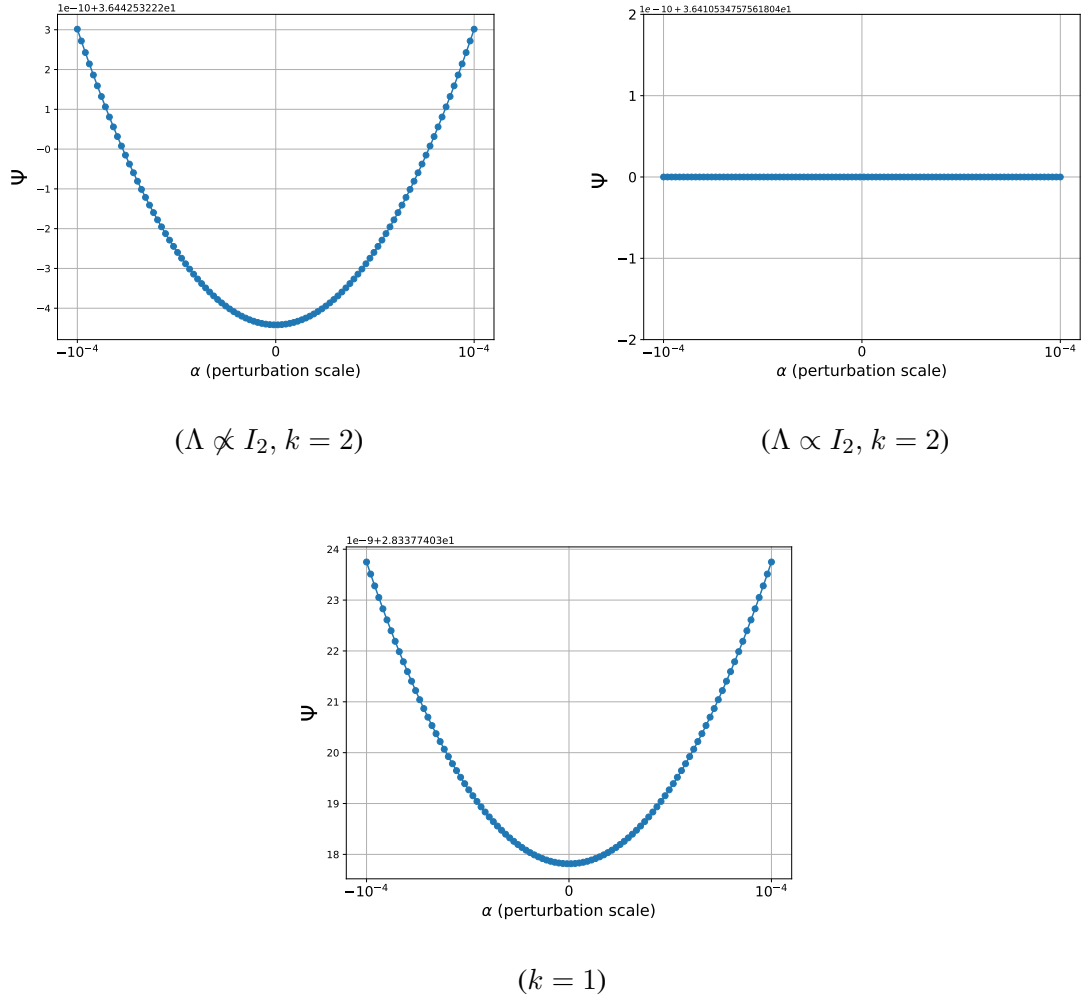


Figure 6. Visualization of Ψ near q^*

3.4.5. Full experimental details for the Monte Carlo study

As discussed in [section 3.3](#), by varying ρ , we expect BPCA to be more precise as $\rho \rightarrow 1$, and thus the numerator in (28) to become smaller. For each dataset, we set τ_0 to the maximum-likelihood estimate under the PPCA model. It is expected (and observed) that values of ρ very close to 1 correspond to larger values of τ_0 . Our numerical experiments suggest that irrespective of the value of τ_0 , the $1/2$ -generalized correlation stays above 2, which we interpret as a consequence of a non-sufficiently fast decay of the reconstruction term $\|\mathbf{X} - \mathbf{Z}\mathbf{W}'\|$ within the integral in (28).

Obtaining the global minimizer. For each dataset $\mathbf{X}_\rho^{(j)}$ where $j \in \{1, 2, \dots, 100\}$, we explicitly obtain the global minimizer $(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*) = (\mu_{\mathbf{W}}^*, \Sigma_{\mathbf{W}}^*, \mu_{\mathbf{Z}}^*, \Sigma_{\mathbf{Z}}^*)$ using Theorem 5 in [15].

CAVI run and fixed point. For each dataset $\mathbf{X}_\rho^{(j)}$ where $j \in \{1, 2, \dots, 100\}$, we initialize $q_{\mathbf{Z}}^{(0)} = \mathcal{N}(0, I_n \otimes I_k)$ and run CAVI with the fixed input value $\tau_0 = \tau_\rho^j$ until convergence. We record the

resulting stationary point

$$(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*) = (\mu_{\mathbf{W}}^*, \Sigma_{\mathbf{W}}^*, \mu_{\mathbf{Z}}^*, \Sigma_{\mathbf{Z}}^*).$$

In our experiments, these stationary points were observed to either coincide with the global analytic solutions in [15] or one of their rotational translates.

Perturbations near the fixed point. We generate small perturbations around this stationary point. For the mean of $q_{\mathbf{W}}^*$ we set $\mu_{\mathbf{W}}^{(\text{pert})} = \mu_{\mathbf{W}}^* + \epsilon \xi_{\mathbf{W}}$ with $\epsilon = 10^{-7}$ and $\xi_{\mathbf{W}}$ a standard matrix-normal random matrix of dimension $d \times k$. We then increase the positive eigenvalues of $\Sigma_{\mathbf{W}}^*$ by ϵ to obtain $\Sigma_{\mathbf{W}}^{(\text{pert})}$, while preserving positive definiteness. The same perturbation scheme is applied to $\mu_{\mathbf{Z}}^*$ and $\Sigma_{\mathbf{Z}}^*$. This ensures that the neighborhood radius r_0 in (27) remains sufficiently small ($\approx 10^{-10}$). We generate 100 independent realizations of $\xi_{\mathbf{W}}$, yielding 100 perturbed points near $(q_{\mathbf{W}}^*, q_{\mathbf{Z}}^*)$.

Proxy for generalized correlation. At each perturbed point $(q_{\mathbf{W}}^{(\text{pert})}, q_{\mathbf{Z}}^{(\text{pert})})$, we compute $\Delta(q_{\mathbf{W}}^{(\text{pert})}, q_{\mathbf{Z}}^{(\text{pert})})$ and form the proxy

$$\text{GCorr}_{1/2}^* := \frac{\Delta(q_{\mathbf{W}}^{(\text{pert})}, q_{\mathbf{Z}}^{(\text{pert})})}{\sqrt{D_{\text{KL},1/2}(q_{\mathbf{W}}^{(\text{pert})} \| q_{\mathbf{W}}^*) D_{\text{KL},1/2}(q_{\mathbf{Z}}^{(\text{pert})} \| q_{\mathbf{Z}}^*)}}.$$

Pathwise maximization along CAVI trajectories. We reinitialize CAVI from each perturbed distribution and follow the trajectory until it re-converges to the same stationary point, recording the maximum $\text{GCorr}_{1/2}^*$ observed along that path. For each fixed ρ , we take the maximum value observed across the simulated CAVI paths as a proxy for the overall 1/2-generalized correlation at that ρ .

Additionally, we observed exponential convergence of CAVI despite initializing at a large KL divergence from a stationary point. Figure 7 illustrates this behavior for the $\mathbf{Z}$ block, and we observed similar results for the $\mathbf{W}$ block. We observed such exponential convergence for every dataset $\mathbf{X}_{\rho}^{(j)}$ where $j \in \{1, 2, \dots, 100\}$.

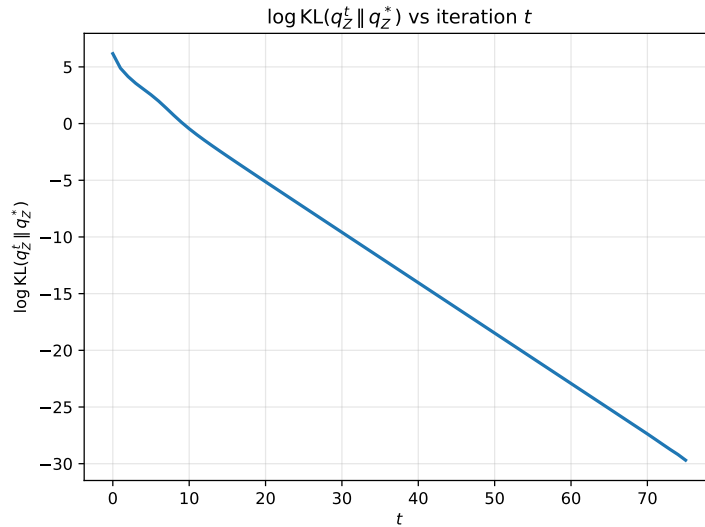


Figure 7. Local convergence of the **Z** block.

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Chapter 4

Future research directions and some open questions

4.1. Some open questions from [chapter 1](#)

In this section we particularly talk about an open question that naturally arises from the work in [chapter 1](#). The question is mentioned in [section 1.8](#) where we ask whether one can obtain rates of convergence or prove something stronger, such as asymptotic normality of the MLE for the PPCA model within the quotient space.

Unfortunately, for a general quotient topological space, obtaining rates seems difficult, since it would require a notion of smoothness (for example, differentiability) on the quotient space, which is not available in general unless the quotient is a Riemannian manifold. In cases where the quotient does admit a natural Riemannian manifold structure, as we show in [subsection 4.1.1](#) for PPCA, one possible route is to adapt recent manifold-based M -estimation results, which establish consistency and asymptotic normality under appropriate regularity assumptions [\[3, 8\]](#).

However, the above route has its limitations, because it applies only when the quotient parameter space admits enough smooth manifold structure; PPCA is tractable precisely because its identifiable quotient has this regularity. A potentially broader alternative is to develop a local asymptotic normality (LAN) expansion in quotient spaces. In regular parametric problems, LAN provides a local quadratic expansion of the log-likelihood ratio around the true parameter and is a standard route to asymptotic normality and efficiency results [\[9, 7\]](#). Motivated by recent geometric analyses of PCA on the Grassmannian, which establish local quadratic control of the objective in geodesic coordinates (Corollary 2, [\[5\]](#)), it may be possible to prove analogous LAN-type expansions, or at least local quadratic inequalities, for broader classes of quotient spaces. Developing such a theory would be ambitious, but it seems like a promising direction for future work.

4.1.1. Riemannian structure on Θ/C

Recall that

$$\Theta := \{(\mathbf{W}, \sigma^2) \in \mathbb{R}^{p \times q} \times \mathbb{R}_{>0} : \text{rank}(\mathbf{W}) = q\}.$$

Since the set of full-column-rank $p \times q$ matrices is open in $\mathbb{R}^{p \times q}$, the set Θ is an open subset of $\mathbb{R}^{p \times q} \times \mathbb{R}$, hence a smooth manifold of dimension

$$\dim(\Theta) = pq + 1.$$

Following [chapter 1](#), we recall

$$C = \{(\mathbf{W}, \sigma^2) \in \Theta : \mathbf{W}\mathbf{W}^T + \sigma^2 I_p = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p\}.$$

We have the following theorem that characterizes all elements in C .

Theorem 13. *Suppose $1 \leq q < p$ be integers. Let $O(q)$ denote the group of orthogonal $q \times q$ matrices. We have that*

$$C = \{(\mathbf{W}_0 \mathbf{R}, \sigma_0^2) : \mathbf{R} \in O(q)\}.$$

In particular, C is exactly the orbit of θ_0 under the right action of $O(q)$ on Θ given by

$$(\mathbf{W}, \sigma^2) \cdot \mathbf{R} := (\mathbf{W}\mathbf{R}, \sigma^2), \quad \mathbf{R} \in O(q).$$

Proof. If $\mathbf{W} = \mathbf{W}_0 \mathbf{R}$ for some $\mathbf{R} \in O(q)$ and $\sigma^2 = \sigma_0^2$, then

$$\mathbf{W}\mathbf{W}^T + \sigma^2 I_p = \mathbf{W}_0 \mathbf{R} \mathbf{R}^T \mathbf{W}_0^T + \sigma_0^2 I_p = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p,$$

so $(\mathbf{W}, \sigma^2) \in C$.

Conversely, let $(\mathbf{W}, \sigma^2) \in C$. Then

$$\mathbf{W}\mathbf{W}^T + \sigma^2 I_p = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 I_p.$$

Suppose first that $\sigma^2 > \sigma_0^2$. Rearranging the identity above gives

$$\mathbf{W}_0 \mathbf{W}_0^T - \mathbf{W}\mathbf{W}^T = (\sigma^2 - \sigma_0^2) I_p.$$

Since $\sigma^2 - \sigma_0^2 > 0$, the matrix on the right-hand side is positive definite. Moreover, $\mathbf{W}\mathbf{W}^T$ is positive semidefinite. Hence, for every nonzero $y \in \mathbb{R}^p$,

$$y^T \mathbf{W}_0 \mathbf{W}_0^T y = y^T (\mathbf{W}_0 \mathbf{W}_0^T - \mathbf{W}\mathbf{W}^T) y + y^T \mathbf{W}\mathbf{W}^T y > 0.$$

Thus $\mathbf{W}_0 \mathbf{W}_0^T$ is positive definite. But a positive definite $p \times p$ matrix must have full rank p , whereas $\text{rank}(\mathbf{W}_0 \mathbf{W}_0^T) = q < p$, a contradiction. Therefore $\sigma^2 > \sigma_0^2$ cannot occur.

Interchanging the roles of $(\mathbf{W}, \sigma^2)$ and $(\mathbf{W}_0, \sigma_0^2)$, the same argument shows that $\sigma^2 < \sigma_0^2$ is also impossible. We conclude that

$$\sigma^2 = \sigma_0^2.$$

Thus, we have that

$$\mathbf{W}\mathbf{W}^T = \mathbf{W}_0\mathbf{W}_0^T.$$

By a standard linear algebra argument we see that there exists an orthogonal matrix $\mathbf{R} \in O(q)$ such that

$$\mathbf{W} = \mathbf{W}_0\mathbf{R}.$$

□

Proposition 9. *The quotient space Θ/C can be identified with the orbit of the right action of $O(q)$ on Θ . Moreover, Θ/C is a smooth manifold of dimension*

$$pq + 1 - \frac{q(q-1)}{2}.$$

Proof. By Theorem 13, each equivalence class is exactly an $O(q)$ -orbit. Thus Θ/C is canonically identified with the orbit space of the above action.

Consider the smooth right action

$$O(q) \times \Theta \rightarrow \Theta, \quad (\mathbf{R}, (\mathbf{W}, \sigma^2)) \mapsto (\mathbf{WR}, \sigma^2).$$

This action is free. Indeed, if $(\mathbf{WR}, \sigma^2) = (\mathbf{W}, \sigma^2)$, then $\mathbf{WR} = \mathbf{W}$, and since $\mathbf{W}$ has full column rank, it follows that $\mathbf{R} = I_q$. The action is also proper because $O(q)$ is compact. By the standard quotient manifold theorem for free and proper Lie group actions; see, for example, [2, Chapter 9], the orbit space Θ/C is a smooth manifold and the canonical projection

$$\pi : \Theta \rightarrow \Theta/C$$

is a smooth submersion.

Finally, since $\dim(\Theta) = pq + 1$ and

$$\dim O(q) = \frac{q(q-1)}{2},$$

we obtain

$$\dim(\Theta/C) = pq + 1 - \frac{q(q-1)}{2}.$$

□

We now equip Θ/C with a natural Riemannian structure. Endow Θ with the Euclidean product metric

$$g_{(\mathbf{W}, \sigma^2)}((\mathbf{A}, a), (\mathbf{B}, b)) := \text{tr}(\mathbf{A}^T \mathbf{B}) + ab,$$

for tangent vectors $(\mathbf{A}, a), (\mathbf{B}, b) \in T_{(\mathbf{W}, \sigma^2)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$. The action of $O(q)$ preserves this metric, since for every $\mathbf{R} \in O(q)$,

$$g_{(\mathbf{WR}, \sigma^2)}((\mathbf{AR}, a), (\mathbf{BR}, b)) = \text{tr}((\mathbf{AR})^T (\mathbf{BR})) + ab = \text{tr}(\mathbf{A}^T \mathbf{B}) + ab.$$

Thus, we have the following corollary.

Corollary 4. *The quotient space Θ/C carries a natural quotient Riemannian metric induced by the Euclidean product metric on Θ . Consequently, Θ/C is a Riemannian manifold.*

4.2. Some open questions from [chapter 2](#)

In this chapter, we developed a robust Bayesian PCA model that allows us to carry out statistical inference in the presence of outliers. The numerical experiments in that work focused mainly on the case where the density function of the error term is regularly varying. However, the theoretical results suggest that using log-regularly varying functions could allow us to achieve whole robustness. In this regard, extending the latent-variable augmentation strategy of Hamura et al. [6] to higher dimensions appears to be a promising direction for future work. More precisely, one question we would like to pursue is whether one can construct a computationally feasible sampler or optimization scheme for multivariate log-regularly varying BPCA, and whether such a method delivers the promised whole robustness in finite samples. Answering this question would provide a natural follow-up to the present work, as it would help close the main gap between theory and practice.

As a second open question, it would also be interesting to design numerical experiments aimed at better understanding the extent to which [Assumption 3](#) is optimal in [chapter 2](#), as suggested in the discussion at the end of [subsection 2.2.3](#).

4.3. Some open questions from [chapter 3](#)

We now discuss two limitations and motivate two open questions from this chapter. A limitation of our work lies in the assumptions of [Theorem 12](#). In particular, we expect the result to hold under a more general version of the condition in (28), with an upper bound that may depend on the problem parameters and on the data. The condition in (28) is derived from the sufficient condition proposed by Bhattacharya et al. [1, eq. (14)], and therefore also reflects a limitation of their abstract framework.

We also expect that the non-singularity assumption on the Hessian in [Theorem 12](#) may not be necessary. Proving a result of the same type without this condition would likely require a substantially different proof technique. Rather than analyzing the local behavior of CAVI through a uniform control of the generalized correlation on a KL ball, one would need to work directly with CAVI trajectories. In this sense, our work highlights the need for stronger theoretical tools in the study of CAVI convergence rates, especially in more general settings.

Another limitation of BPCA is that the number of principal components, namely k , is treated as a hyperparameter and is therefore assumed to be fixed by the user. In practice, however, it is often desirable to learn this hyperparameter in a data-driven way. A natural extension of our

work would thus be to establish exponential convergence of CAVI in the setting where k is selected using the ELBO-based model-selection procedure proposed by Chérif-Abdellatif [4]. Such a result would make the methodology more practical and would provide a more complete theoretical understanding of BPCA in realistic applications.

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Chapter 5

Conclusion

In this thesis, we studied different probabilistic variants of PCA and a common issue we have always dealt with is non-identifiability. A central message of the thesis is that non-identifiability should not be viewed only as a technical obstacle. Rather, it forces us to identify the right object on which statistical inference should be carried out. Starting from a general topological point of view, we gradually narrowed our focus to PPCA and BPCA highlighting this right object in the subsequent chapters.

In [chapter 1](#), we developed a general framework based on quotient topological spaces in order to study consistency of maximum likelihood estimator in non-identifiable models, and we applied it to PPCA. This allowed us to give a rigorous justification of maximum likelihood estimation in a model where the parameter is only recoverable up to an equivalence class. More broadly, this chapter shows that quotient constructions provide a natural language for statistical inference in the presence of symmetries in model parameterization.

In [chapter 2](#), we then presented a robust Bayesian extension of PPCA and studied how to perform robust statistical inference when the data contain outliers. The proposed robust Bayesian PCA model is based on heavy-tailed errors, we also established theoretical results regarding robustness properties. On the computational side, we showed that the model remains tractable through a latent-variable augmentation. We complemented the theory with numerical experiments comparing Gibbs sampling and CAVI, and with a statistical study of principal component inference through an identifiable low-rank reconstruction. In this sense, the chapter provided both theory and computational methods on how to proceed when outliers are present.

Finally, in [chapter 3](#), we contributed to the growing literature on the theoretical analysis of CAVI. We focused on Bayesian PCA with normal errors, which is a particularly challenging probabilistic model because it combines latent variables, non-identifiability, and multimodality. In this setting, we obtained theoretical results shedding light on the local convergence behaviour of CAVI and supported the analysis with numerical evidence.

The thesis also asks several open research questions, which are discussed in [chapter 4](#) in detail. These include stronger asymptotic results on quotient spaces, new computationally feasible robust Bayesian PCA models based on log-regularly varying errors, a better understanding of the sharpness of the assumptions used in [Theorem 10](#), and further progress on the convergence analysis of CAVI for Bayesian PCA and for non-log-concave models, in general. We hope that the present work provides a useful step toward a broader theory of inference and computation for non-identifiable statistical models.